LMGB · SESSION COMPANION · INDIA · GURUGRAM

Location analytics, from intuition to evidence

Every idea is built on the board first, from something you already know, and then run on real data from Gurugram, India in the browser workbench.

Tool: **quantng.github.io/location-lab** · country guide: **quantng.github.io/location-lab/india/**

Chrome or Edge on a laptop. Nothing to install. Files you upload never leave your laptop.

THE PLACE

Gurugram in one slide

Gurugram sits 30 km south-west of central New Delhi in the state of Haryana, part of the National Capital Region (NCR). In thirty years it grew from farmland into one of India's main corporate hubs: global IT and consulting offices, BPO campuses, malls and gated high-rises.

Growth followed the roads and the metro, not a master plan, so services cluster sharply along a few corridors. Cyber City and Cyber Hub form the office and dining core; the Yellow Line and Rapid Metro carry commuters; MDI's campus lies in the Sukhrali area near the city centre.

Currency	Indian rupee (₹)
Admin units	Districts of the NCR (Gurugram, Delhi districts, Faridabad, Gautam Buddha Nagar …)
Business core	Cyber City · Cyber Hub · Golf Course Road · Sohna Road
Transit	Delhi Metro Yellow Line, Rapid Metro Gurgaon; the Old Gurugram line is under construction
Why it matters here	Dense, uneven, market-driven: ideal for testing clustering and hotspots

THE DATA

What is in the catalogue

Cafés & restaurants (459, OpenStreetMap): name, type (cafe, restaurant, fast_food), cuisine.

Metro stations (275 across the NCR). **NCR districts** (25 polygons, geoBoundaries). **MDI campus** outline. **NDVI**: Sentinel-2 greenness around MDI, 25 Oct 2024, as a Cloud-Optimised GeoTIFF.

Every vector layer ships as GeoJSON (plain text), GeoParquet (compressed columns) and FlatGeobuf (with a built-in spatial index). Pick the format in the Add data window and compare the load size.

Layer	GeoJSON	GeoParquet	FlatGeobuf
Cafés & restaurants	102 KB	32 KB	80 KB
MDI campus	1 KB	10 KB	2 KB
NCR districts	137 KB	97 KB	86 KB
Metro stations	51 KB	22 KB	43 KB

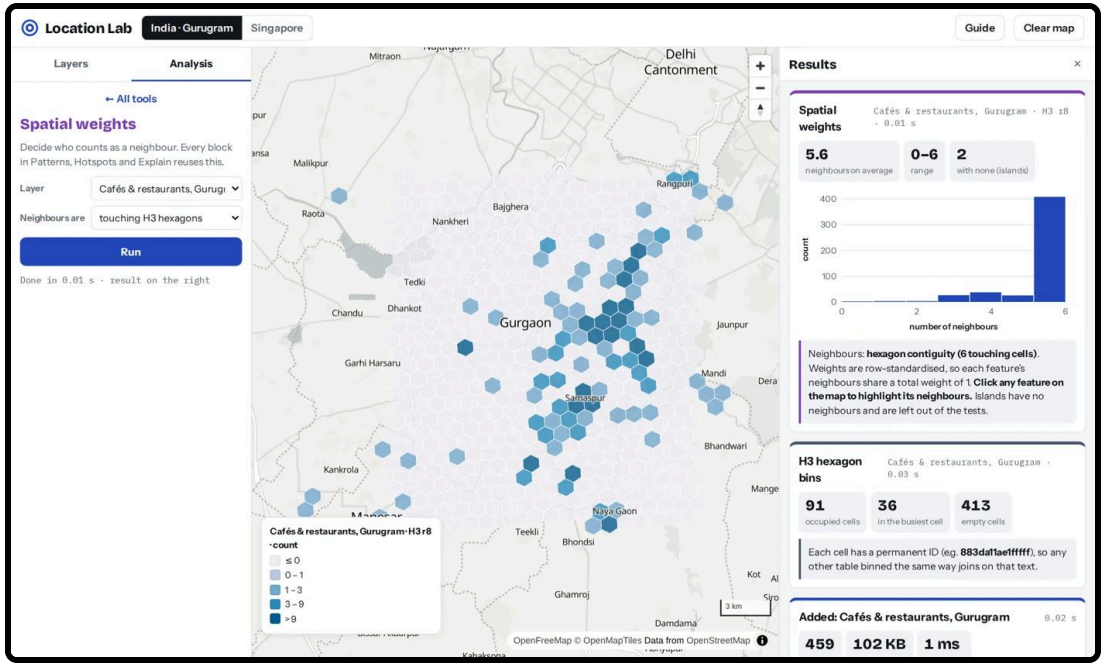
THE TOOL

Three moves: add, drop, read

1 · Add data. Layers tab → **+ Add data** → choose the file format → click a dataset. It appears as a layer card; use the card to colour by any field, zoom, open the table or download.

2 · Drop an analysis. Analysis tab → drag a tool onto the map or onto a layer card (or just click it). Check the settings, press **Run**.

3 · Read the result. The map restyles, the legend updates, and the Results drawer on the right holds the numbers, the chart and a one-line reading. Click any feature to see its values.



THE SESSION IN ONE LINE

Is “where” random? If not, why not?

Part	Question	Tools
1 · Spatial data recap	Load, measure, filter, join, bin, smooth, read a satellite image	Nearest · SQL · Box→distance · Join · H3 · Heatmap · NDVI
2 · Neighbours and lag	Who counts as “near”, and how are they doing?	Weights · Spatial lag
3 · Point patterns	Clustered, dispersed or random?	Point pattern test
4 · Autocorrelation	Do similar values sit together, and where?	Moran’s I · LISA · Gi*
5 · Clusters	Which points form one market?	DBSCAN
6 · Surfaces	What is the value between samples?	Sample · IDW · Kriging
7 · Explain	Which factors, and do they vary by place?	Distance · OLS · Lag · Error · GWR

PART 1

Spatial data recap

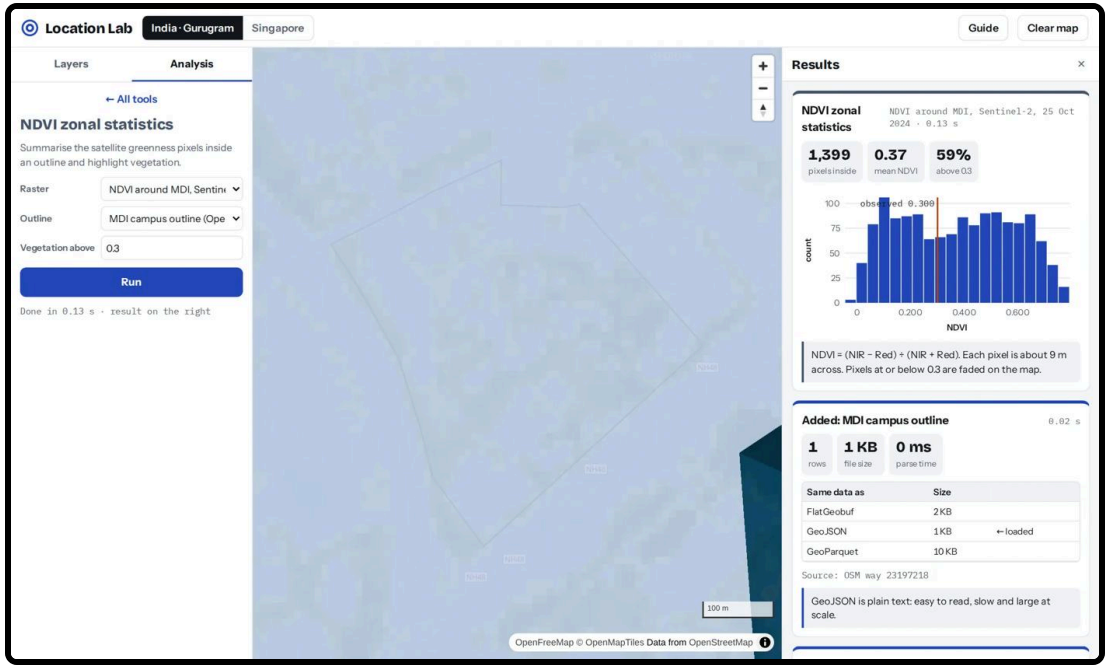
The path we take on the board:

Load → **Measure** → **Filter** → **Join** → **Bin** → **Smooth** → **Pixels**

Seven moves you already know

Last session you learned the grammar of spatial data. A file carries **geometry** (where) and **attributes** (what). You **measured** distance on a curved Earth, **filtered** cheaply with a bounding box before the exact test, **joined** points to the polygon they fall in, **binned** points into H3 hexagons, **smoothed** them into a heatmap and **summarised** satellite pixels inside an outline.

Today's question is new: once the points are on the map, is their arrangement telling us something, or could it be chance?



Where business already uses this

IN THE NEWS · THE WEEK · 31 MAY 2025

India Post launches DIGIPIN, a 4 m × 4 m code for every spot in India

A national grid with a permanent code per cell, built from latitude and longitude: the same idea as H3 hexagons, applied to addressing.

https://www.theweek.in/news/sci-tech/2025/05/31/what-is-digipin-india-new-technology-postal-geo-address-system-department-of-posts.html

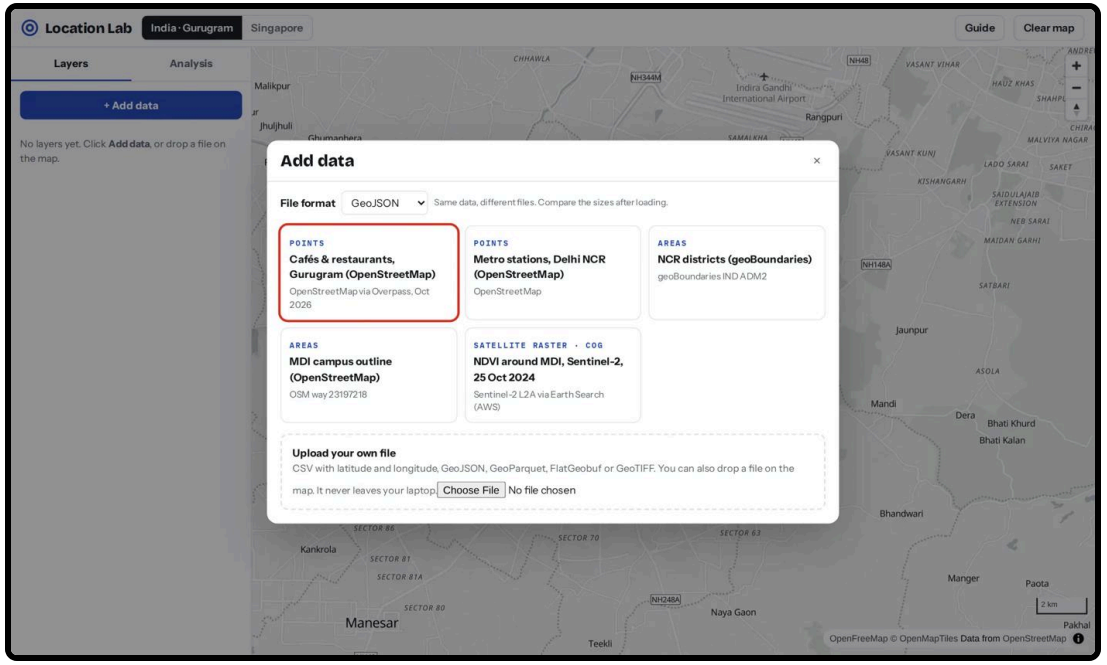
LAB 1 · INDIA · GURUGRAM

1 · Spatial data in seven moves

Recap of the last session on one dataset: load, measure, filter, join, bin, smooth and read a satellite image.

Open **quantng.github.io/location-lab**, choose **India · Gurugram** in the top bar, then follow the steps.

1. Load the 459 cafés and restaurants.
2. Explore the table: fields, blanks, colour by type.
3. Nearest places to MDI, with 1, 3 and 5 km rings.
4. Ask the same question in SQL.
5. Box filter, then exact 3 km distance (150 → 124).
6. Load the NCR district outlines.
7. Spatial join: which district is each place in?
8. Bin into H3 hexagons and raise them in 3D.
9. Smooth the points into a heatmap.
10. Load the NDVI satellite raster (COG).
11. Load the MDI campus outline.
12. Zonal statistics: how green is the campus?



Load the 459 cafés and restaurants.

DO THIS

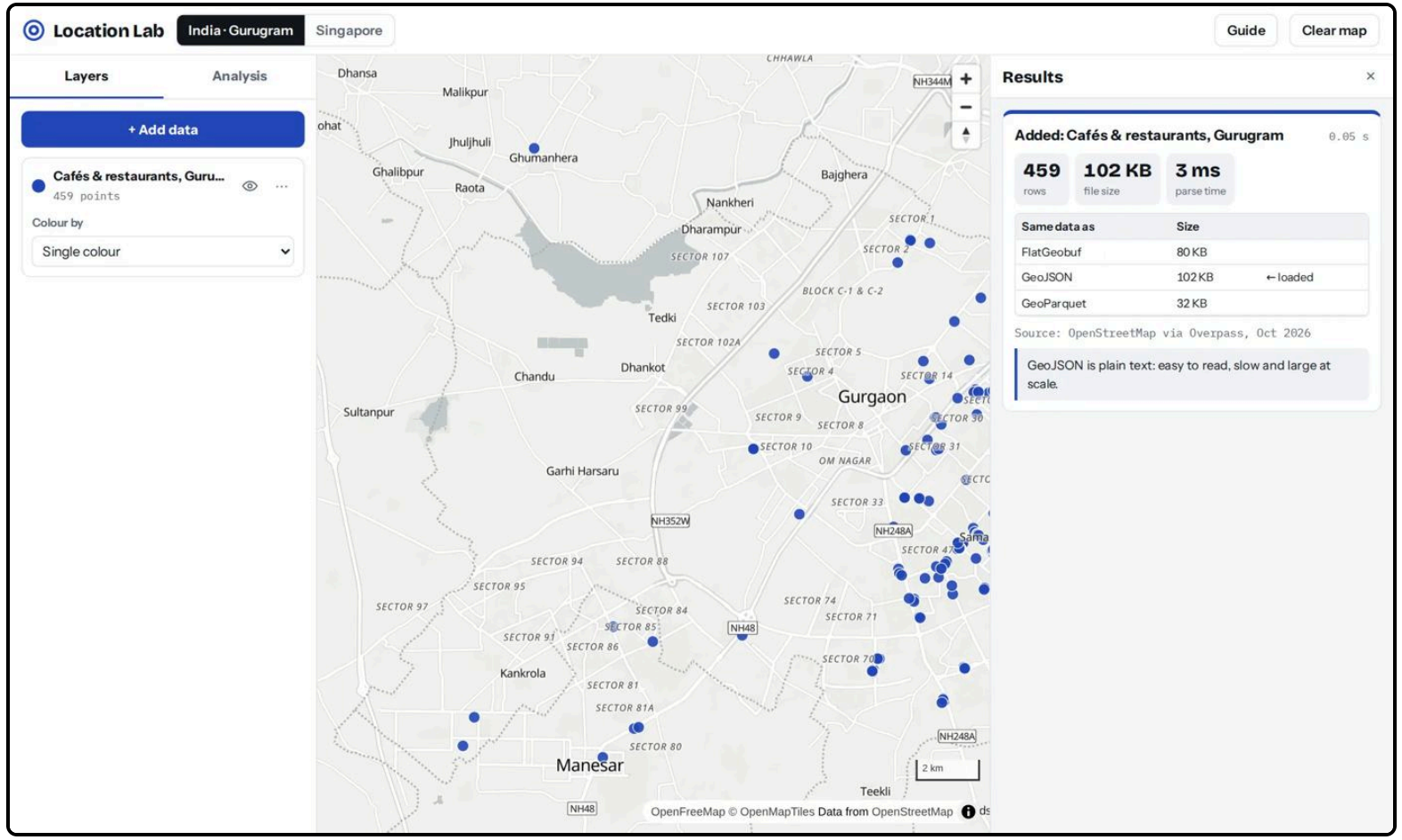
Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	Cafés & restaurants, Gurugram (OpenStreetMap)
File format	GeoJSON

YOU SHOULD SEE

459	102 KB	3 ms
rows	file size	parse time

> GeoJSON is plain text: easy to read, slow and large at scale.



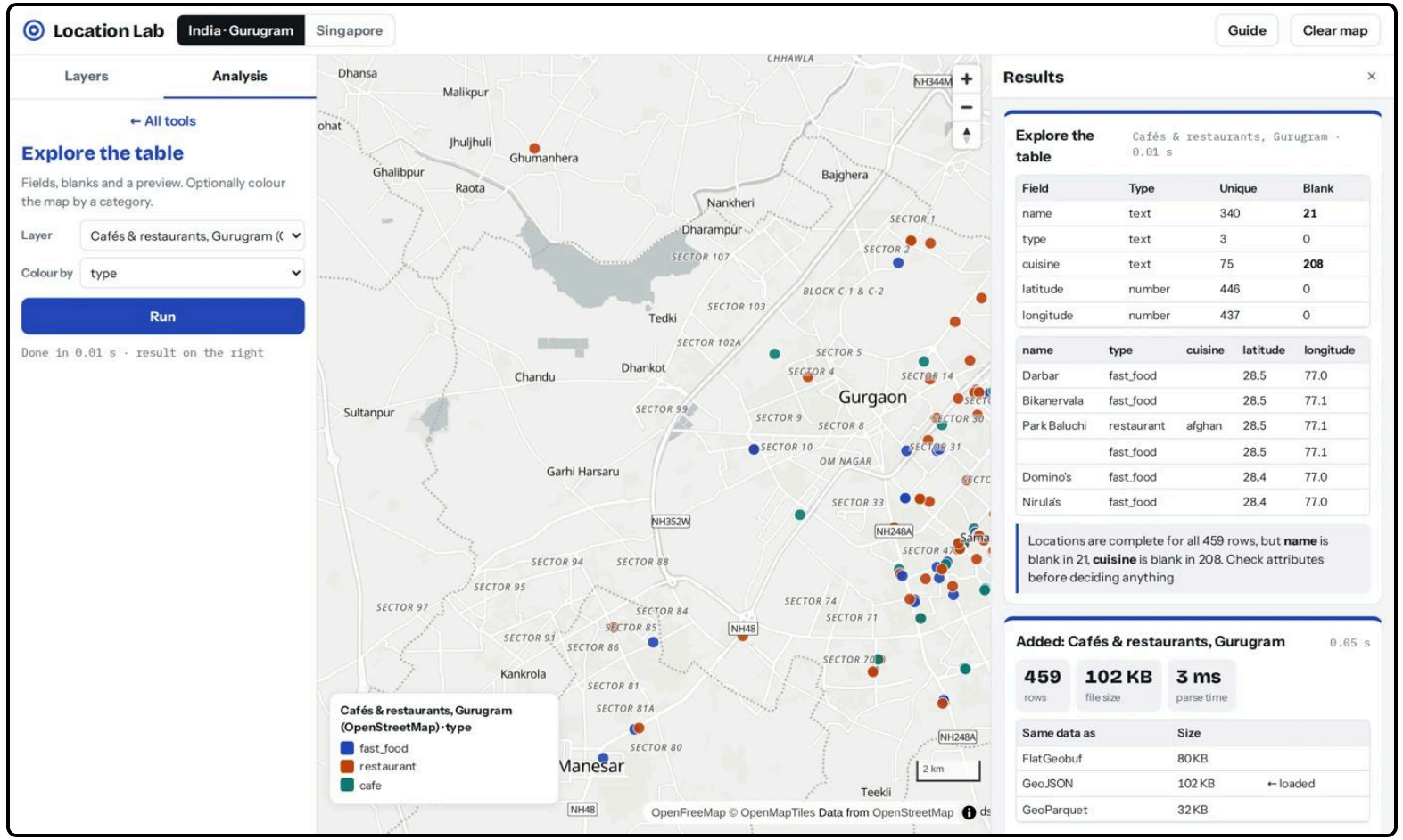
Explore the table: fields, blanks, colour by type.

DO THIS

Analysis tab → drag **Explore the table** onto the map (or click it), check the settings, press **Run**.

Layer	Cafés & restaurants, Gurugram (OpenStreetMap)
Colour by	type

> Locations are complete for all 459 rows, but name is blank in 21, cuisine is blank in 208. Check attributes before deciding anything.



Nearest places to MDI, with 1, 3 and 5 km rings.

DO THIS

Analysis tab → drag **Nearest to a point** onto the map (or click it), check the settings, press **Run**.

Layer	Cafés & restaurants, Gurugram (OpenStreetMap)
From	MDI Gurgaon campus centre
Show nearest	5

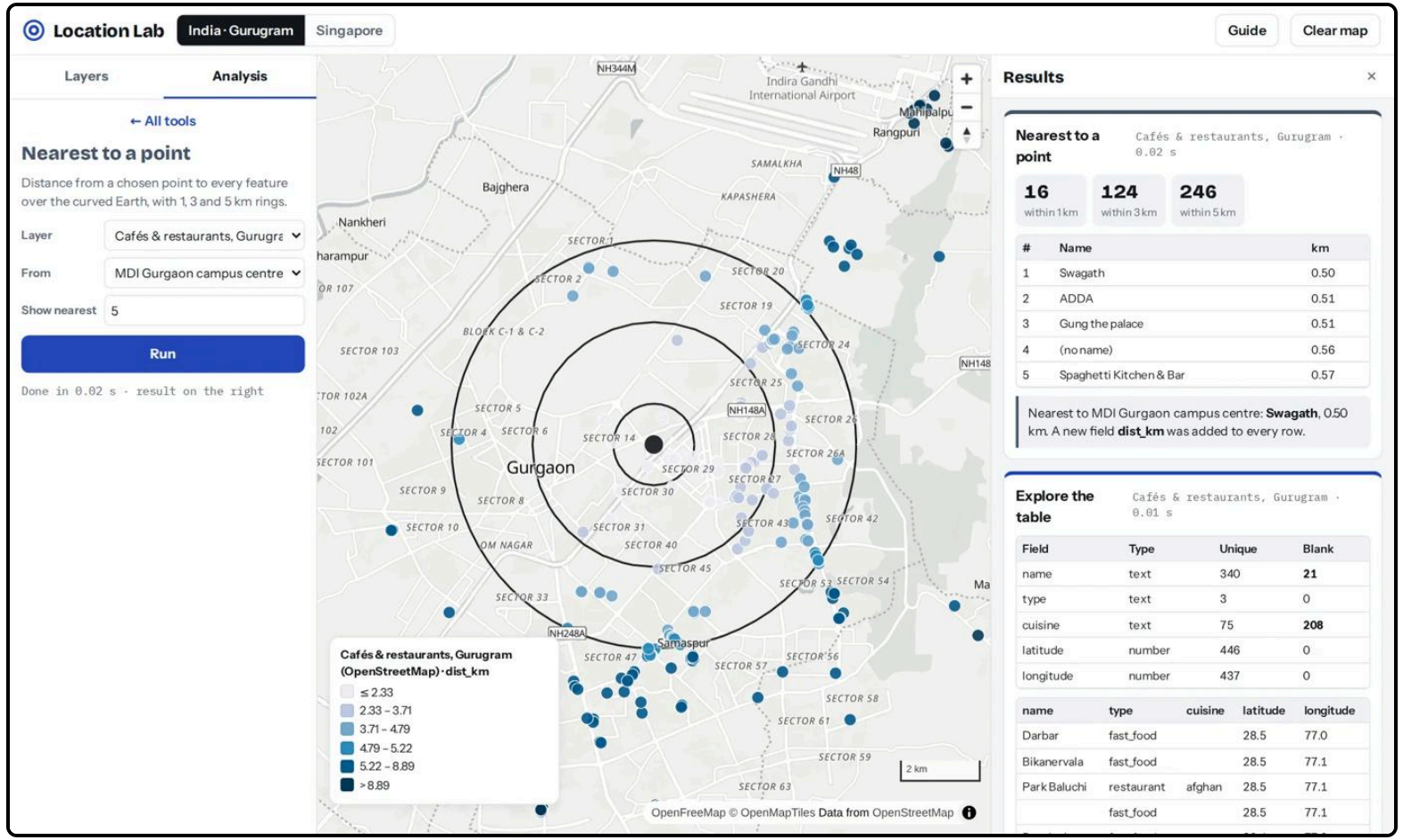
YOU SHOULD SEE

16 within 1 km

124 within 3 km

246 within 5 km

> Nearest to MDI Gurgaon campus centre: Swagath, 0.50 km. A new field dist_km was added to every row.

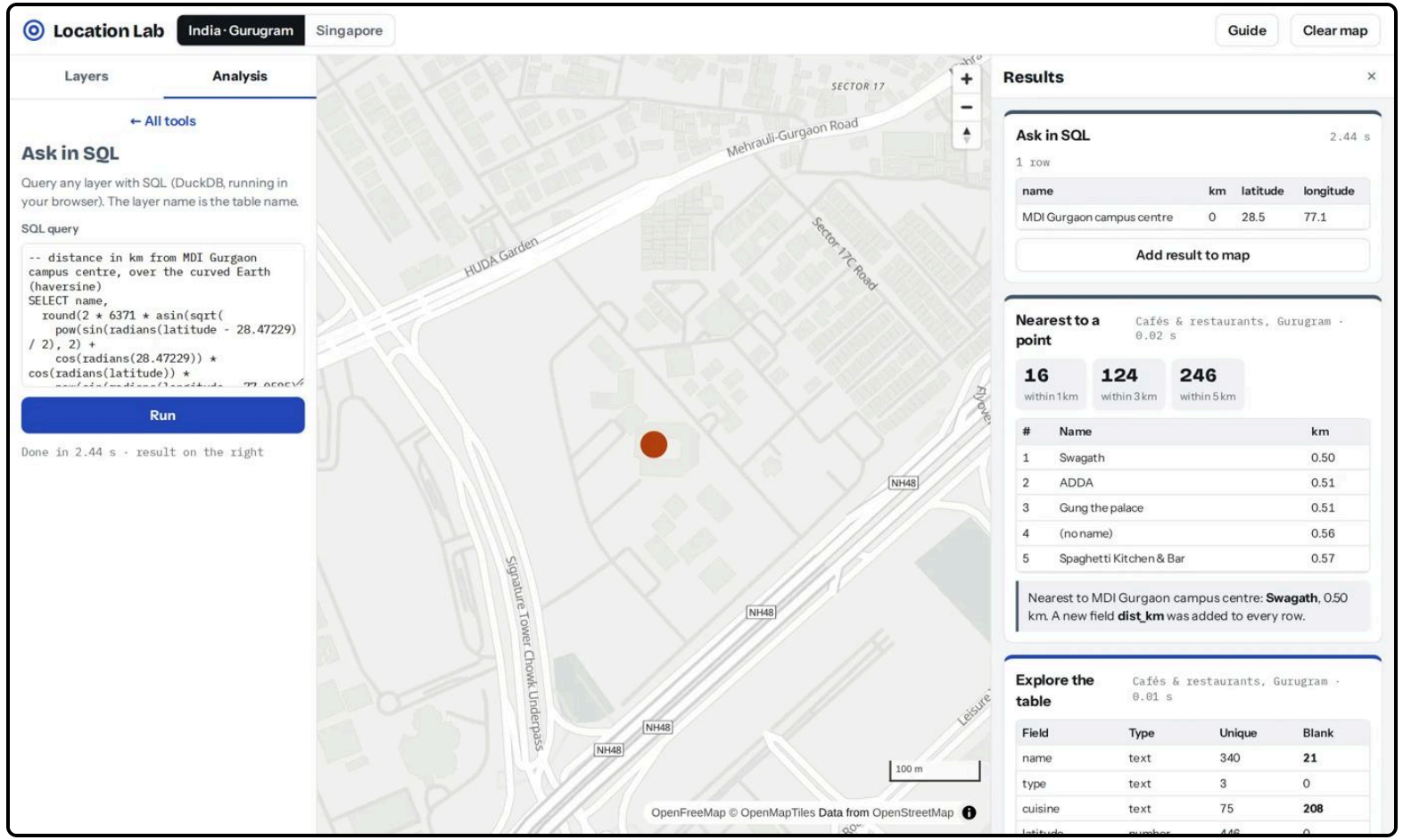


Ask the same question in SQL.

DO THIS

Analysis tab → drag **Ask in SQL** onto the map (or click it), check the settings, press **Run**.

SQL query	(query in the box)



Box filter, then exact 3 km distance (150 → 124).

DO THIS

Analysis tab → drag **Box filter, then distance** onto the map (or click it), check the settings, press **Run**.

Layer	Cafés & restaurants, Gurugram (OpenStreetMap)
Centre	MDI Gurgaon campus centre
Radius (km)	3

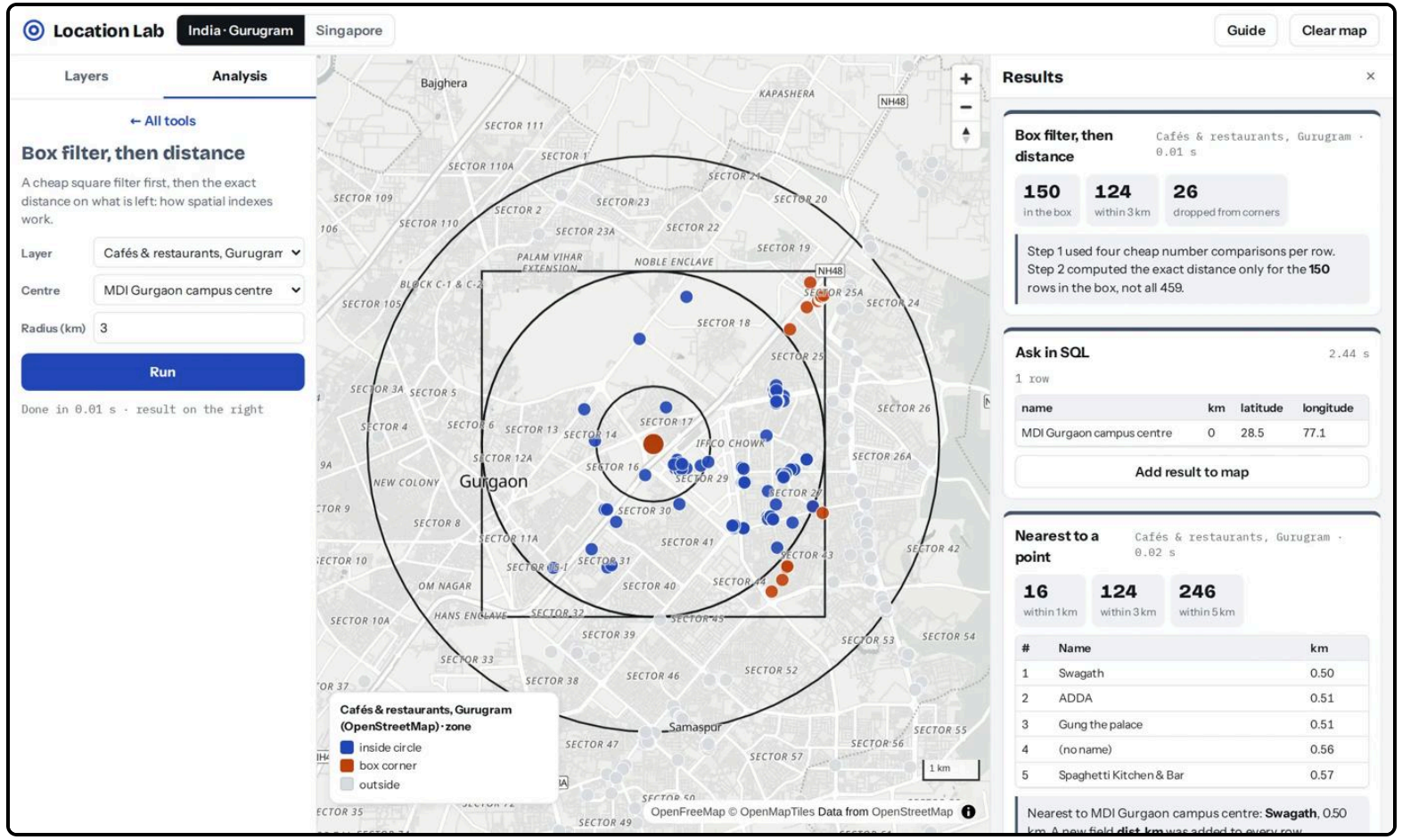
YOU SHOULD SEE

150
in the box

124
within 3 km

26
dropped from corners

> Step 1 used four cheap number comparisons per row. Step 2 computed the exact distance only for the 150 rows in the box, not all 459.



Load the NCR district outlines.

DO THIS

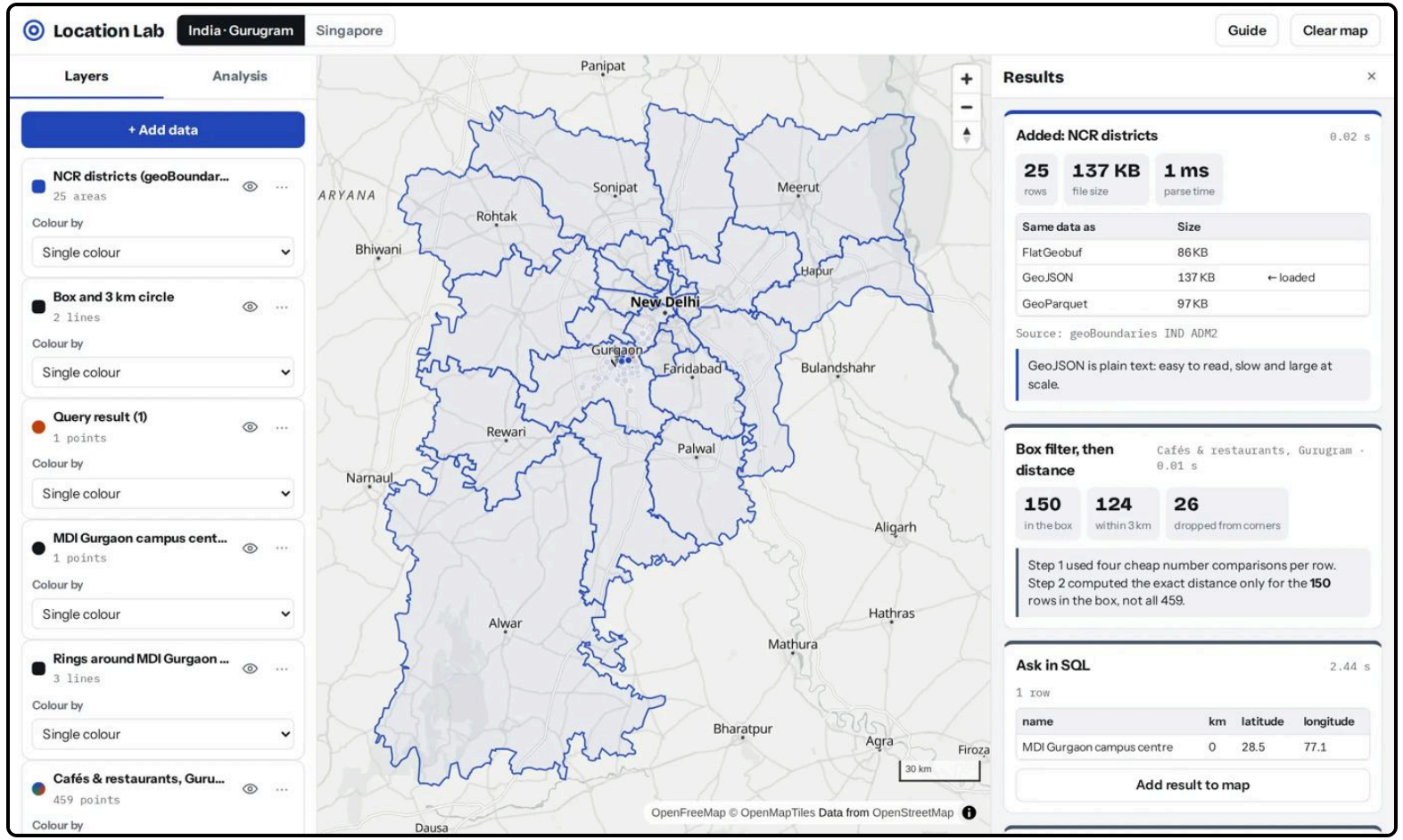
Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	NCR districts (geoBoundaries)
File format	GeoJSON

YOU SHOULD SEE

25 rows | **137 KB** file size | **1 ms** parse time

> GeoJSON is plain text: easy to read, slow and large at scale.



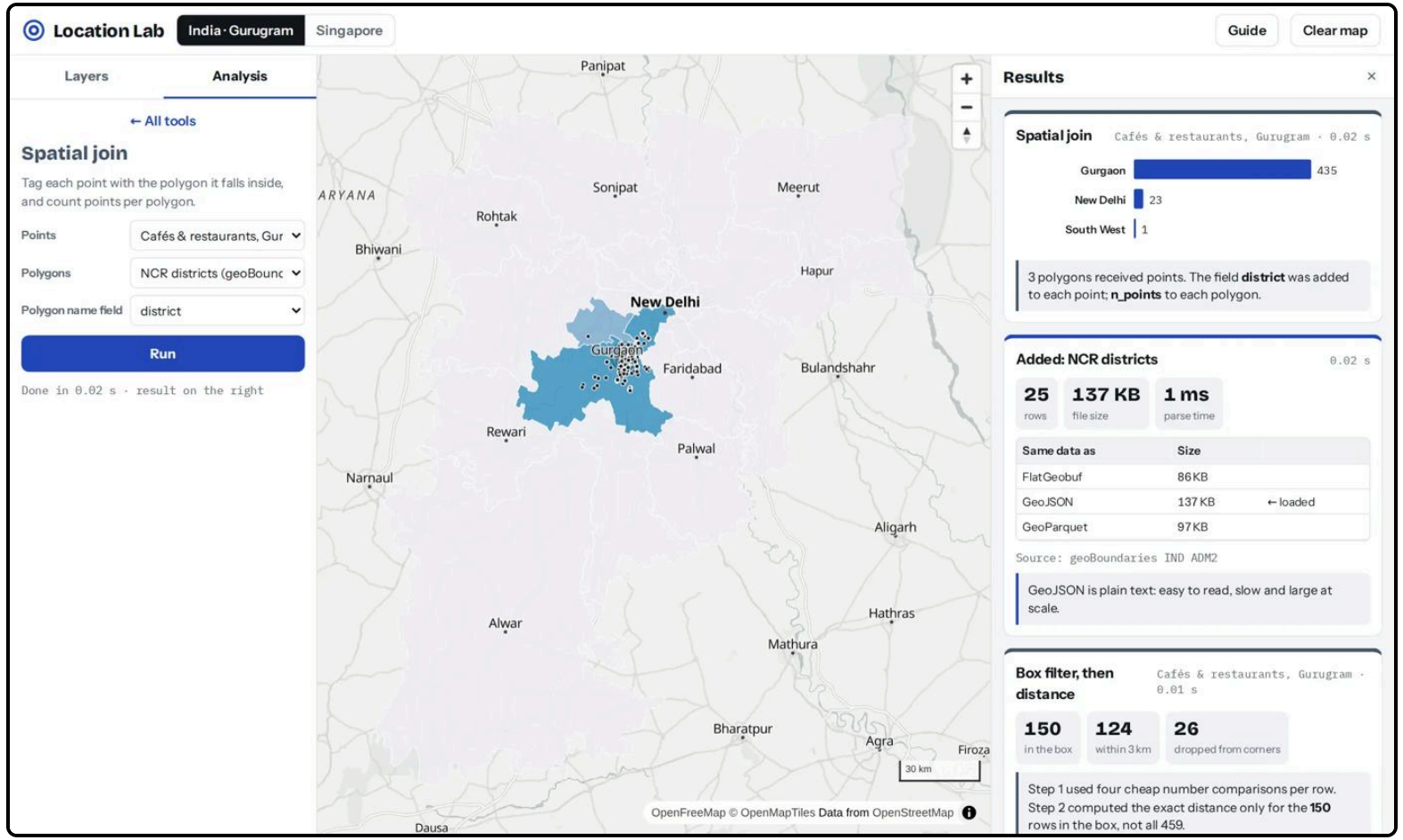
Spatial join: which district is each place in?

DO THIS

Analysis tab → drag **Spatial join** onto the map (or click it), check the settings, press **Run**.

Points	Cafés & restaurants, Gurugram (OpenStreetMap)
Polygons	NCR districts (geoBoundaries)
Polygon name field	district

> 3 polygons received points. The field district was added to each point; n_points to each polygon.



Bin into H3 hexagons and raise them in 3D.

DO THIS

Analysis tab → drag **H3 hexagon bins** onto the map (or click it), check the settings, press **Run**.

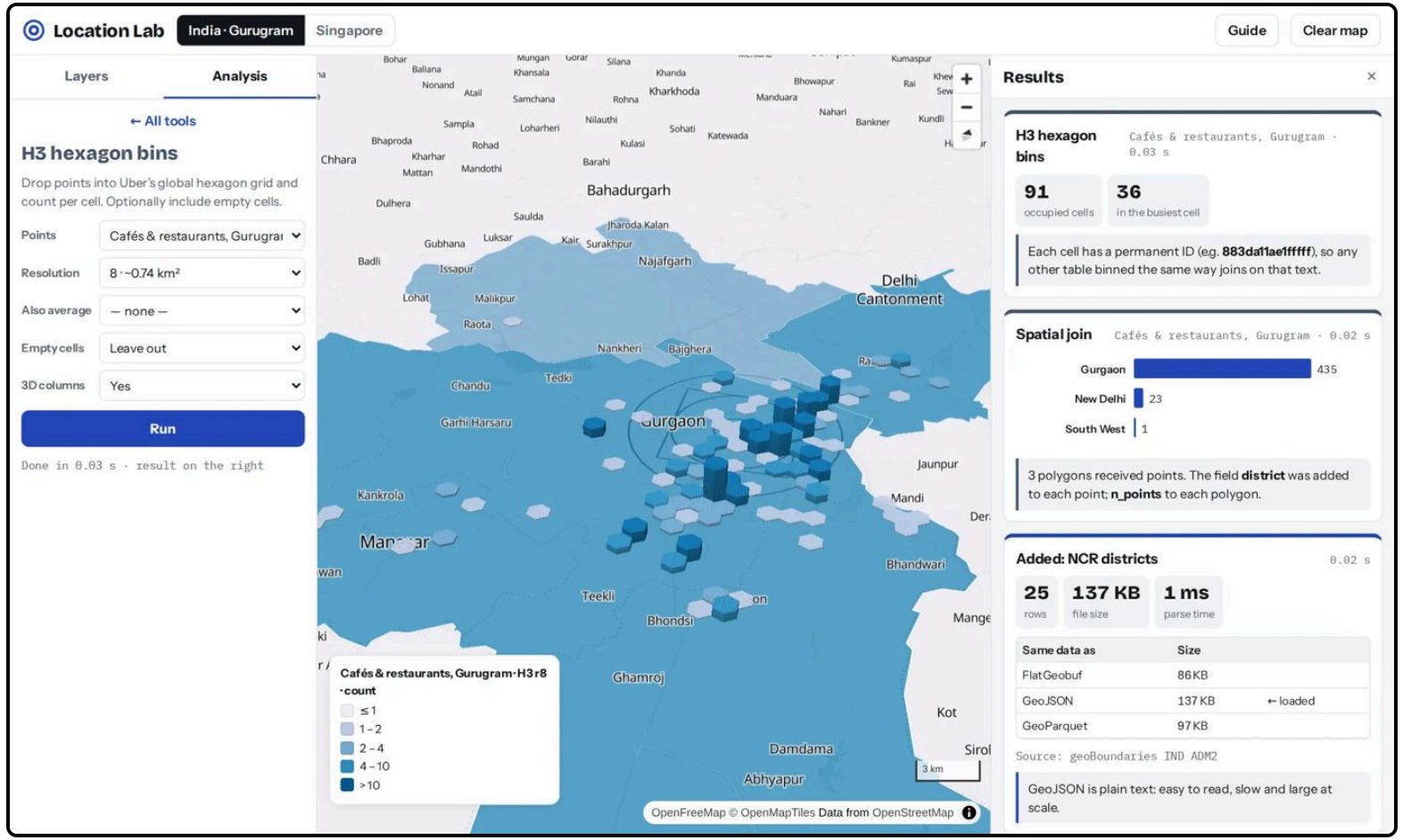
Points	Cafés & restaurants, Gurugram (OpenStreetMap)
Resolution	8 · ~0.74 km²
Also average	— none —
Empty cells	Leave out
3D columns	Yes

YOU SHOULD SEE

91 occupied cells

36 in the busiest cell

> Each cell has a permanent ID (e.g. 883da11ae1fffff), so any other table binned the same way joins on that text.



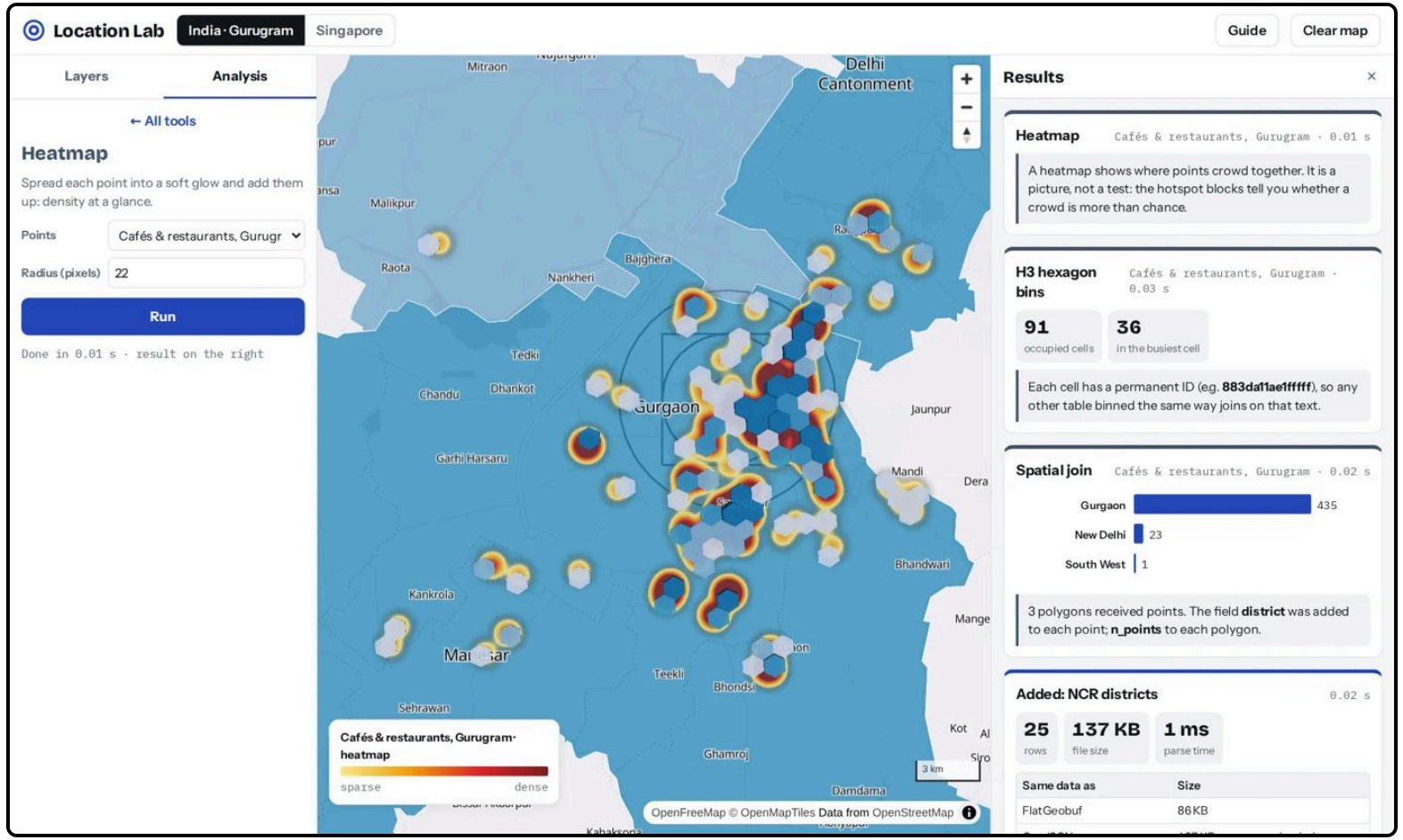
Smooth the points into a heatmap.

DO THIS

Analysis tab → drag **Heatmap** onto the map (or click it), check the settings, press **Run**.

Points	Cafés & restaurants, Gurugram (OpenStreetMap)
Radius (pixels)	22

> A heatmap shows where points crowd together. It is a picture, not a test: the hotspot blocks tell you whether a crowd is more than chance.



Load the NDVI satellite raster (COG).

DO THIS

Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	NDVI around MDI, Sentinel-2, 25 Oct 2024
File format	Cloud-Optimised GeoTIFF

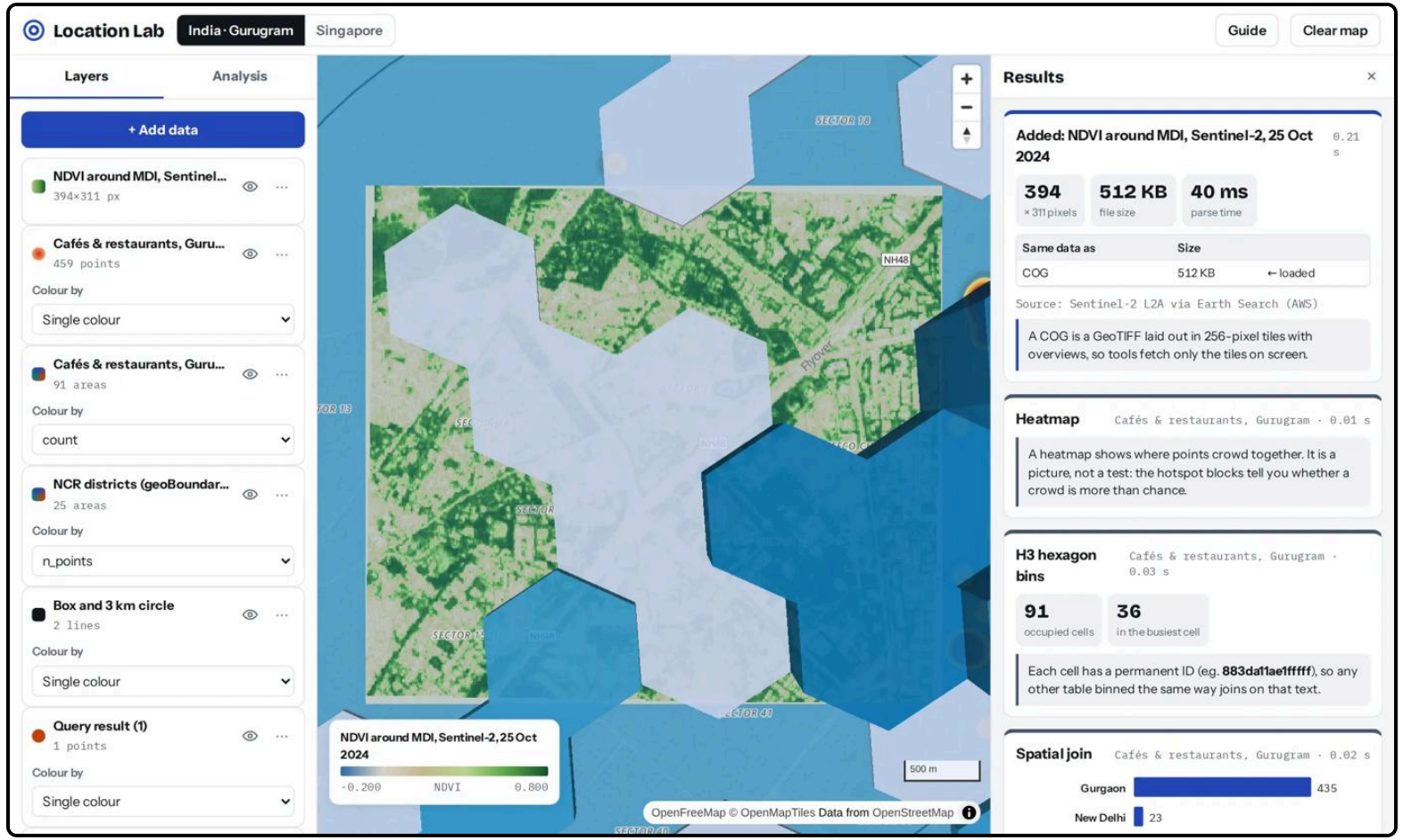
YOU SHOULD SEE

394
× 311 pixels

512 KB
file size

40 ms
parse time

> A COG is a GeoTIFF laid out in 256-pixel tiles with overviews, so tools fetch only the tiles on screen.



Load the MDI campus outline.

DO THIS

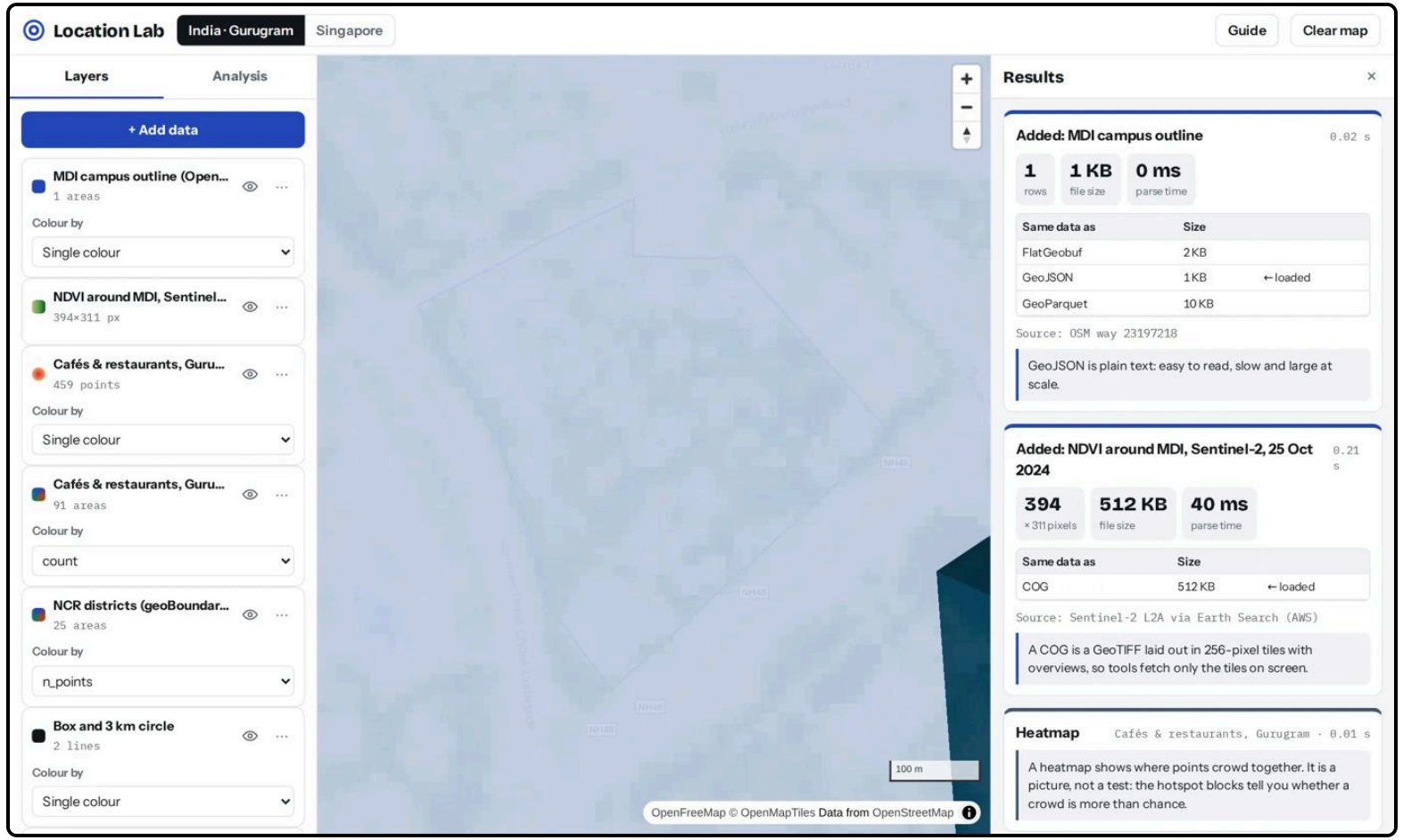
Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	MDI campus outline (OpenStreetMap)
File format	GeoJSON

YOU SHOULD SEE

1 rows | **1 KB** file size | **0 ms** parse time

> GeoJSON is plain text: easy to read, slow and large at scale.



Zonal statistics: how green is the campus?

DO THIS

Analysis tab → drag **NDVI zonal statistics** onto the map (or click it), check the settings, press **Run**.

Raster	NDVI around MDI, Sentinel-2, 25 Oct 2024
Outline	MDI campus outline (OpenStreetMap)
Vegetation above	0.3

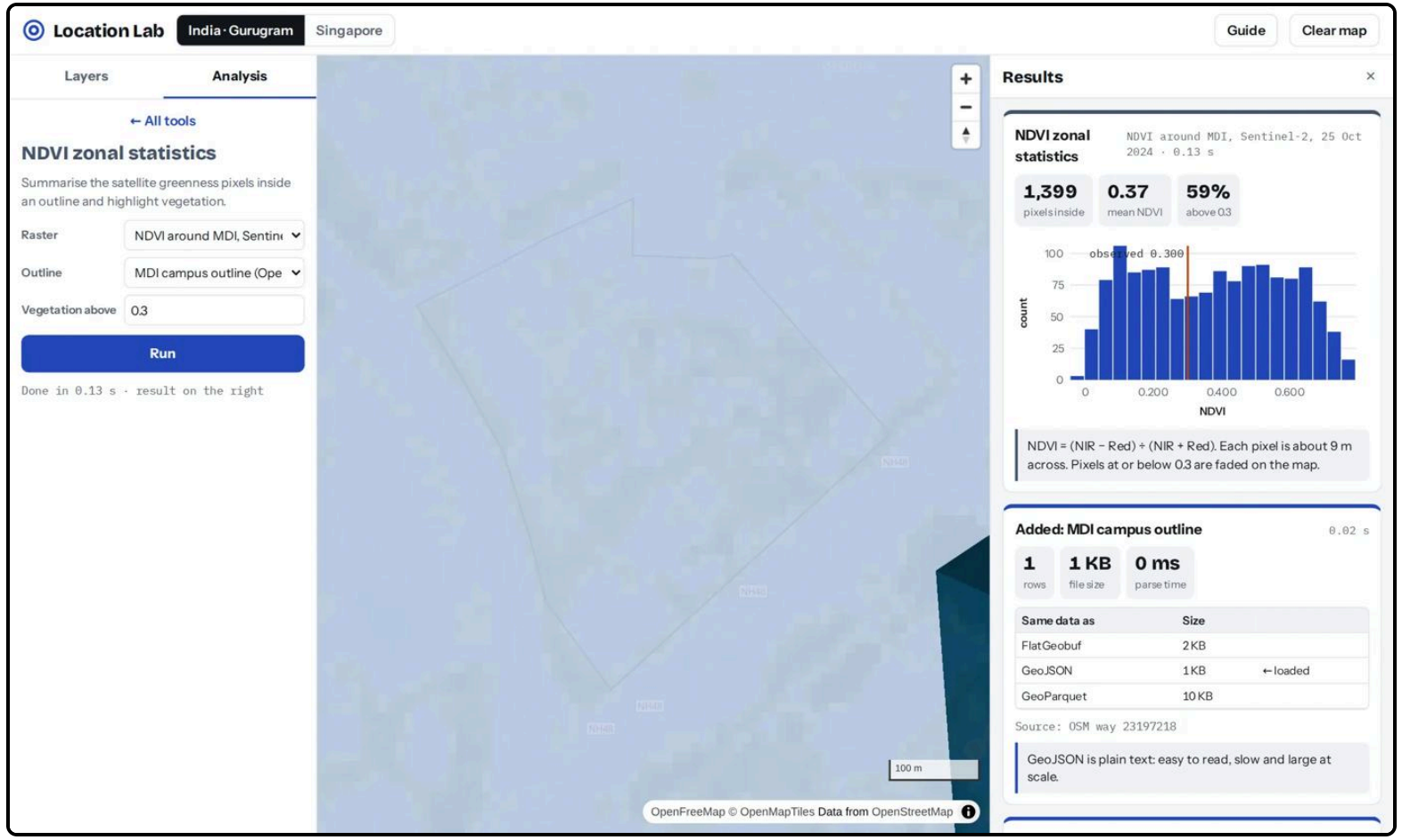
YOU SHOULD SEE

1,399 pixels inside

0.37 mean NDVI

59% above 0.3

> NDVI = (NIR − Red) ÷ (NIR + Red). Each pixel is about 9 m across. Pixels at or below 0.3 are faded on the map.



Three questions

1. A 3 km box returns more cafés than the exact 3 km test. Which number is right?
2. Why does GeoParquet load faster than GeoJSON?
3. What does an NDVI mean of 0.37 for the MDI campus tell you?

Answers

1. **A 3 km box returns more cafés than the exact 3 km test. Which number is right?**
   The exact test. The box is a fast first filter that over-includes its corners.
2. **Why does GeoParquet load faster than GeoJSON?**
   It stores each column together and compresses it, so the file is several times smaller and parses faster.
3. **What does an NDVI mean of 0.37 for the MDI campus tell you?**
   Moderate vegetation on average: green, but not dense canopy.

PART 2

Neighbours and spatial lag

The path we take on the board:

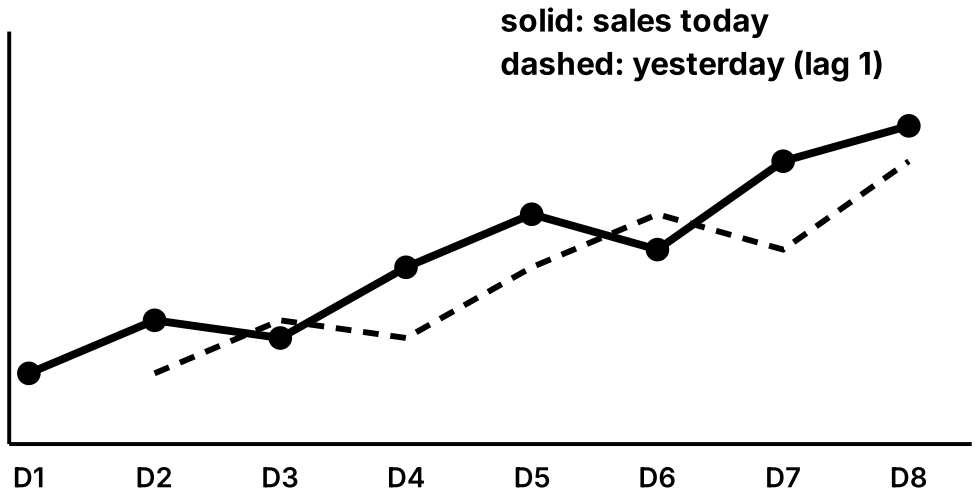
Temporal lag → **Space has no direction** → **Rook · queen · hex** → **Band · k nearest** →
Weights W → **Spatial lag Wy**

Start with time: what is a “lag”?

Take daily sales. Shift the series by one day and lay it beside the original: today’s value next to yesterday’s. That shifted copy is the **lag**, written y_{t-1}.

If today tends to look like yesterday, the two series move together and their correlation is high. Economists call this **autocorrelation**: a variable correlated with its own past.

Time is easy because it has one direction and one obvious neighbour: the day before.



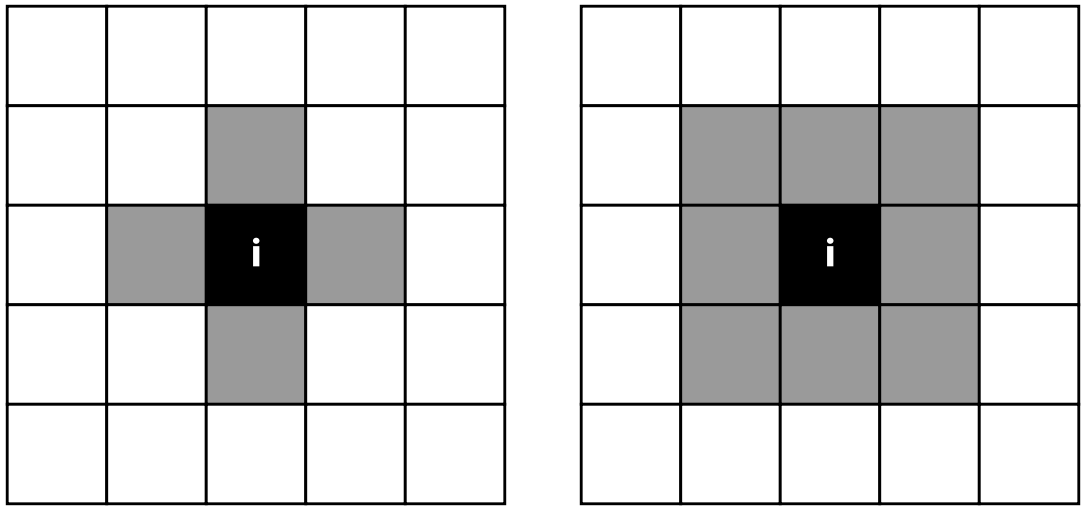
Sales over 8 days and the same series shifted by one day

Space has no “yesterday”

A café on Golf Course Road has neighbours to the north, south, east and west, near and far. There is no single “previous” location. So before we can lag anything in space, we must **decide who counts as a neighbour**.

With a grid, chess gives two answers. The **rook** moves along rows and columns: four neighbours that share an edge. The **queen** also moves diagonally: eight neighbours that share an edge or a corner.

The same rules apply to districts or planning areas: rook means sharing a border, queen means touching even at a single point.



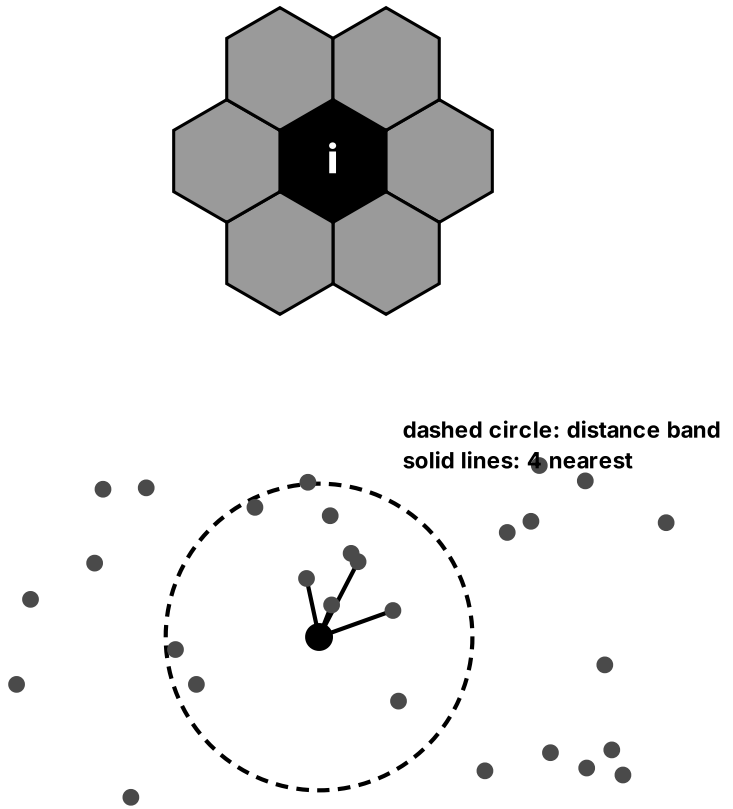
Rook (left): 4 neighbours · Queen (right): 8 neighbours

Hexagons, distance bands, and k nearest

Hexagons remove the rook/queen ambiguity: every H3 cell has exactly six neighbours, all at the same distance, all sharing a full edge.

For **points** there are no borders, so we use distance. A **distance band** makes everyone within d metres a neighbour; in sparse areas some points get none (islands). **k nearest** gives each point exactly k neighbours, however crowded or empty its area.

In Gurugram we bin cafés into hexagons and use hexagon contiguity.



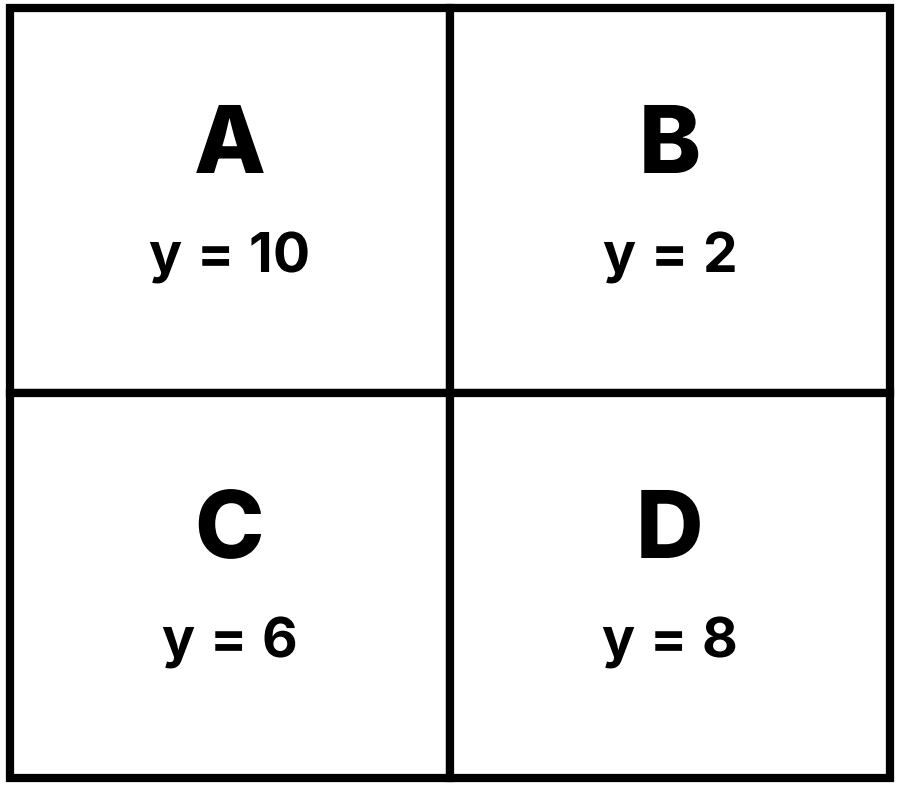
Top: six hexagon neighbours · Bottom: band vs 4 nearest

Write it down: the weights matrix W

Four areas in a 2 × 2 grid with values A = 10, B = 2, C = 6, D = 8. Under the rook rule, A touches B and C; B touches A and D; and so on. Put a 1 where two areas are neighbours and 0 elsewhere: that table is **W**.

	A	B	C	D
A	0	½	½	0
B	½	0	0	½
C	½	0	0	½
D	0	½	½	0

Dividing each row by its number of neighbours (**row-standardising**) makes every row sum to 1, so multiplying by W gives an **average**.



Rook neighbours: A–B, A–C, B–D, C–D

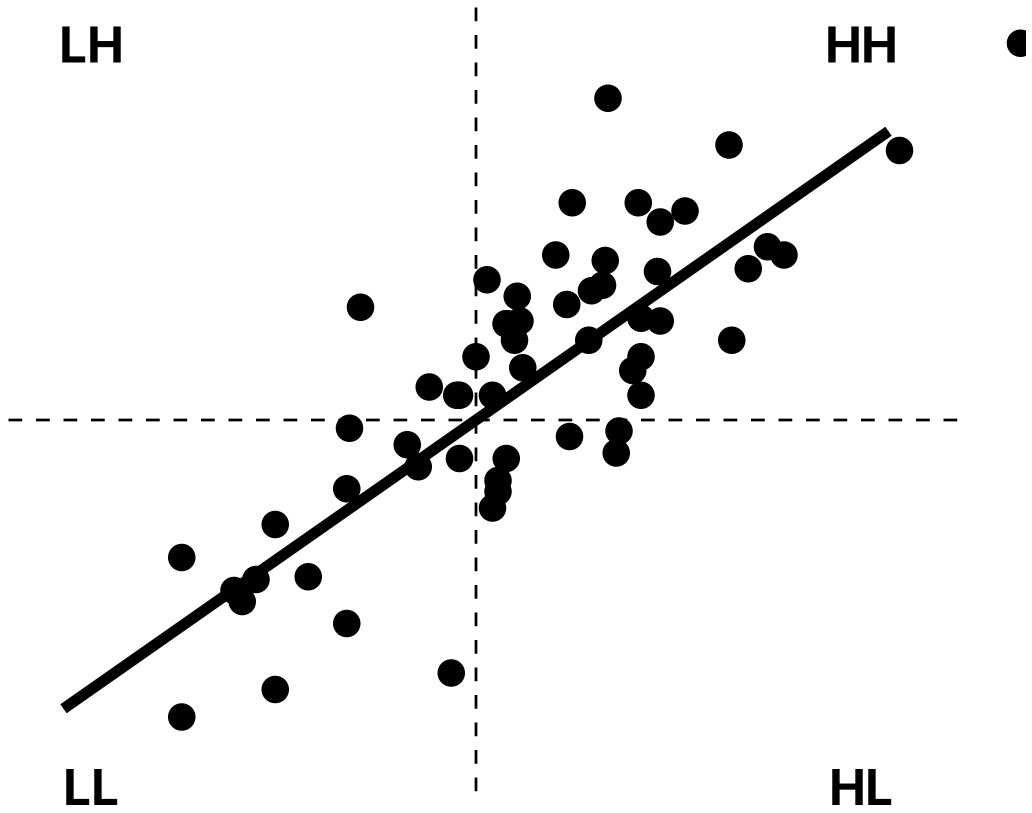
The spatial lag: “how is my neighbourhood doing?”

The spatial lag is the space version of yesterday’s value: the weighted average of a feature’s neighbours.

$$\text{lag}(y)_i = \Sigma_j \, w_{ij} \, y_j = (Wy)_i$$

A: (2 + 6) ÷ 2 = **4** · B: (10 + 8) ÷ 2 = **9** · C: (10 + 8) ÷ 2 = **9** · D: (2 + 6) ÷ 2 = **4**.

A is high (10) but its neighbourhood is low (4); B is low (2) but surrounded by high values (9). Plot each value against its lag: a rising cloud means similar values sit together; a falling one, like our 2 × 2 example, is a checkerboard.



Rising cloud: neighbours resemble each other

Where business already uses this

IN THE NEWS · PROPUBLICA · 29 OCT 2015

Uber prices each city zone separately; a block’s walk can halve the fare

Neighbouring zones can surge very differently. Defining “neighbour” and watching the lag across zones is a pricing question.

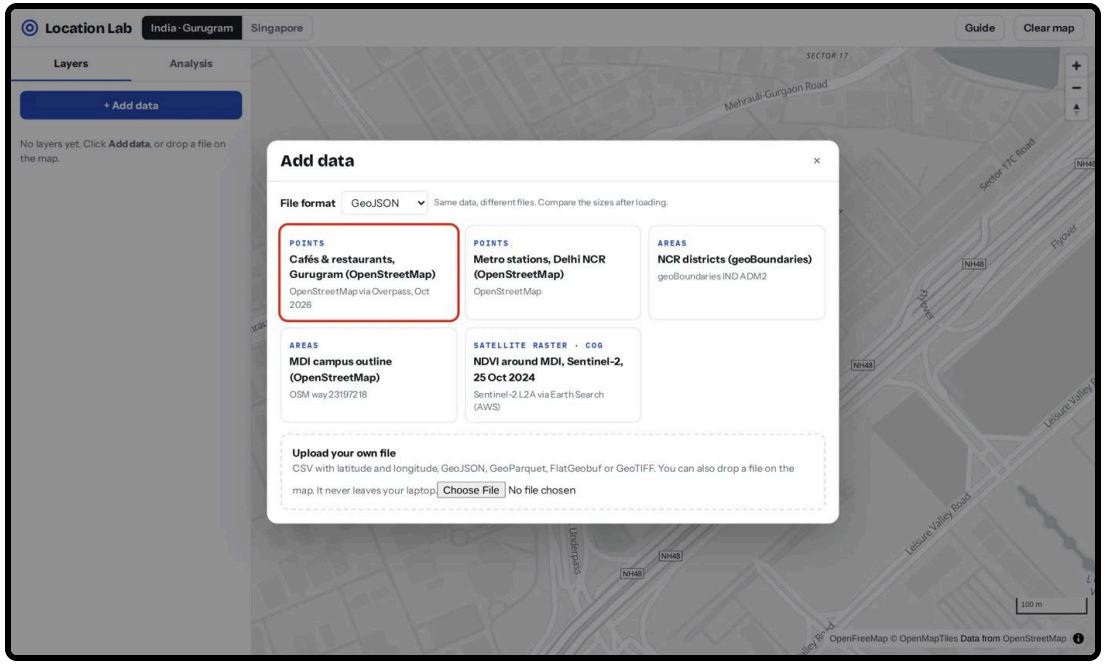
https://www.propublica.org/article/uber-surge-pricing-may-not-lead-to-a-surge-in-drivers

2 · Neighbours and spatial lag

Before testing any pattern, decide who counts as a neighbour. Everything after this reuses that choice.

Open **quantng.github.io/location-lab**, choose **India · Gurugram** in the top bar, then follow the steps.

1. Load the cafés.
2. Count cafés per hexagon, including empty hexagons.
3. Build contiguity weights: each hexagon's 6 touching cells. Click a hexagon to see them.
4. Spatial lag: the average café count of each hexagon's neighbours.



LAB 2 · STEP 1 OF 4

Load the cafés.

DO THIS

Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	Cafés & restaurants, Gurugram (OpenStreetMap)
File format	GeoJSON

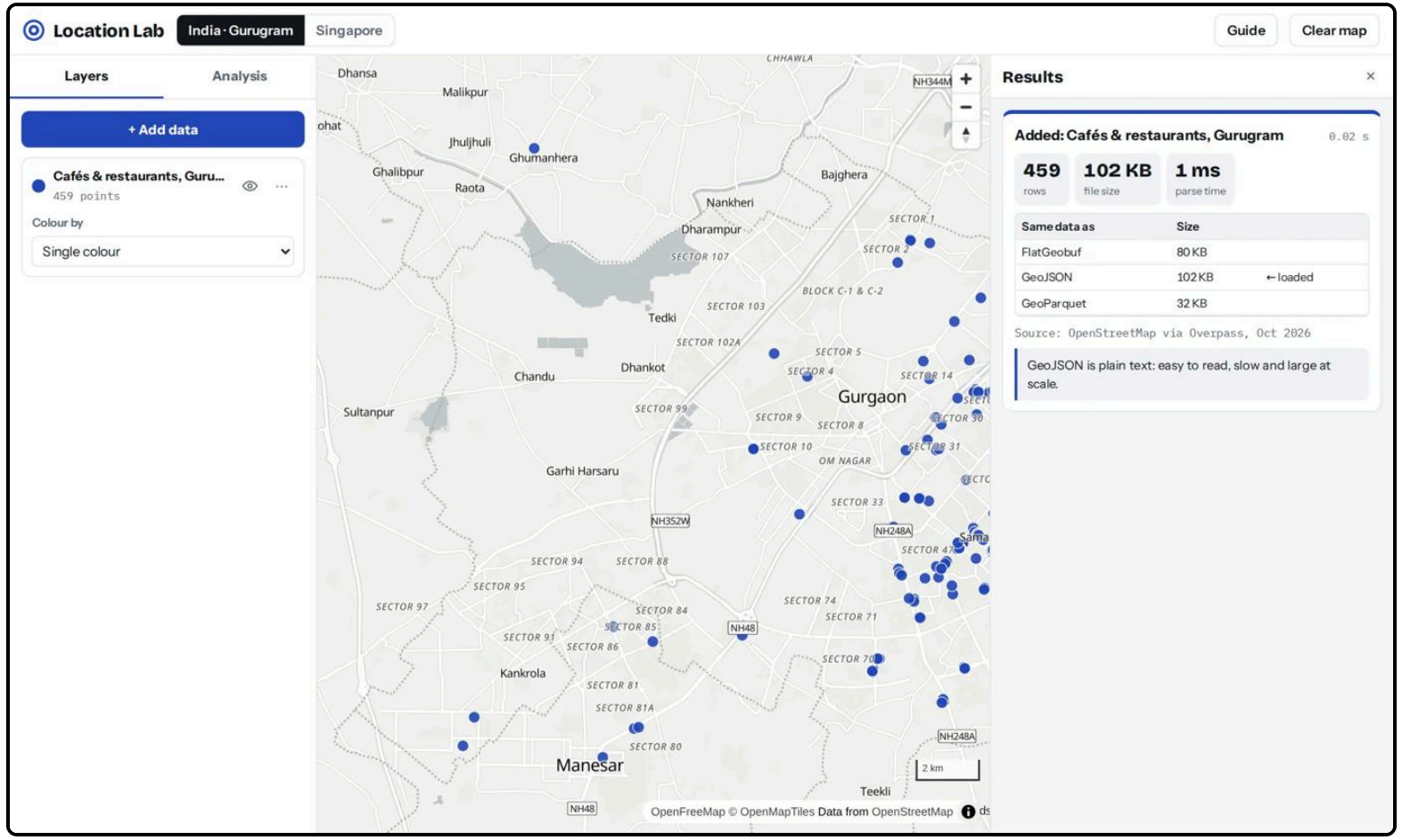
YOU SHOULD SEE

459 rows

102 KB file size

1 ms parse time

> GeoJSON is plain text: easy to read, slow and large at scale.



LAB 2 · STEP 2 OF 4

Count cafés per hexagon, including empty hexagons.

DO THIS

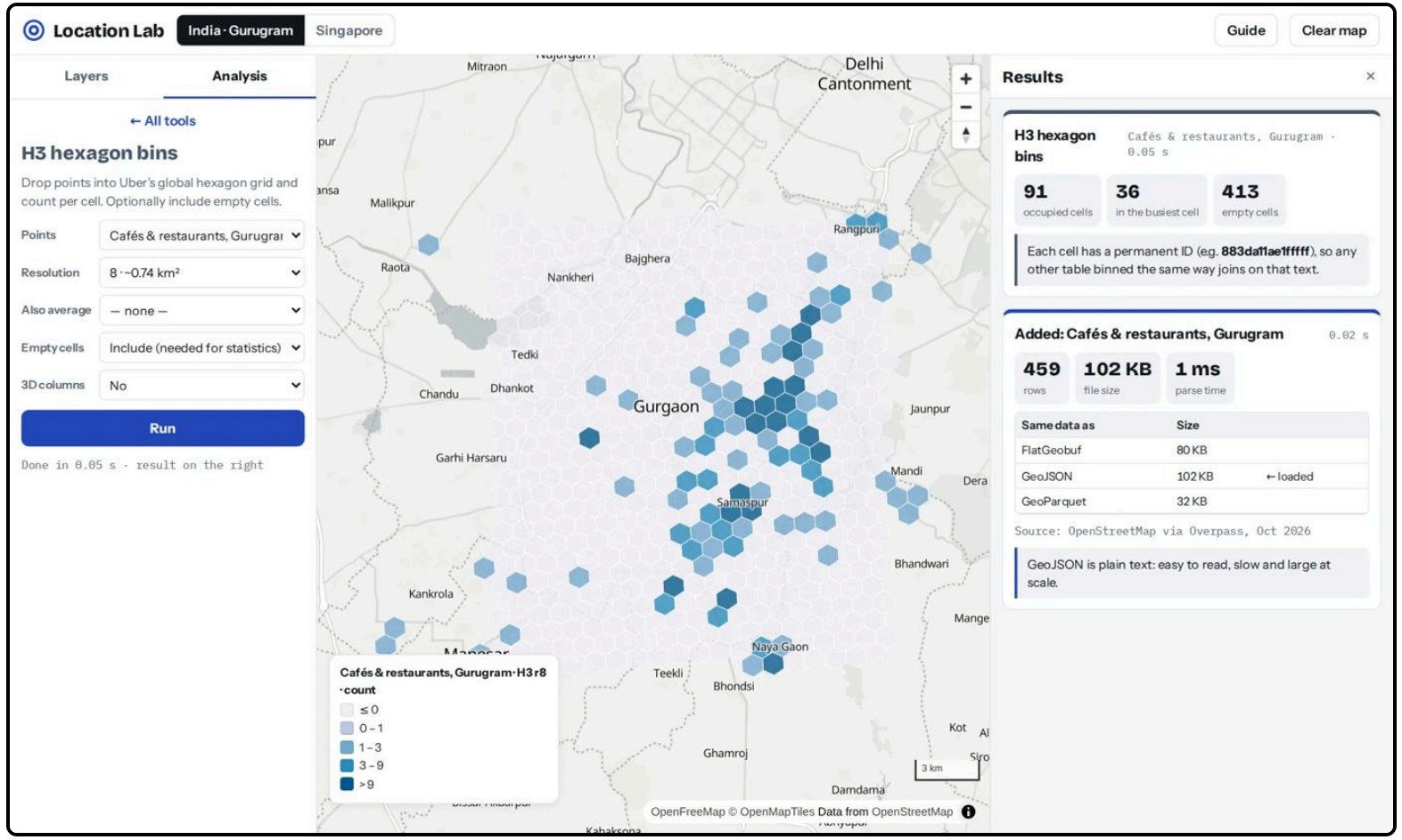
Analysis tab → drag **H3 hexagon bins** onto the map (or click it), check the settings, press **Run**.

Points	Cafés & restaurants, Gurugram (OpenStreetMap)
Resolution	8 · ~0.74 km²
Also average	— none —
Empty cells	Include (needed for statistics)
3D columns	No

YOU SHOULD SEE

91	36	413
occupied cells	in the busiest cell	empty cells

> Each cell has a permanent ID (e.g. 883da11ae1fffff), so any other table binned the same way joins on that text.



Build contiguity weights: each hexagon's 6 touching cells. Click a hexagon to see them.

DO THIS

Analysis tab → drag **Spatial weights** onto the map (or click it), check the settings, press **Run**.

Layer	Cafés & restaurants, Gurugram · H3 r8
Neighbours are	touching H3 hexagons

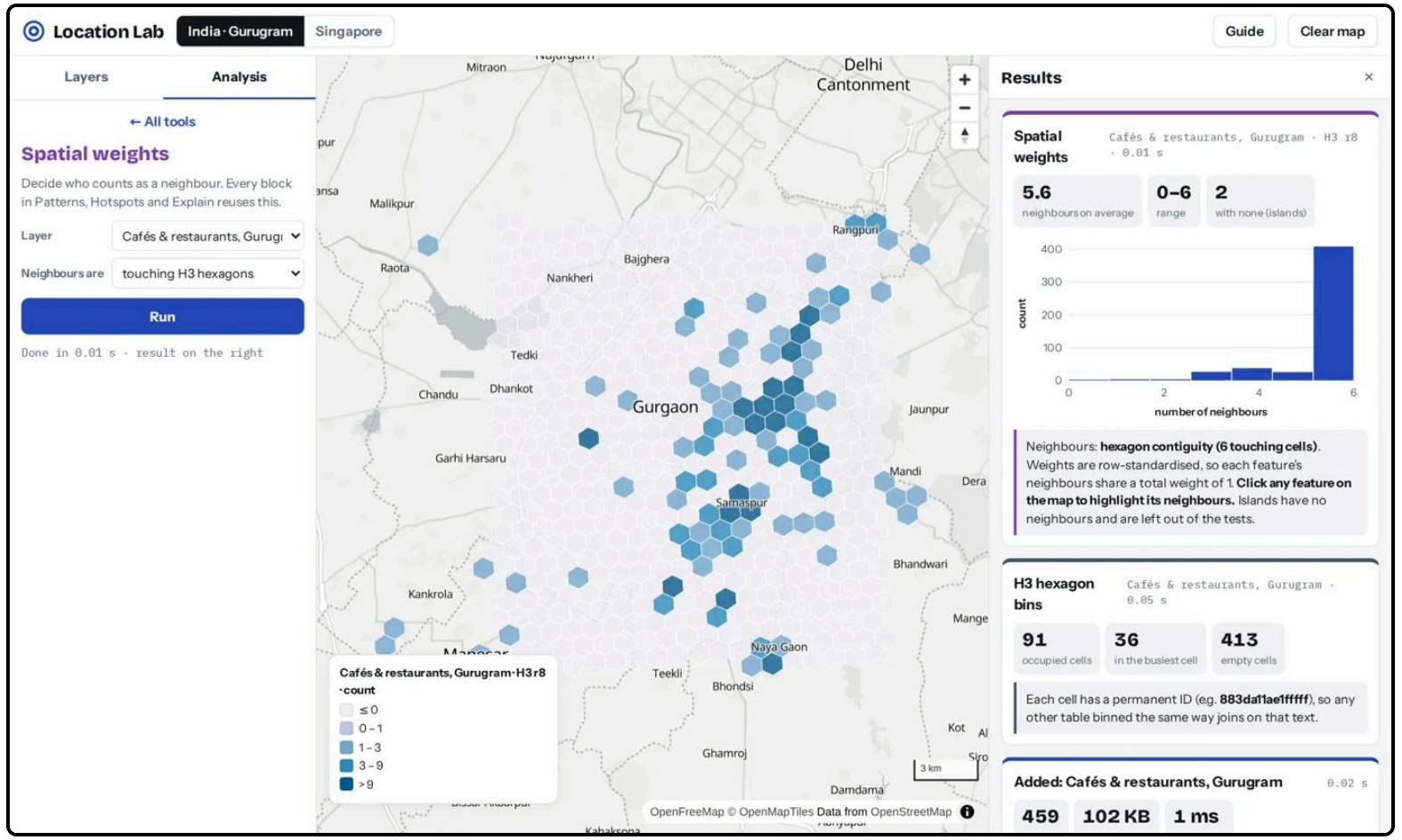
YOU SHOULD SEE

5.6
neighbours on average

0–6
range

2
with none (islands)

> Neighbours: hexagon contiguity (6 touching cells). Weights are row-standardised, so each feature's neighbours share a total weight of 1. Click any feature on the map to highlight its...



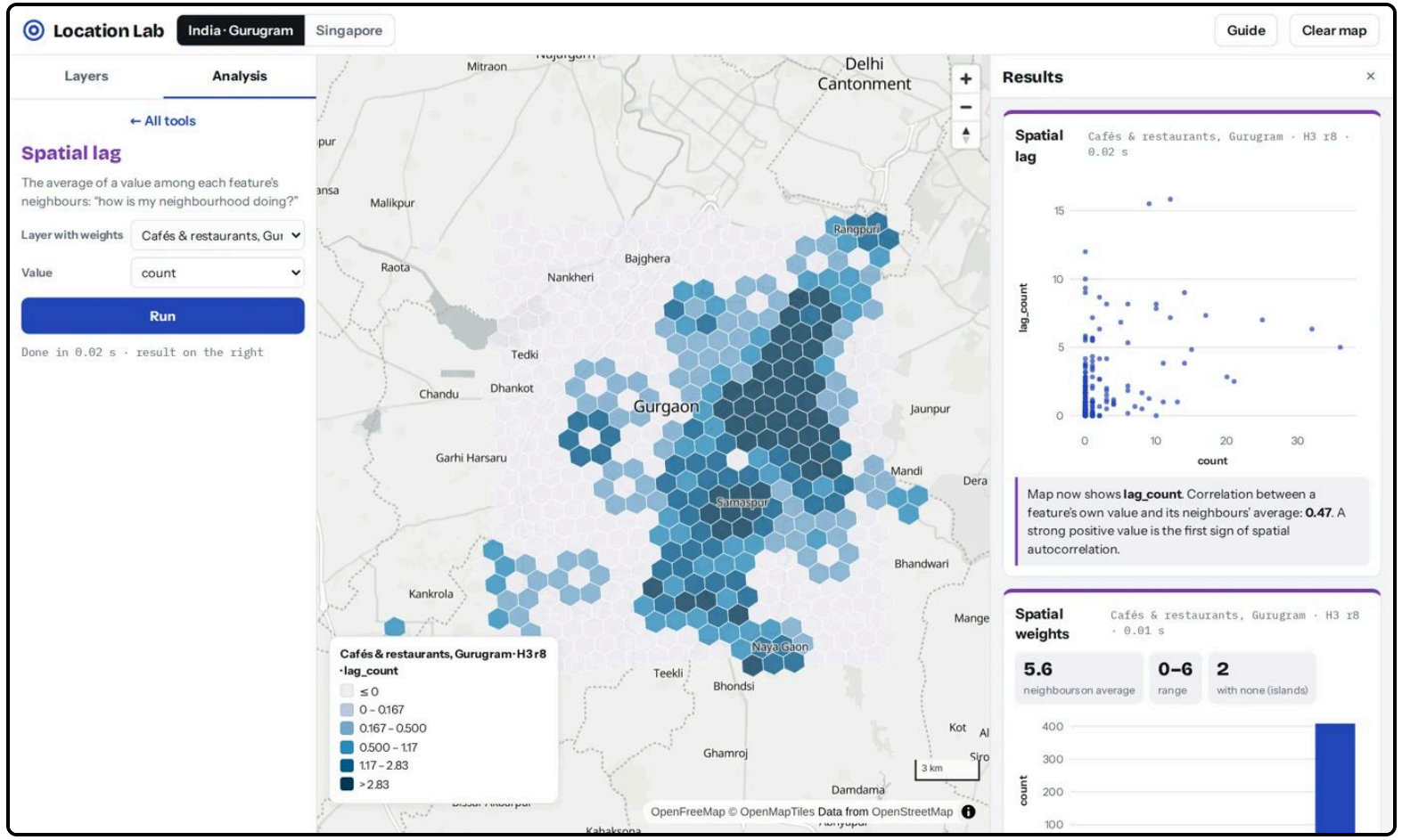
Spatial lag: the average café count of each hexagon's neighbours.

DO THIS

Analysis tab → drag **Spatial lag** onto the map (or click it), check the settings, press **Run**.

Layer with weights	Cafés & restaurants, Gurugram · H3 r8
Value	count

> Map now shows lag_count. Correlation between a feature's own value and its neighbours' average: 0.47. A strong positive value is the first sign of spatial autocorrelation.



Three questions

1. What is the temporal lag of a series, and what is its spatial equivalent?
2. A hexagon’s 6 neighbours have 0, 0, 2, 4, 6 and 0 cafés. What is its spatial lag?
3. Under the queen rule, how many neighbours does an interior grid cell have, and under the rook rule?

Answers

1. **What is the temporal lag of a series, and what is its spatial equivalent?**
   The series shifted one period back; in space, the weighted average of a feature's neighbours (Wy).
2. **A hexagon's 6 neighbours have 0, 0, 2, 4, 6 and 0 cafés. What is its spatial lag?**
   (0+0+2+4+6+0) ÷ 6 = 2 cafés.
3. **Under the queen rule, how many neighbours does an interior grid cell have, and under the rook rule?**
   Queen: 8 (edges and corners). Rook: 4 (edges only).

PART 3

Point patterns

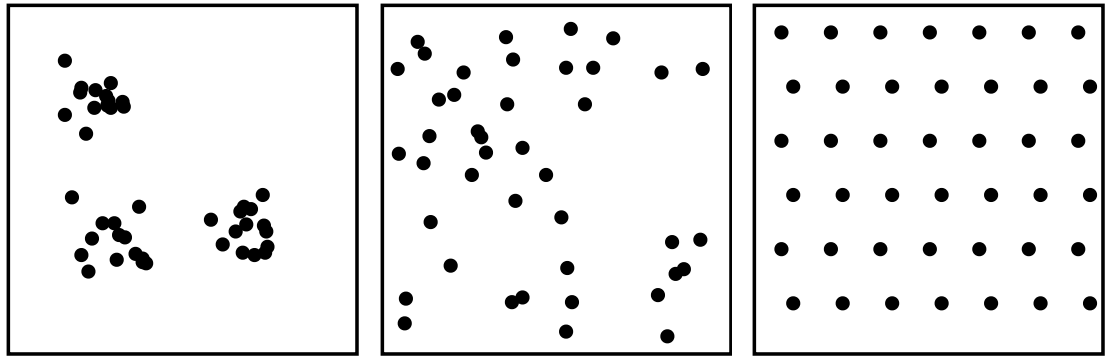
The path we take on the board:

Randomness (CSR) → **Three patterns** → **Nearest-neighbour R** → **Ripley's L + envelope**

What would “random” look like?

Before calling cafés “clustered” we need a benchmark. **Complete spatial randomness (CSR)** means every location in the study area is equally likely and points ignore each other, like rain on a pavement.

Real patterns depart from CSR in two directions. **Clustered**: points attract (food courts, malls, competition for the same footfall). **Dispersed**: points repel (franchise territories, planned services, pharmacies with exclusive catchments).



Clustered R < 1 | Random R ≈ 1 | Dispersed R > 1

Nearest-neighbour ratio R

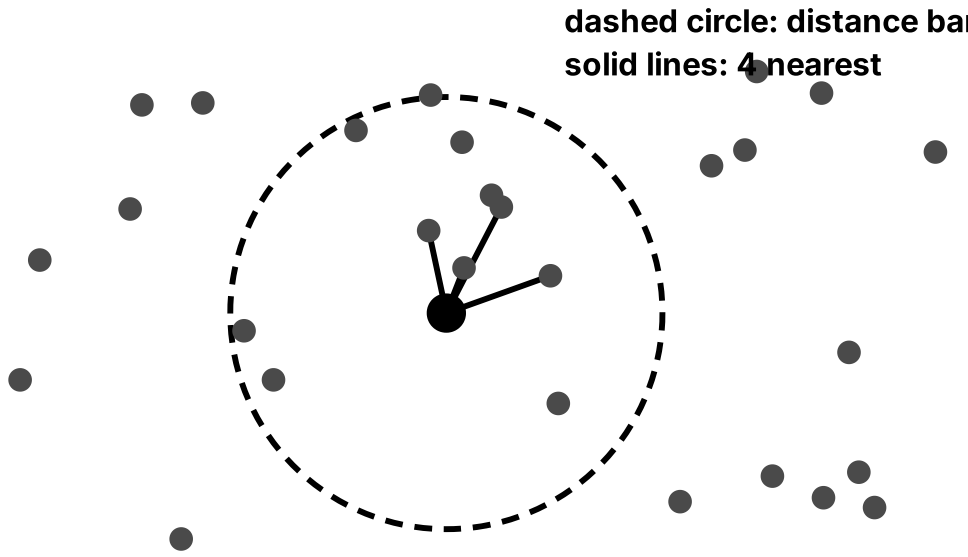
For every point, measure the distance to its nearest neighbour and average. Under CSR with n points in area A, theory gives the expected average:

$$E[d] = 0.5 \div \sqrt{(n / A)}$$

$$R = \text{observed mean } d \div E[d]$$

R < 1: clustered. R ≈ 1: random. R > 1: dispersed. A z-score tells you whether the gap from 1 is larger than chance.

Gurugram cafés: 144 m observed vs 487 m expected, so **R = 0.30**: the nearest café is less than a third as far as chance would put it.



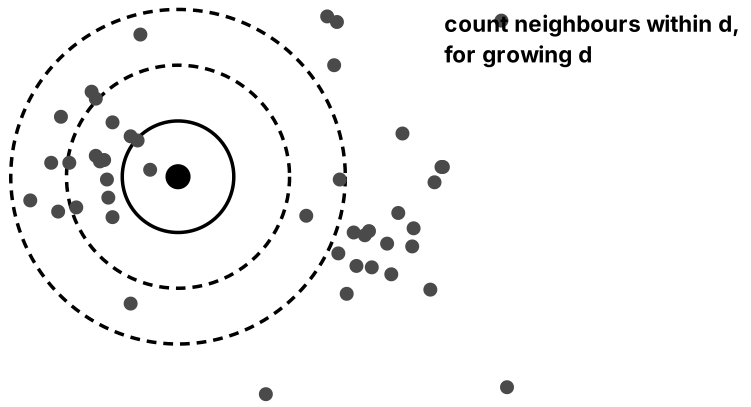
Each point’s nearest-neighbour distance, averaged

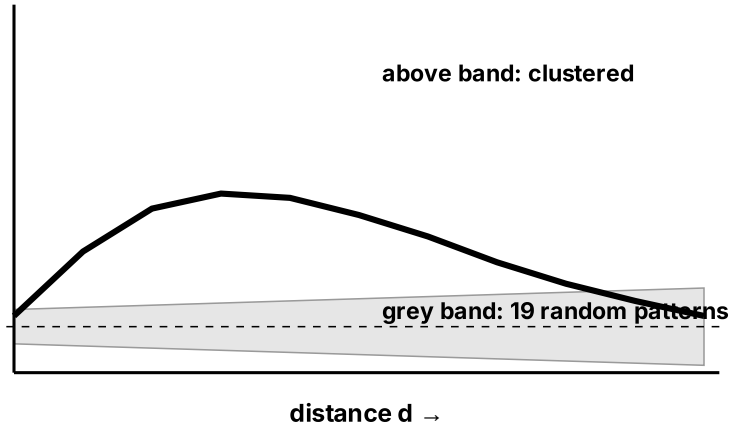
One scale is not enough: Ripley’s K and L

R looks only at the closest neighbour. A pattern can be clustered at 200 m (shops in a mall) and dispersed at 3 km (malls spaced across the city).

Ripley’s idea: draw circles of growing radius d around every point and count how many others fall inside. Compare with the count CSR would give. The **L function** rescales this so that random is a flat line at zero.

To judge chance, simulate 19 random patterns in the same area and draw their range as a grey band. Where the real line rises above the band, the pattern is clustered at that scale.





Where business already uses this

IN THE NEWS · APPAREL RESOURCES, CITING CLSA · 26 AUG 2026

Flipkart Minutes overtakes Instamart in dark stores across India’s top 10 cities

3,536 dark stores in 10 cities, 225 of them in Gurgaon. Are they clustered where demand is, or spread for coverage? That is a point-pattern question.

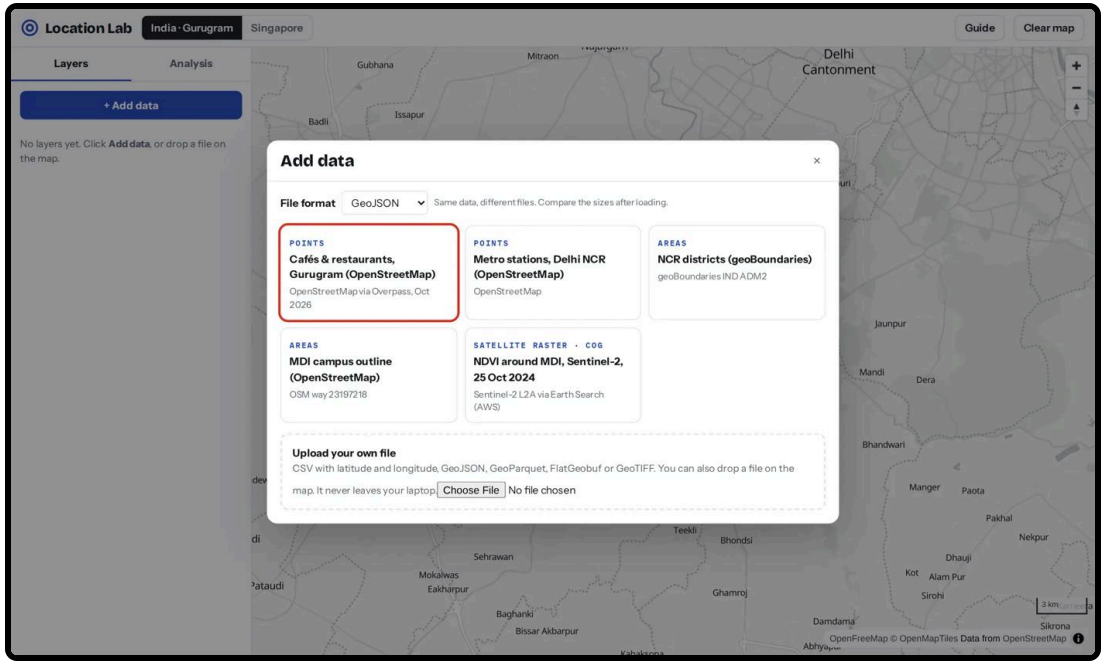
https://apparelresources.com/business-news/retail/flipkart-minutes-overtakes-instamart-dark-stores-across-top-10-cities/

3 · Is the pattern random?

Compare the real cafés with what pure chance would give, at many distances.

Open **quantng.github.io/location-lab**, choose **India · Gurugram** in the top bar, then follow the steps.

1. Load the cafés.
2. Nearest-neighbour ratio and Ripley's L against 19 random patterns.



LAB 3 · STEP 1 OF 2

Load the cafés.

DO THIS

Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	Cafés & restaurants, Gurugram (OpenStreetMap)
File format	GeoJSON

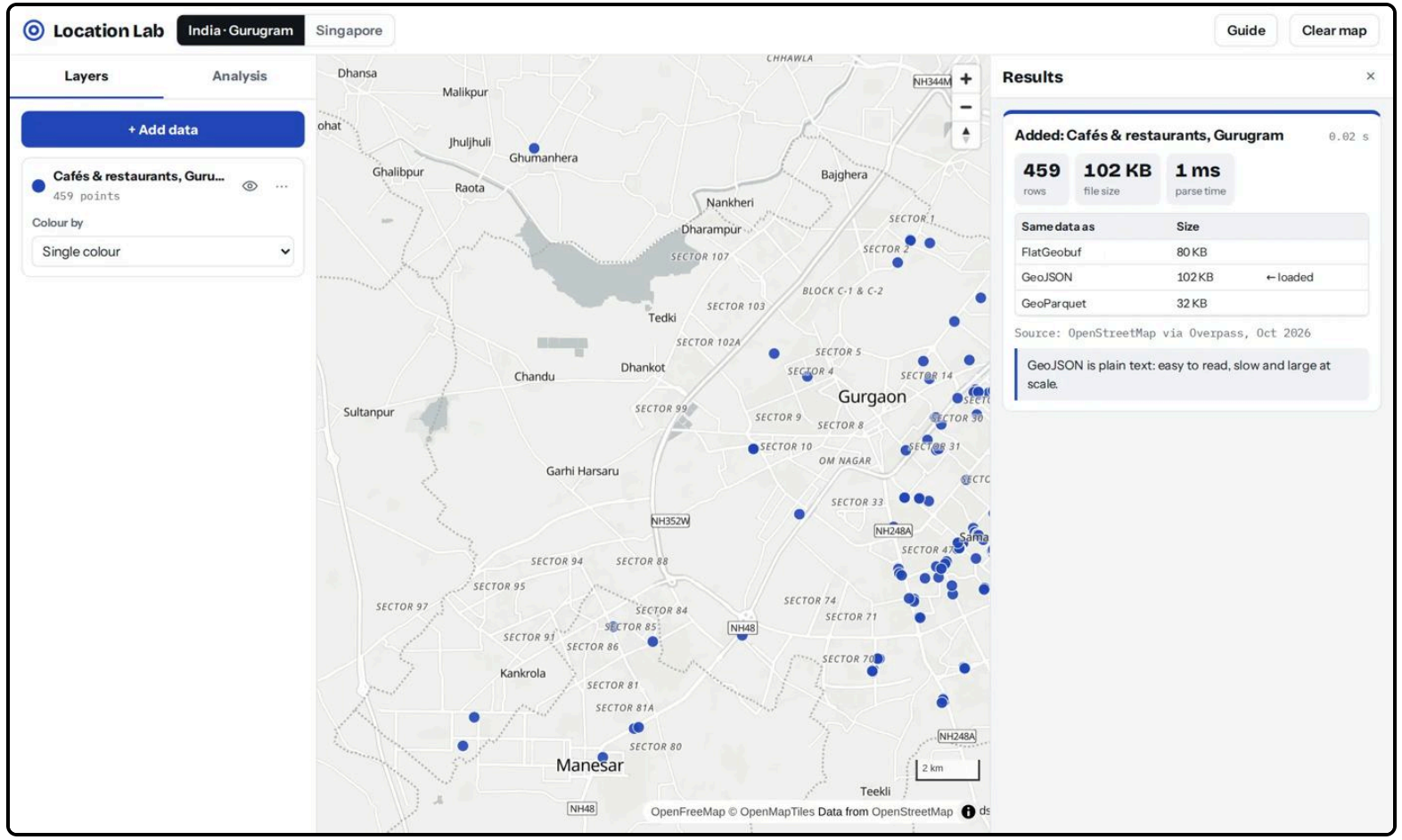
YOU SHOULD SEE

459 rows

102 KB file size

1 ms parse time

> GeoJSON is plain text: easy to read, slow and large at scale.



Nearest-neighbour ratio and Ripley’s L against 19 random patterns.

DO THIS

Analysis tab → drag **Point pattern test** onto the map (or click it), check the settings, press **Run**.

Points	Cafés & restaurants, Gurugram (OpenStreetMap)

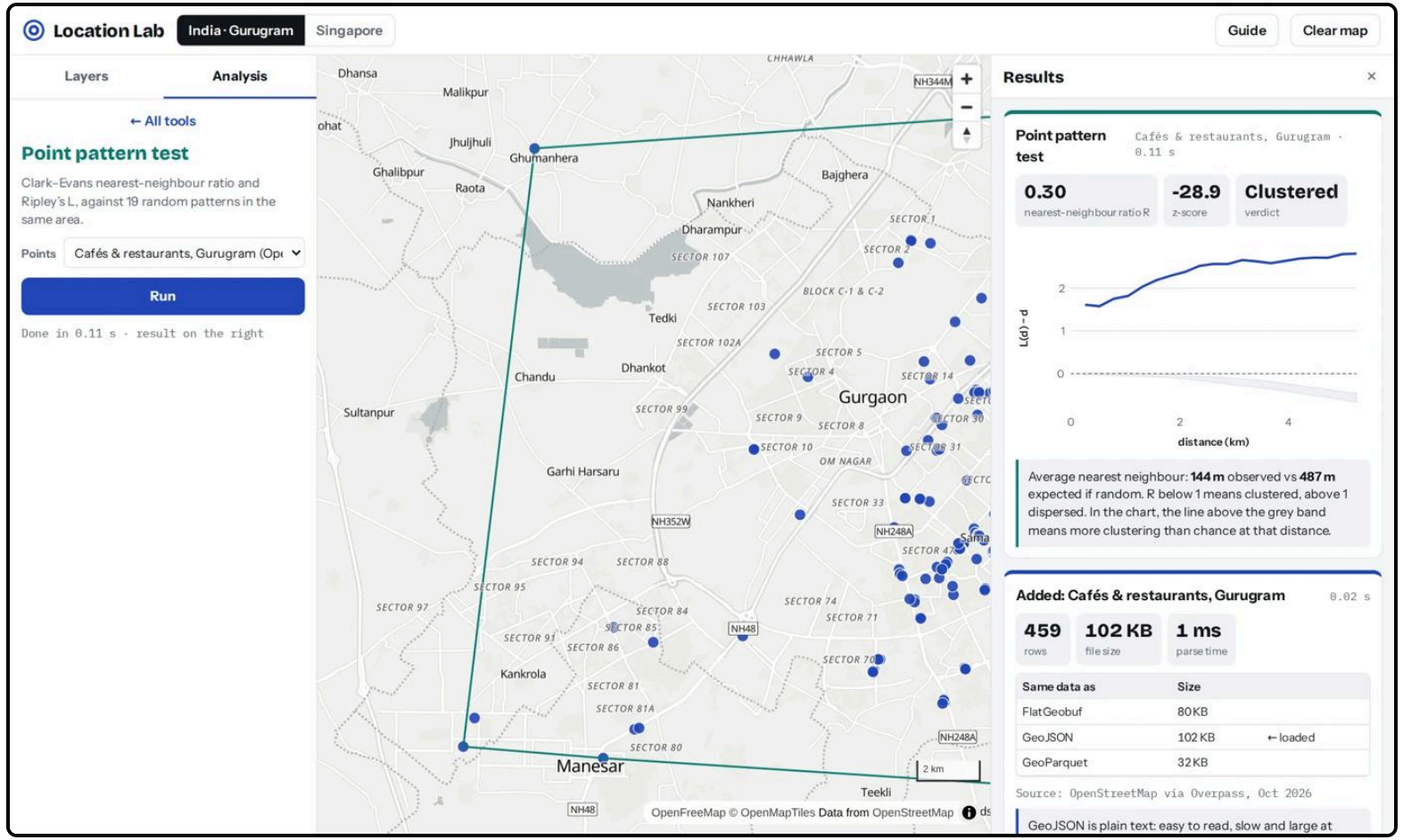
YOU SHOULD SEE

0.30
nearest-neighbour ratio R

-28.9
z-score

Clustered
verdict

> Average nearest neighbour: 144 m observed vs 487 m expected if random. R below 1 means clustered, above 1 dispersed. In the chart, the line above the grey band means more clustering than...



Three questions

1. Gurugram cafés give R = 0.30. Clustered or dispersed?
2. Why is the grey band in the L-plot needed?
3. Give one business reason for clustering and one for dispersion.

Answers

1. **Gurugram cafés give R = 0.30. Clustered or dispersed?**
   Strongly clustered: the nearest café is under a third of the distance chance would give.
2. **Why is the grey band in the L-plot needed?**
   It shows the range of 19 random patterns in the same area; only departures beyond it are more than chance.
3. **Give one business reason for clustering and one for dispersion.**
   Clustering: shared footfall in dining strips. Dispersion: franchise territories or planned services.

PART 4

Autocorrelation and hotspots

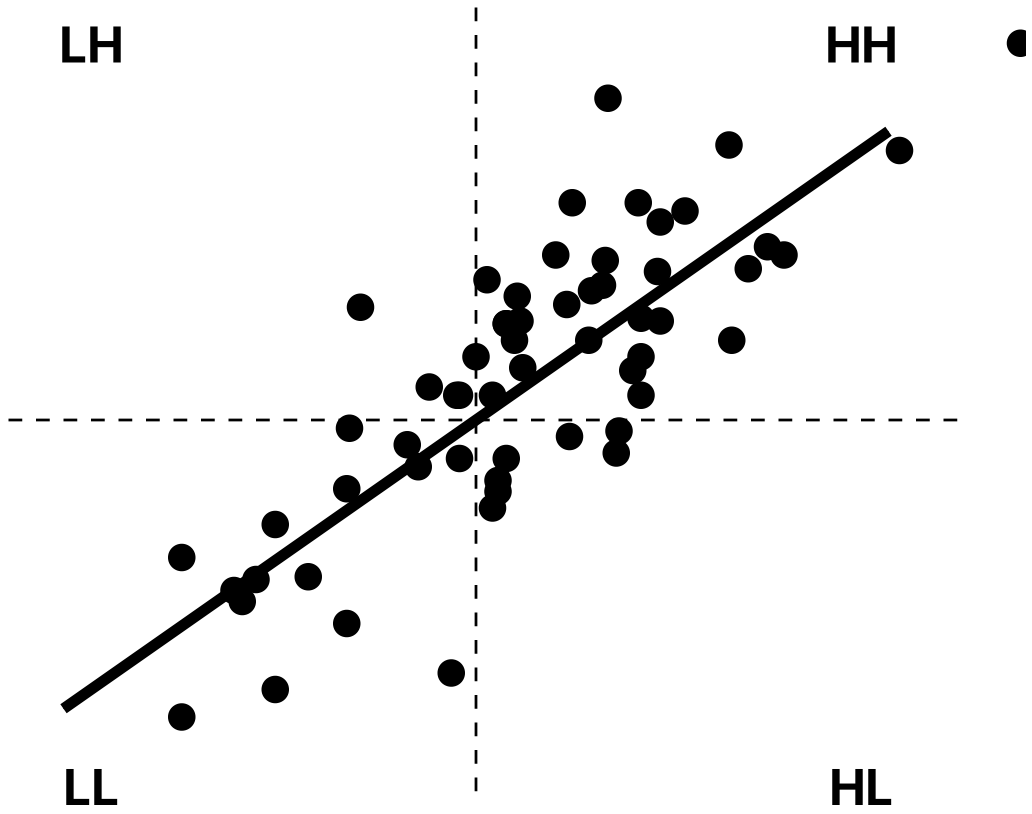
The path we take on the board:

Correlation → **Moran's I** → **Shuffle test** → **LISA** → **Gi***

Recall correlation

Correlation measures whether two variables move together. Standardise each (subtract the mean, divide by the spread), multiply them pairwise, and average. Positive: high goes with high. Negative: high goes with low. Zero: no linear link.

We already saw that a time series can be correlated with its own lag. The obvious next step: correlate a map with **its spatial lag**.



positive correlation: the line rises

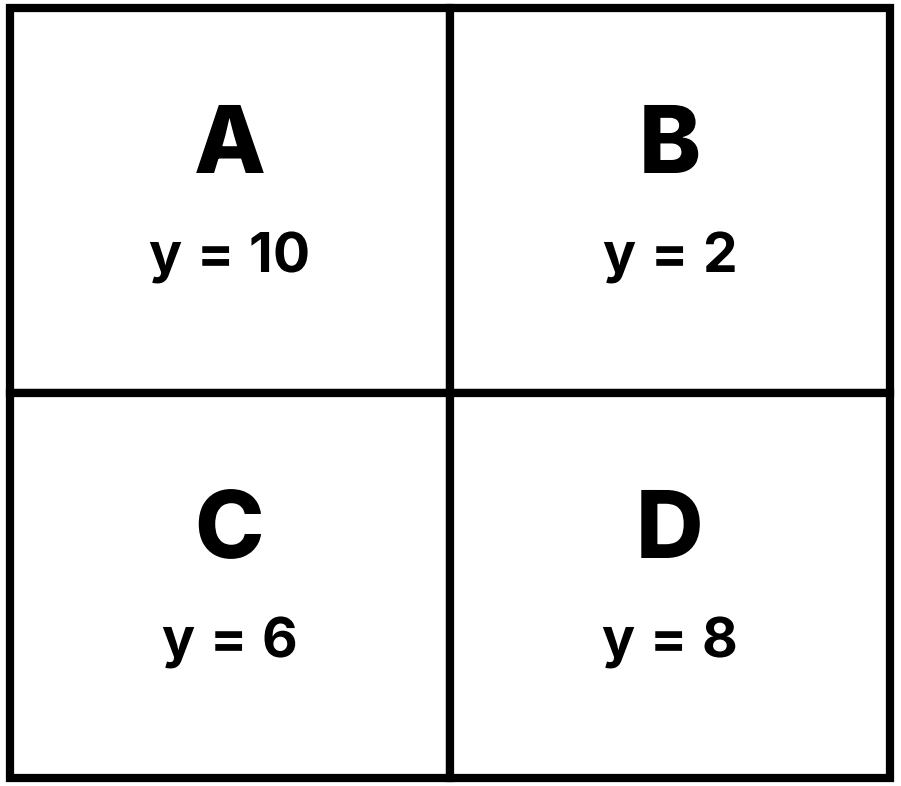
Moran’s I: a map correlated with its own lag

With $z_i = y_i - \bar{y}$ and row-standardised weights:

$$I = \Sigma_i \, z_i \cdot (Wz)_i \div \Sigma_i \, z_i^2$$

It is the slope of the line in the plot of z against its lag. Our 2 × 2 example: mean 6.5, so z = (3.5, −4.5, −0.5, 1.5) and Wz = (−2.5, 2.5, 2.5, −2.5). Numerator −25, denominator 35, so **I = −0.71**: a checkerboard, neighbours unlike each other.

Positive I means similar values cluster (Tobler: near things are more related). Near zero: random.



High A beside low B and C: negative I

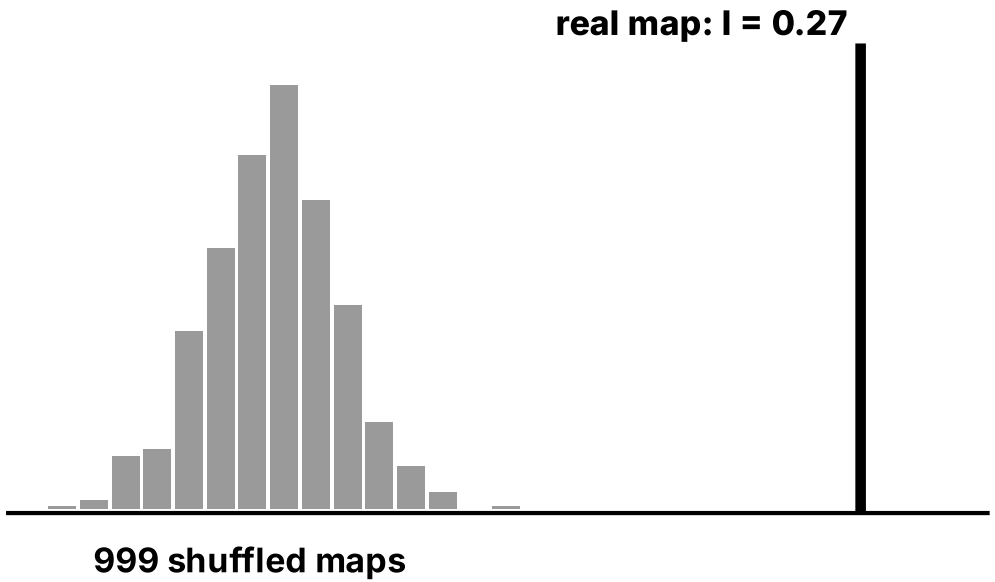
AUTOCORRELATION · STEP 3

Is it chance? Shuffle the map

Keep the values, keep the neighbours, but scatter the values randomly across locations, then recompute I. Do it 999 times. That histogram is what I looks like when location does not matter.

If the real I sits beyond almost every shuffle, the pseudo p-value is (number of shuffles at least as extreme + 1) ÷ 1,000. None out of 999 gives **p = 0.001**, the smallest possible.

Gurugram café counts per hexagon: **I = 0.273**, p = 0.001.

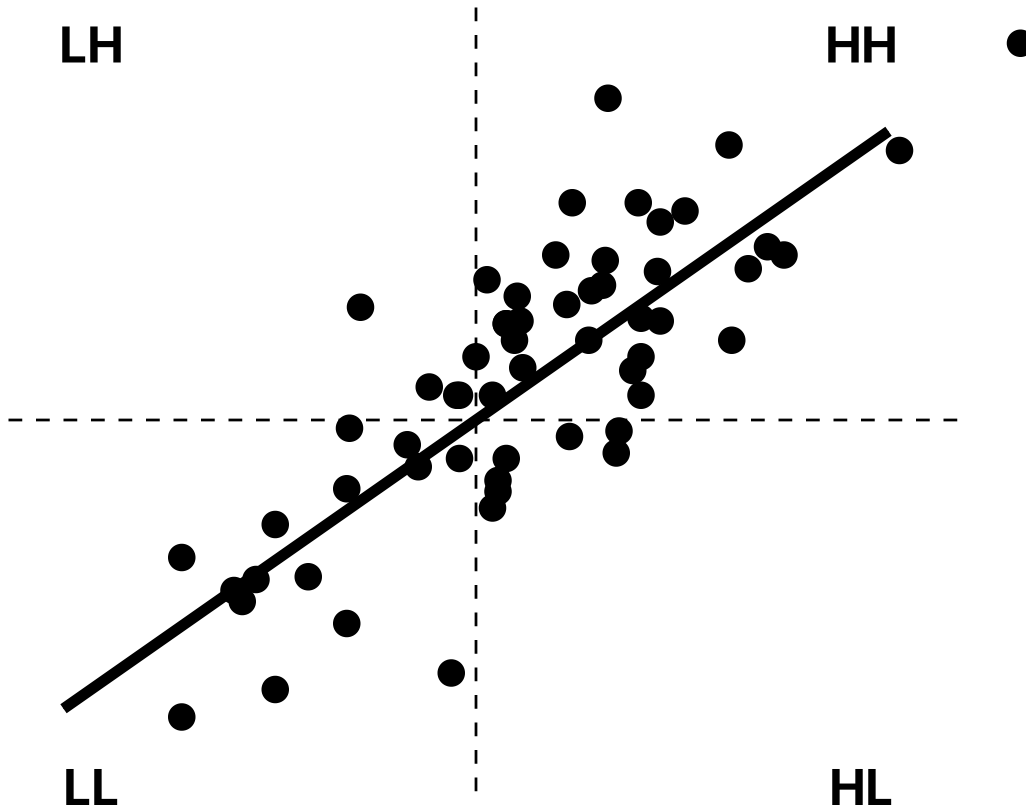


From one number to every place: LISA

Moran's I is an average over the whole map. **Local Moran (LISA)** keeps each place's own term, $I_i = z_i \cdot (Wz)_i$, and tests it by shuffling only that place's neighbours (conditional permutation).

The four quadrants of the Moran scatter give the labels. **High-High**: hotspot. **Low-Low**: coldspot. **High-Low**: a strong performer among weak neighbours. **Low-High**: a weak spot inside a strong area, often an opening or a problem worth checking.

Gurugram: 24 High-High hexagons.



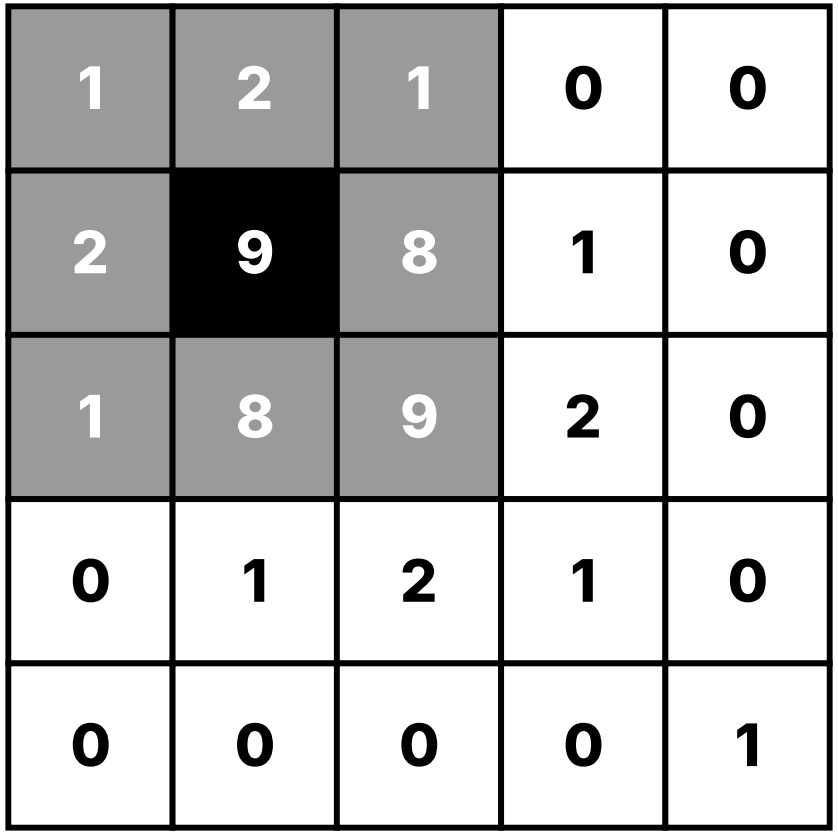
quadrants: HH · LL · HL · LH

Getis-Ord Gi*: is the local total too big?

Gi* draws a window around each place (itself included), adds up the values inside, and compares that local sum with what the window would hold if values were spread at random. The result is a z-score.

In the grid, the black cell's window holds 41 against an expected 17.6 (nine cells × the map mean of 1.96). That is a hotspot.

z ≥ 1.65, 1.96, 2.58 means 90, 95, 99% confidence. A heatmap shows density; Gi* says whether the density is more than chance.



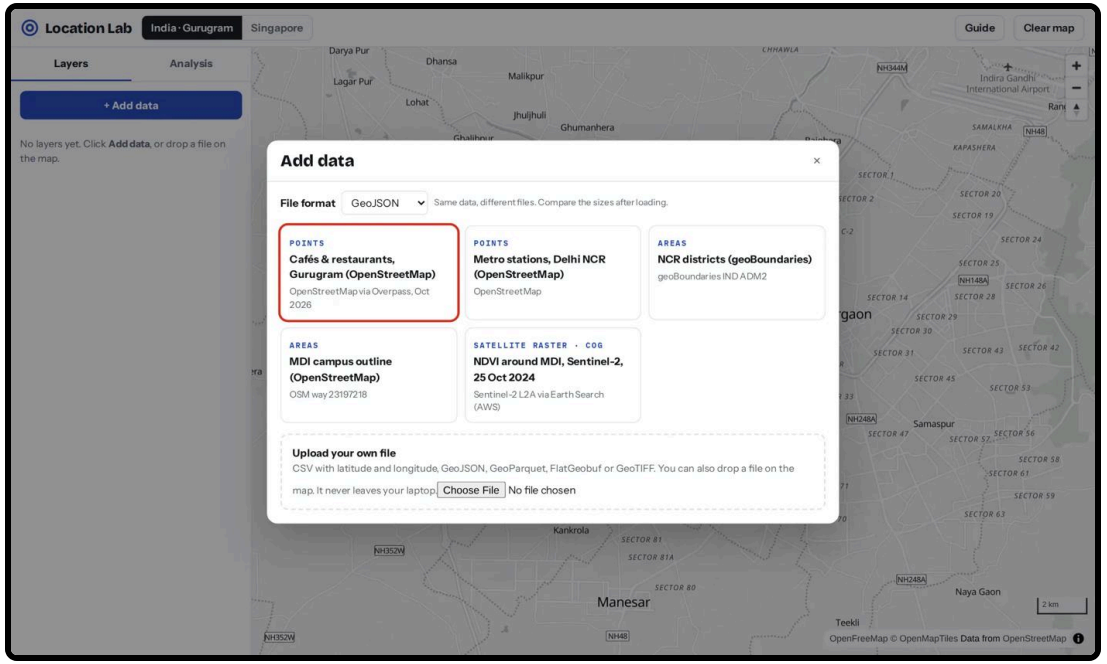
Window of 9 cells around the black cell

4 · Autocorrelation and hotspots

Do busy hexagons sit next to busy hexagons? Where exactly are the significant hotspots?

Open **quantng.github.io/location-lab**, choose **India · Gurugram** in the top bar, then follow the steps.

1. Load the cafés.
2. Count per hexagon, empty cells included.
3. Contiguity weights.
4. Global Moran's I with 999 random shuffles.
5. Local Moran (LISA): hotspots, coldspots and outliers.
6. Getis-Ord Gi*: hotspots at 90/95/99% confidence.



Load the cafés.

DO THIS

Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	Cafés & restaurants, Gurugram (OpenStreetMap)
File format	GeoJSON

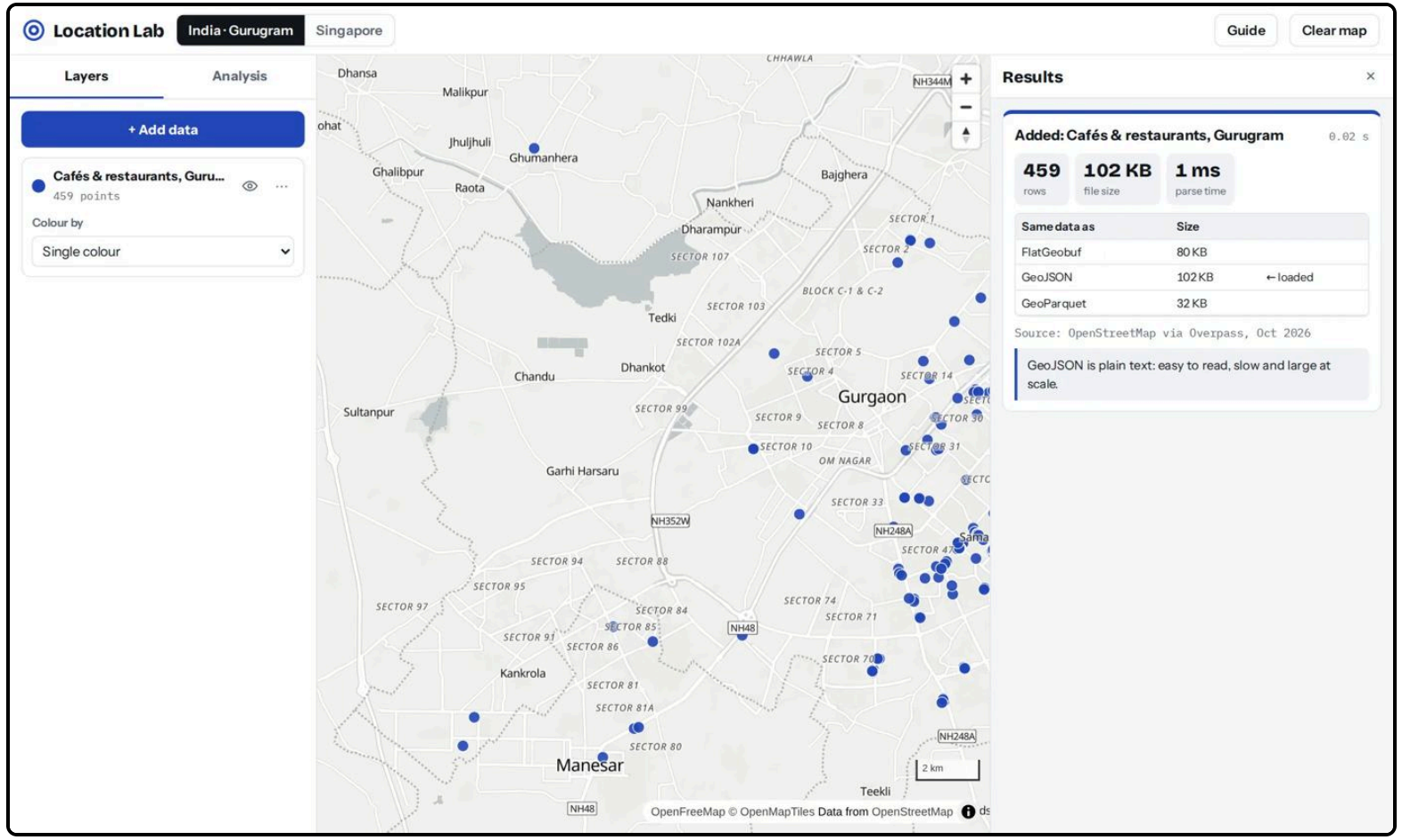
YOU SHOULD SEE

459 rows

102 KB file size

1 ms parse time

> GeoJSON is plain text: easy to read, slow and large at scale.



Count per hexagon, empty cells included.

DO THIS

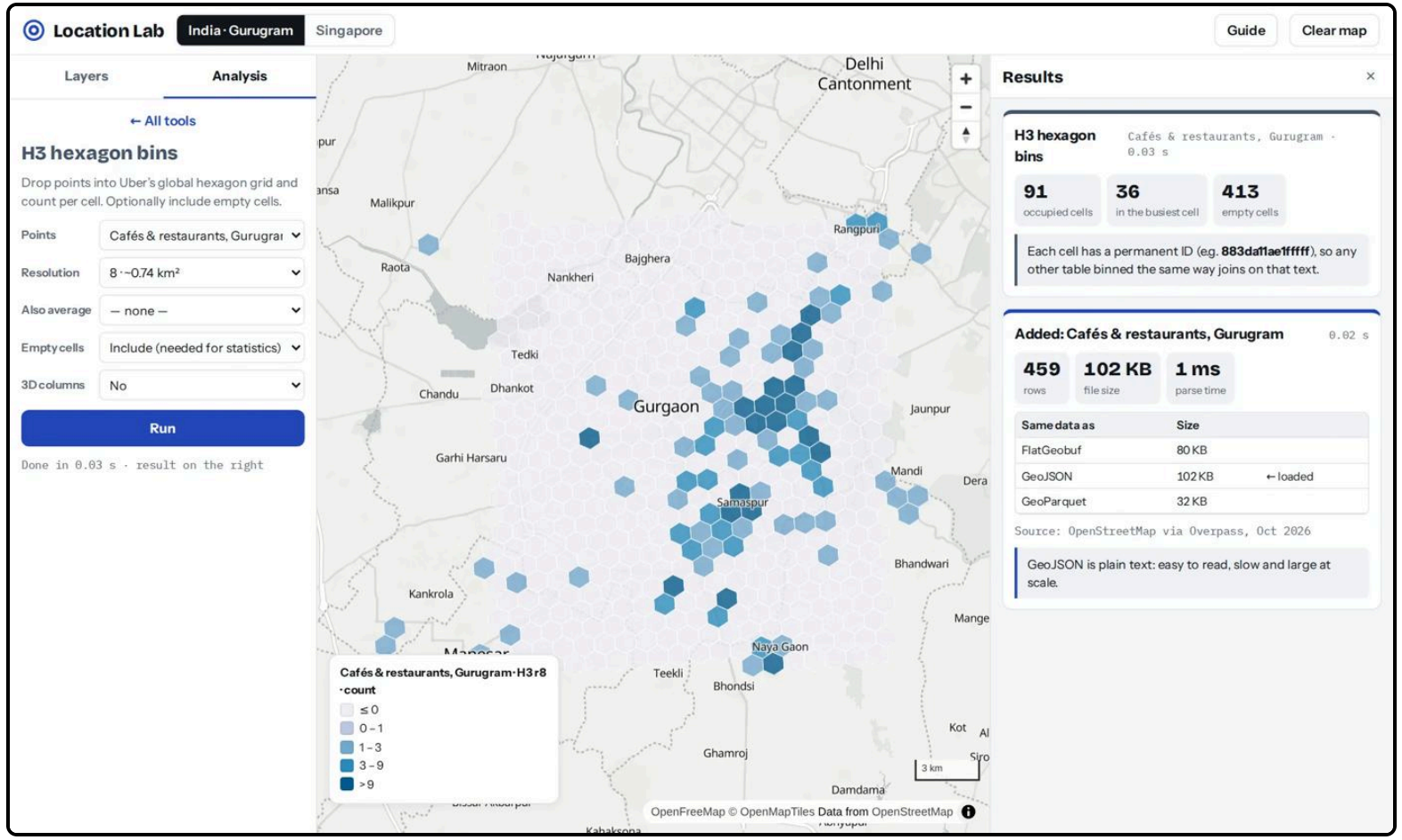
Analysis tab → drag **H3 hexagon bins** onto the map (or click it), check the settings, press **Run**.

Points	Cafés & restaurants, Gurugram (OpenStreetMap)
Resolution	8 · ~0.74 km²
Also average	— none —
Empty cells	Include (needed for statistics)
3D columns	No

YOU SHOULD SEE

91 occupied cells	**36** in the busiest cell	**413** empty cells

> Each cell has a permanent ID (e.g. 883da11ae1fffff), so any other table binned the same way joins on that text.



Contiguity weights.

DO THIS

Analysis tab → drag **Spatial weights** onto the map (or click it), check the settings, press **Run**.

Layer	Cafés & restaurants, Gurugram · H3 r8
Neighbours are	touching H3 hexagons

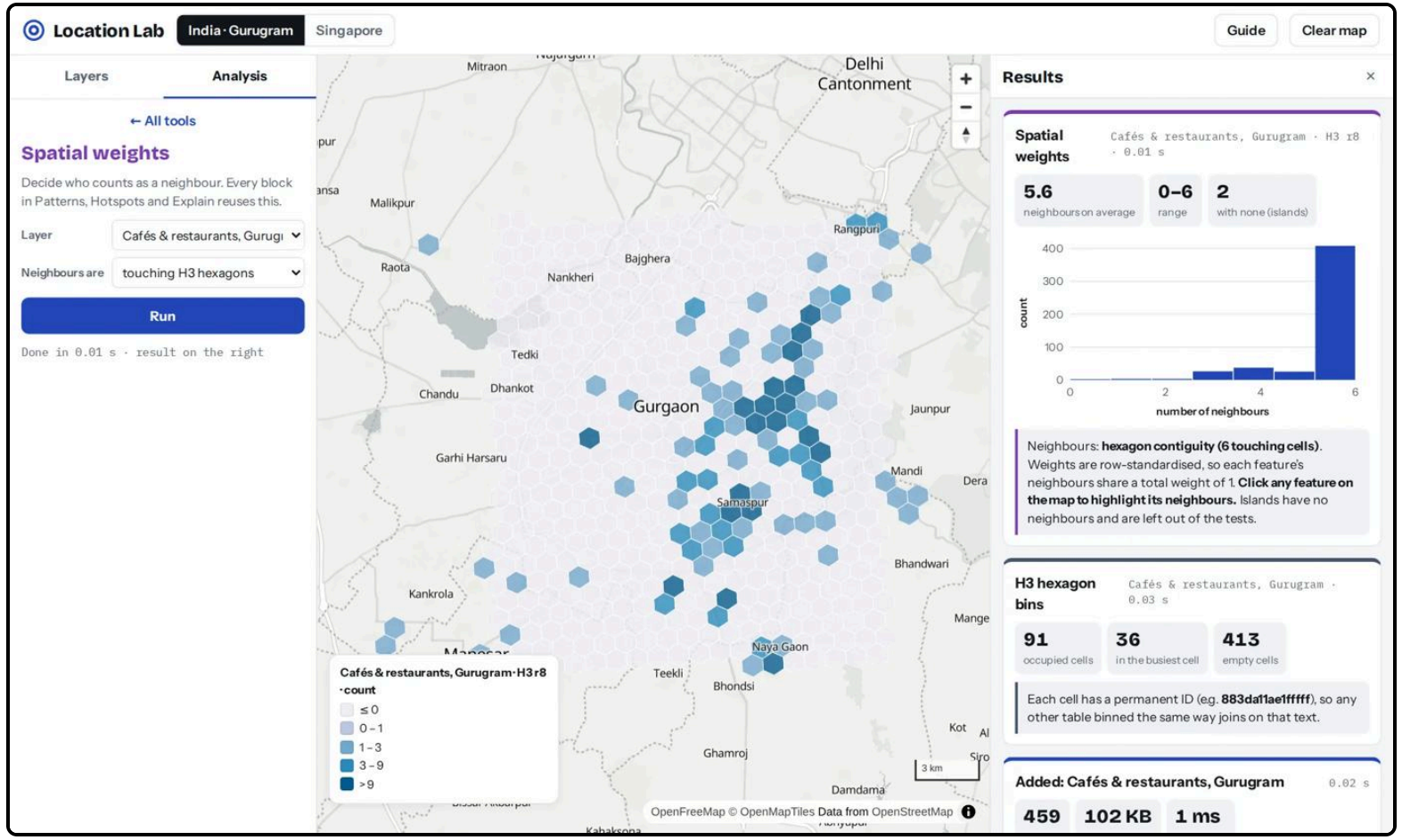
YOU SHOULD SEE

5.6
neighbours on average

0–6
range

2
with none (islands)

> Neighbours: hexagon contiguity (6 touching cells). Weights are row-standardised, so each feature's neighbours share a total weight of 1. Click any feature on the map to highlight its…



Global Moran’s I with 999 random shuffles.

DO THIS

Analysis tab → drag **Global Moran’s I** onto the map (or click it), check the settings, press **Run**.

Layer with weights	Cafés & restaurants, Gurugram · H3 r8
Value	count
Shuffles	999

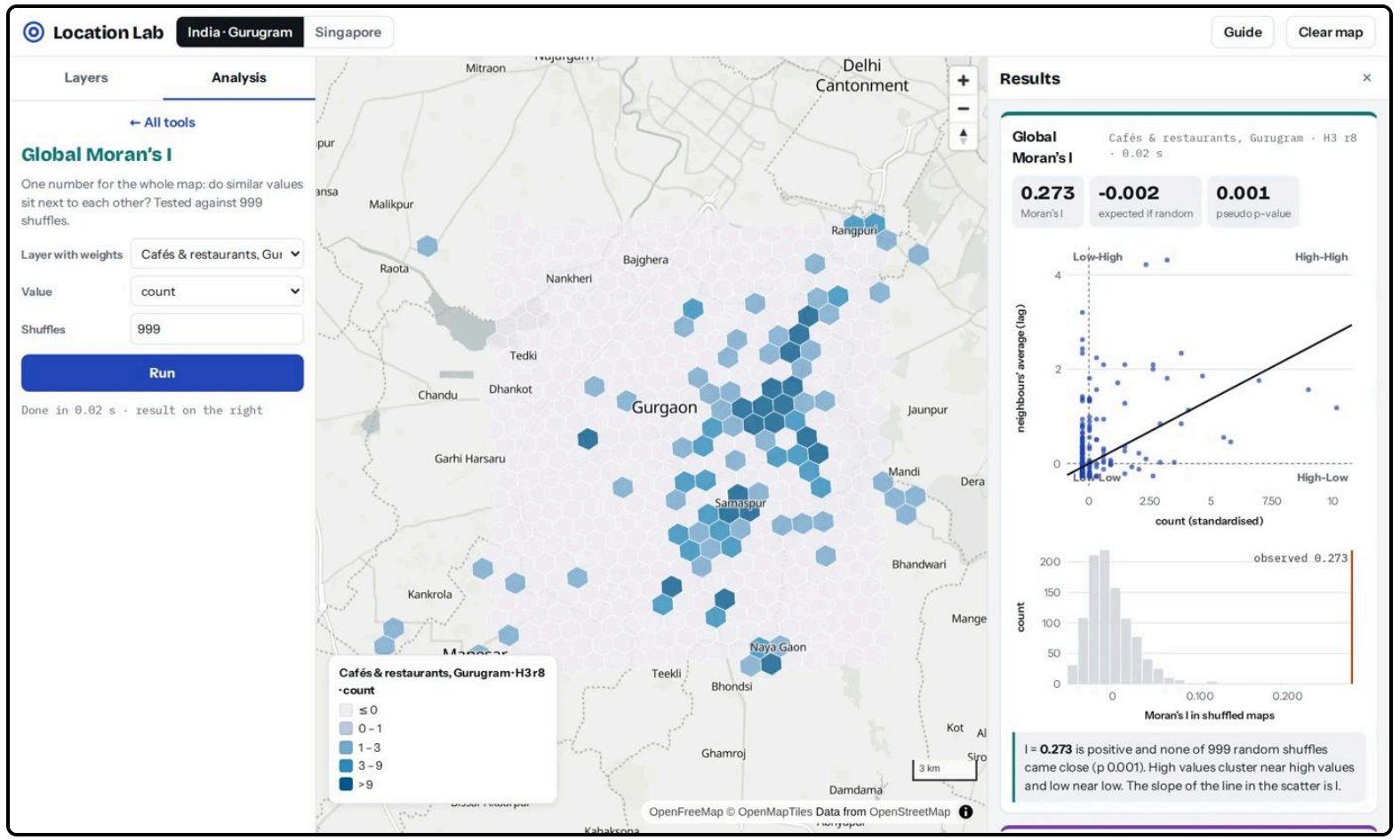
YOU SHOULD SEE

0.273
Moran’s I

-0.002
expected if random

0.001
pseudo p-value

> I = 0.273 is positive and none of 999 random shuffles came close (p 0.001). High values cluster near high values and low near low. The slope of the line in the scatter is I.



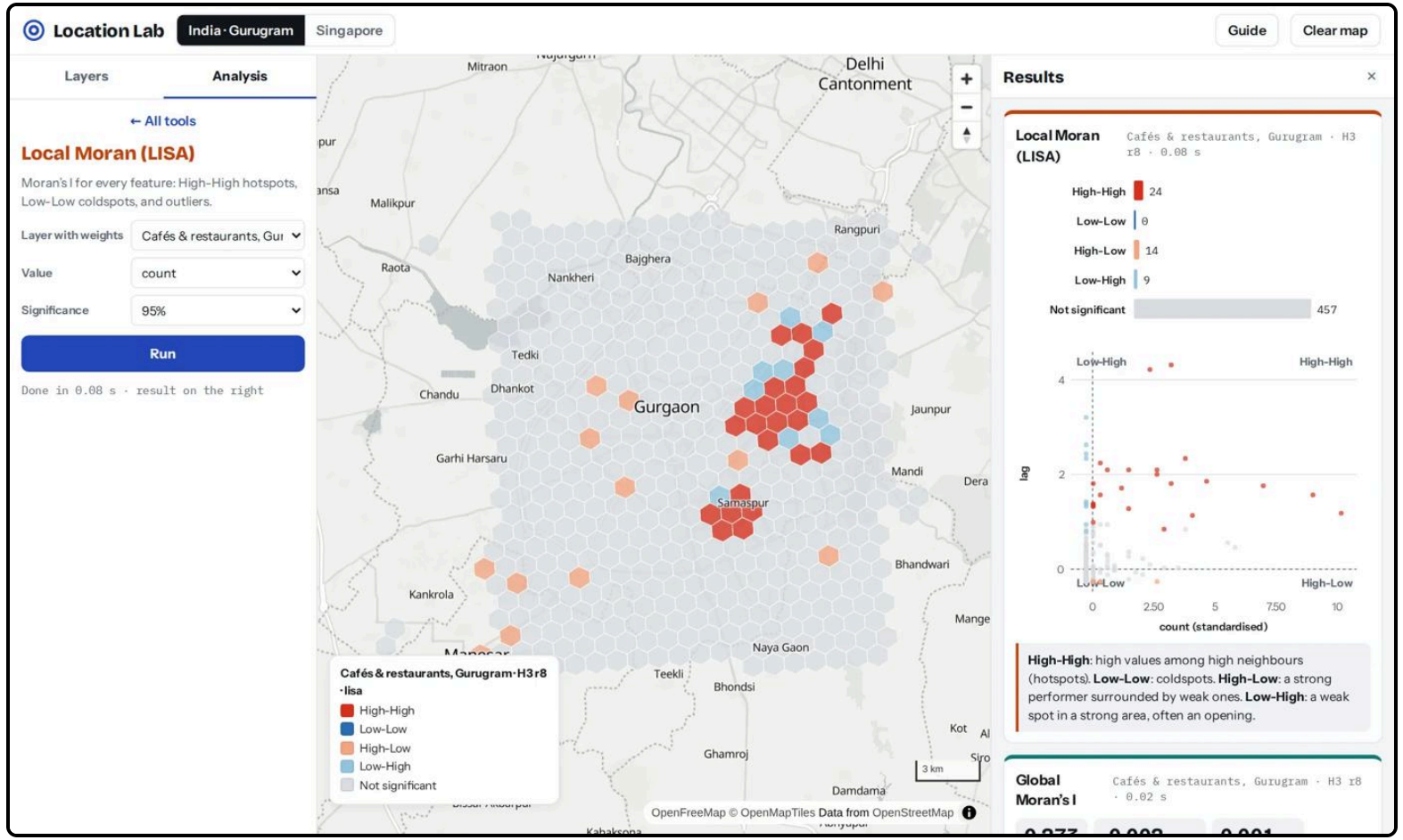
Local Moran (LISA): hotspots, coldspots and outliers.

DO THIS

Analysis tab → drag **Local Moran (LISA)** onto the map (or click it), check the settings, press **Run**.

Layer with weights	Cafés & restaurants, Gurugram · H3 r8
Value	count
Significance	95%

> High-High: high values among high neighbours (hotspots). Low-Low: coldspots. High-Low: a strong performer surrounded by weak ones. Low-High: a weak spot in a strong area, often an opening.



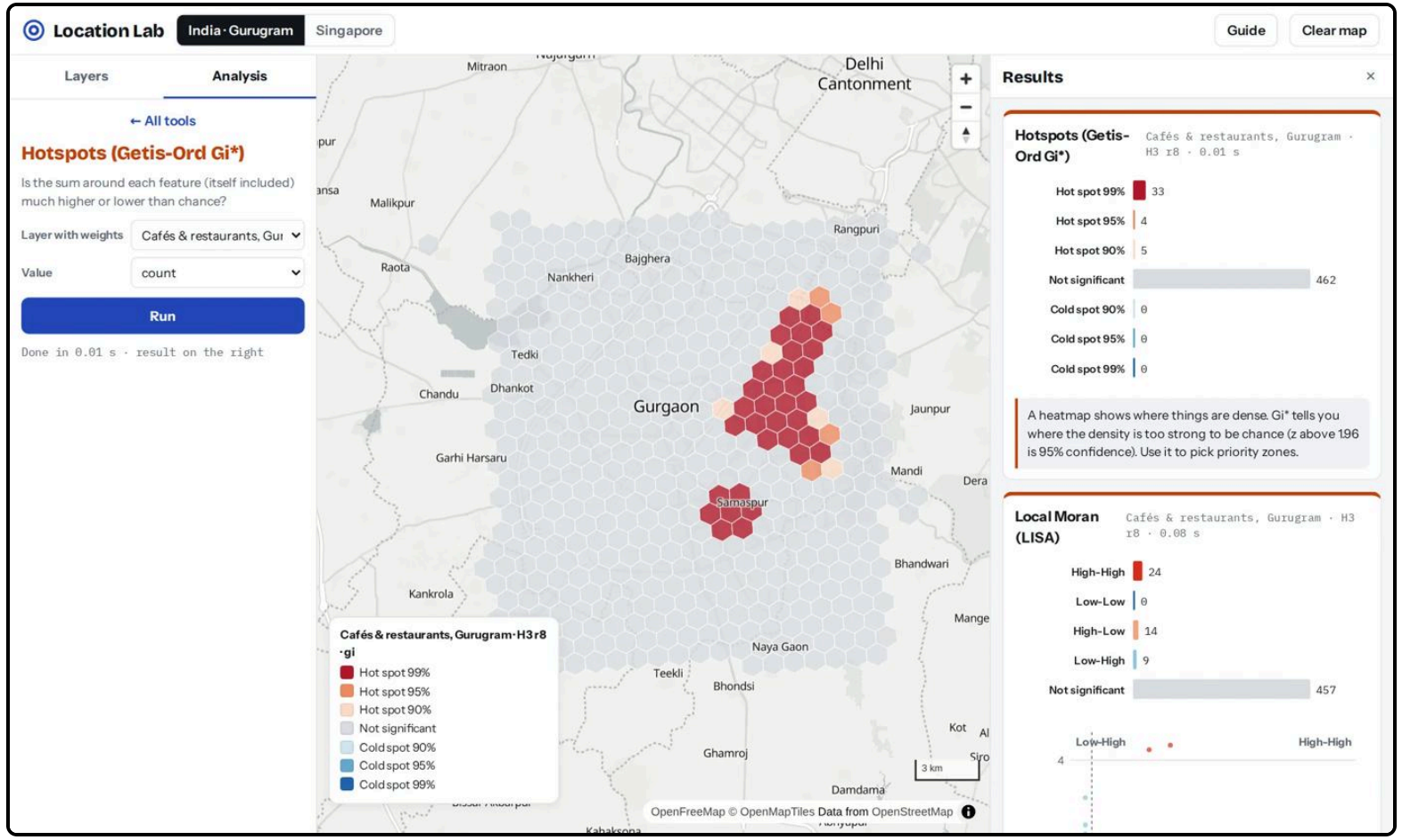
Getis-Ord Gi*: hotspots at 90/95/99% confidence.

DO THIS

Analysis tab → drag **Hotspots (Getis-Ord Gi*)** onto the map (or click it), check the settings, press **Run**.

Layer with weights	Cafés & restaurants, Gurugram · H3 r8
Value	count

> A heatmap shows where things are dense. Gi* tells you where the density is too strong to be chance (z above 1.96 is 95% confidence). Use it to pick priority zones.



Three questions

1. Moran’s I = 0.273 with p = 0.001. Interpret.
2. What does a Low-High hexagon mean for a café chain?
3. Why is a heatmap not enough to choose priority zones?

Answers

1. **Moran's I = 0.273 with p = 0.001. Interpret.**
   Positive autocorrelation: hexagons with many cafés sit next to others with many. No shuffle came close.
2. **What does a Low-High hexagon mean for a café chain?**
   A gap inside a busy area: possibly an opening, or a site-specific barrier worth checking.
3. **Why is a heatmap not enough to choose priority zones?**
   It shows density, not significance. Gi* tests whether a concentration exceeds chance.

PART 5

Clusters for territories

The path we take on the board:

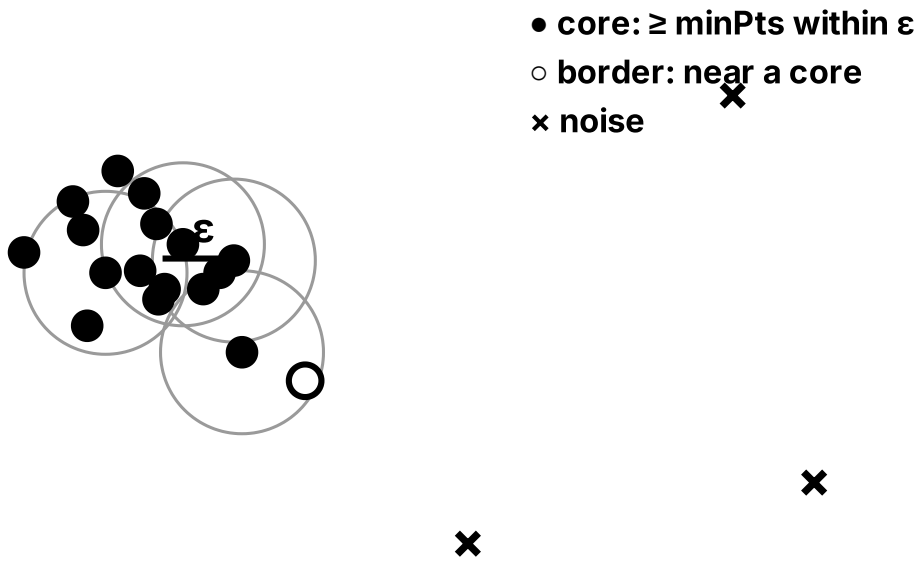
Density → **Core · border · noise** → **ε and minPts**

Density, not distance to a centre

k-means assumes round clusters and asks you for k in advance. Markets are not round: they follow roads, malls and metro lines.

DBSCAN asks two business questions instead. **ε**: how close is “the same place”? **minPts**: how many points make a market? A point with at least minPts others within ε is a **core**. Points within ε of a core are **border** points. Everything else is **noise**.

Clusters grow by chaining cores together, so they can take any shape, and the number of clusters comes out of the data.

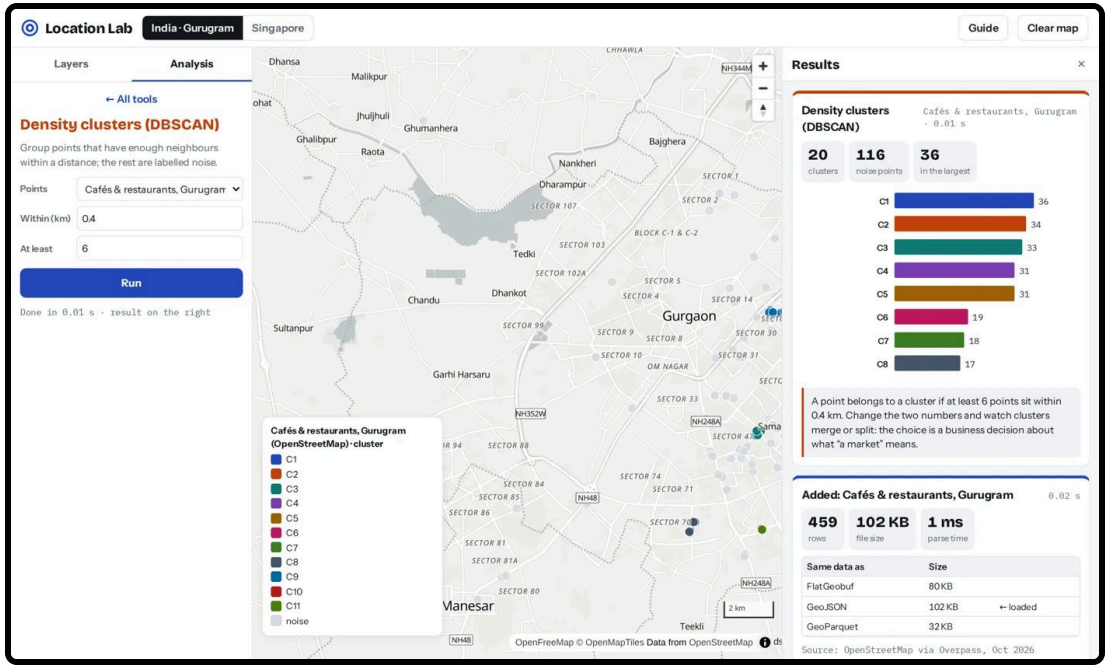


Choosing ε and minPts

Set ε from the decision: a delivery radius, a walking distance, a block spacing. Set minPts from what counts as a viable cluster. Larger ε or smaller minPts merges clusters and shrinks noise; the opposite splits them.

Gurugram, ε = 400 m and minPts = 6: **20 clusters** and **116 noise** cafés, the isolated spots outside every dining strip.

Noise is not useless: an isolated outlet with strong sales may mark an underserved catchment.



Where business already uses this

IN THE NEWS · OUTLOOK BUSINESS · 20 JAN 2025

Blinkit crosses 1,000 dark stores and targets 2,000

Each dark store serves a cluster of nearby demand. DBSCAN-style clustering on orders is how such territories are drawn.

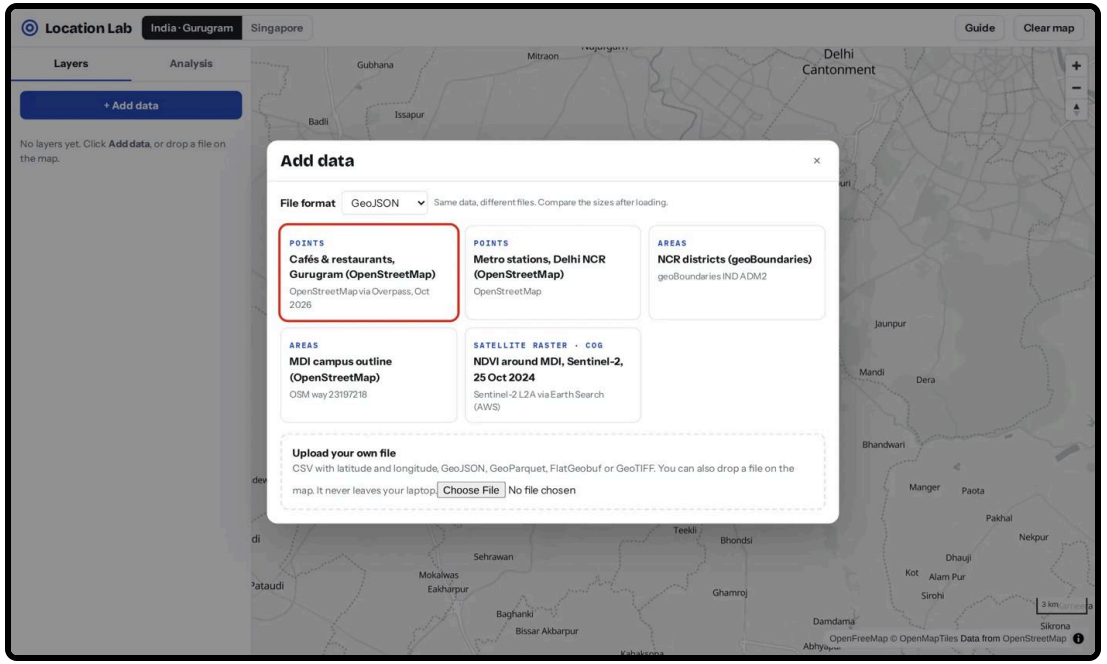
https://www.outlookbusiness.com/start-up/e-commerce/blinkit-crosses-1000-dark-store-milestone-as-quick-commerce-race-heats-up

5 · Clusters for territories

Group nearby outlets into market clusters and flag isolated ones.

Open **quantng.github.io/location-lab**, choose **India · Gurugram** in the top bar, then follow the steps.

1. Load the cafés.
2. DBSCAN: clusters of at least 6 places within 400 m.



LAB 5 · STEP 1 OF 2

Load the cafés.

DO THIS

Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	Cafés & restaurants, Gurugram (OpenStreetMap)
File format	GeoJSON

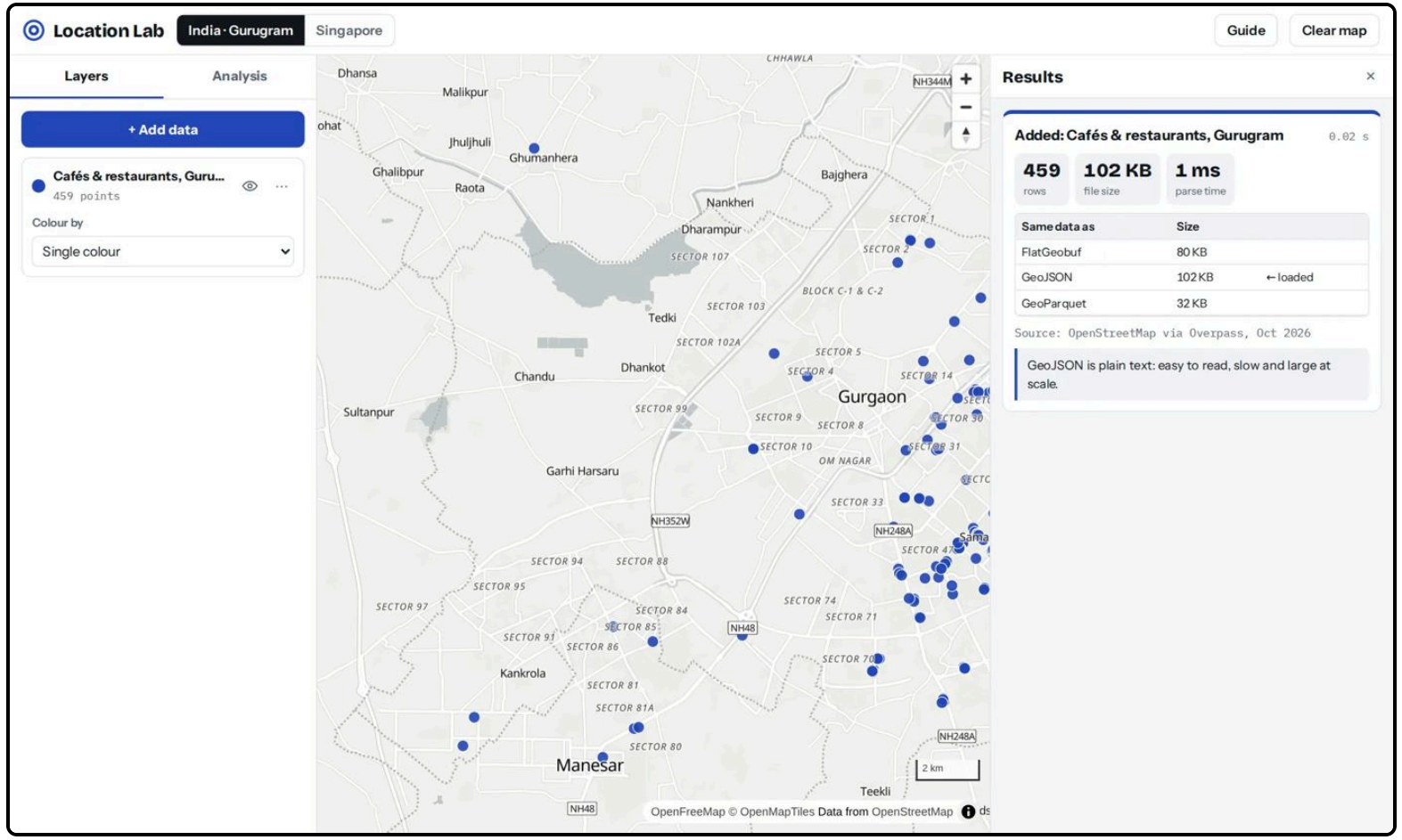
YOU SHOULD SEE

459 rows

102 KB file size

1 ms parse time

> GeoJSON is plain text: easy to read, slow and large at scale.



DBSCAN: clusters of at least 6 places within 400 m.

DO THIS

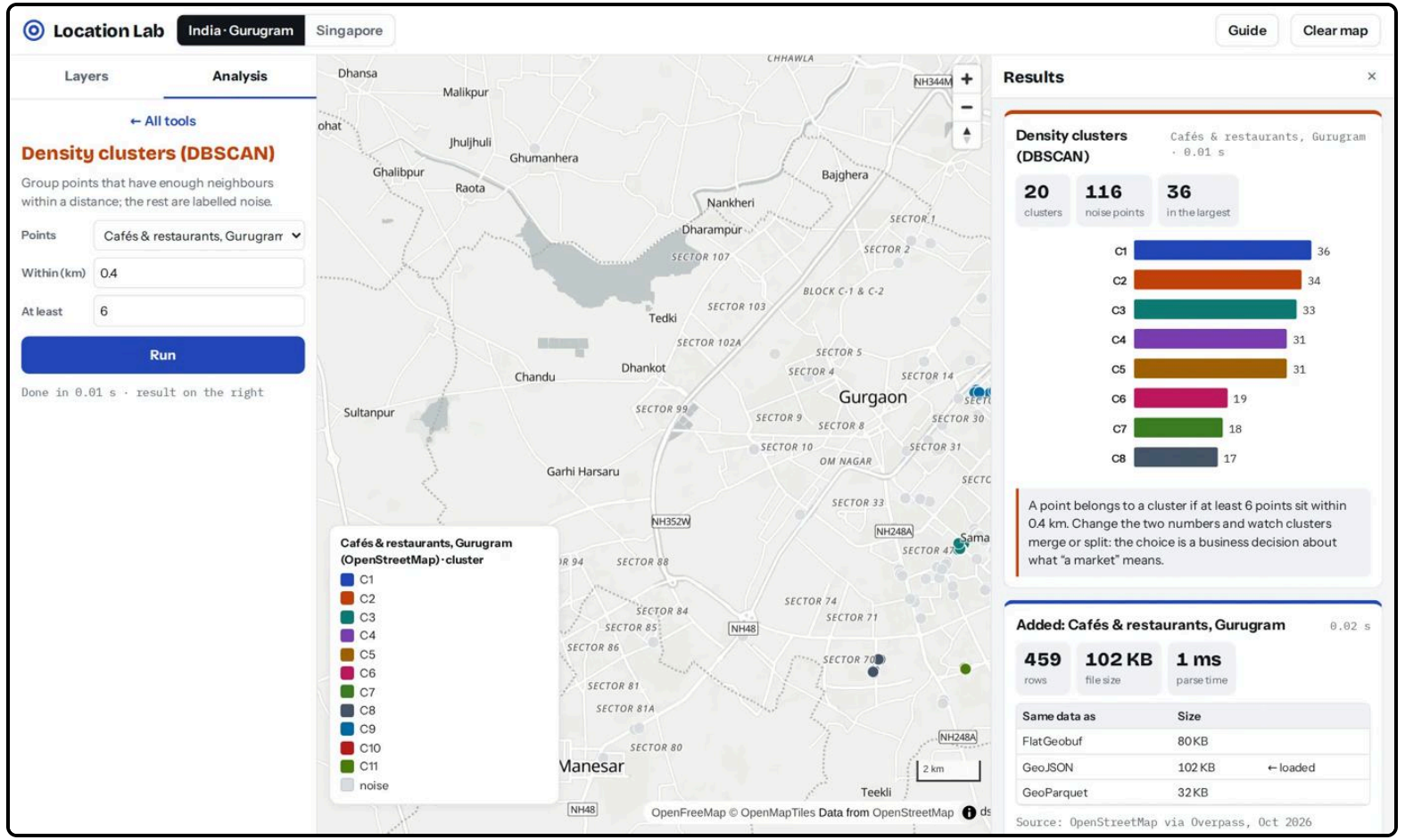
Analysis tab → drag **Density clusters (DBSCAN)** onto the map (or click it), check the settings, press **Run**.

Points	Cafés & restaurants, Gurugram (OpenStreetMap)
Within (km)	0.4
At least	6

YOU SHOULD SEE

20	116	36
clusters	noise points	in the largest

> A point belongs to a cluster if at least 6 points sit within 0.4 km. Change the two numbers and watch clusters merge or split: the choice is a business decision about what “a market” means.



Three questions

1. DBSCAN labels 116 cafés as noise. How can you reduce that?
2. Why prefer DBSCAN to k-means for dining clusters?
3. Name a business choice that sets ε.

Answers

1. **DBSCAN labels 116 cafés as noise. How can you reduce that?**
   Increase ε or lower minPts.
2. **Why prefer DBSCAN to k-means for dining clusters?**
   Clusters follow roads and malls and take any shape; DBSCAN needs no k and flags isolated outlets.
3. **Name a business choice that sets ε.**
   The walking or delivery distance that counts as one market.

PART 6

From samples to surfaces

The path we take on the board:

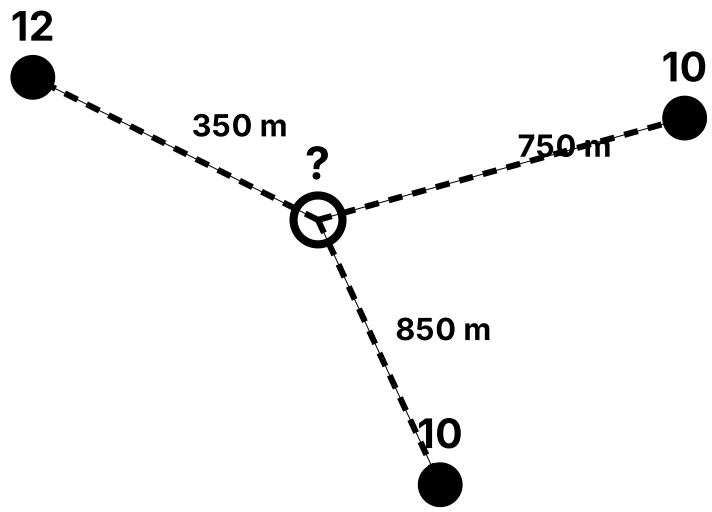
IDW → Power → Variogram → Kriging + error → Hold-out check

Guess the value between samples: IDW

You know the value at three places and want it at a fourth. Closer samples should count more. **Inverse distance weighting** gives each sample weight 1 ÷ distance^{p} and takes the weighted average.

Samples 12, 10, 10 at 350, 750, 850 m. With p = 1: (12/350 + 10/750 + 10/850) ÷ (1/350 + 1/750 + 1/850) = **11.1**. With p = 2 the nearest sample dominates more: **11.4**.

IDW is simple and exact at the samples, but it cannot tell you how sure it is.

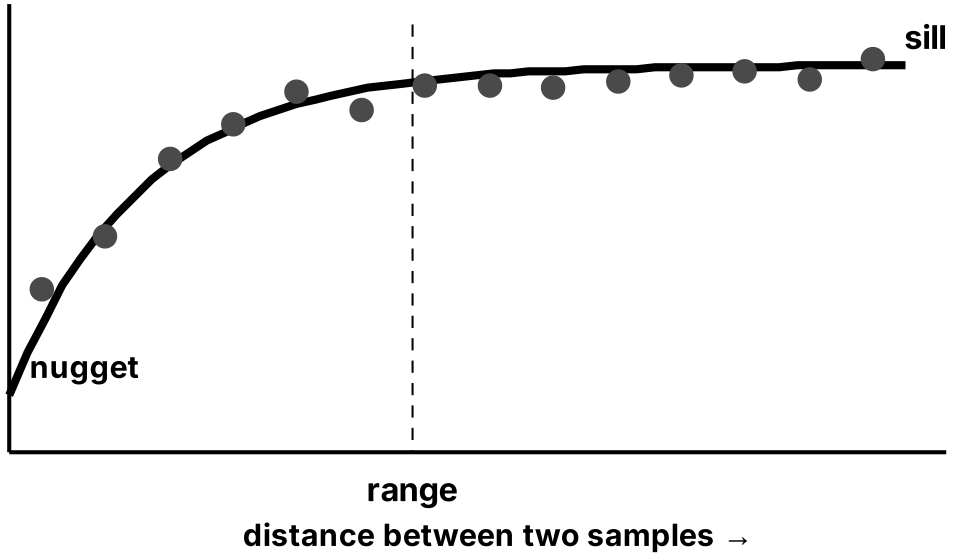


The variogram: how fast similarity fades

Take every pair of samples, compute half the squared difference of their values, and plot it against their distance. Close pairs are similar (low); far pairs are not (high).

Three numbers describe the curve. **Nugget**: the difference at almost zero distance (measurement noise, micro-variation). **Sill**: the plateau, the total variance. **Range**: the distance where the curve flattens, beyond which samples tell you nothing about each other.

Gurugram NDVI: range ≈ **0.54 km**.

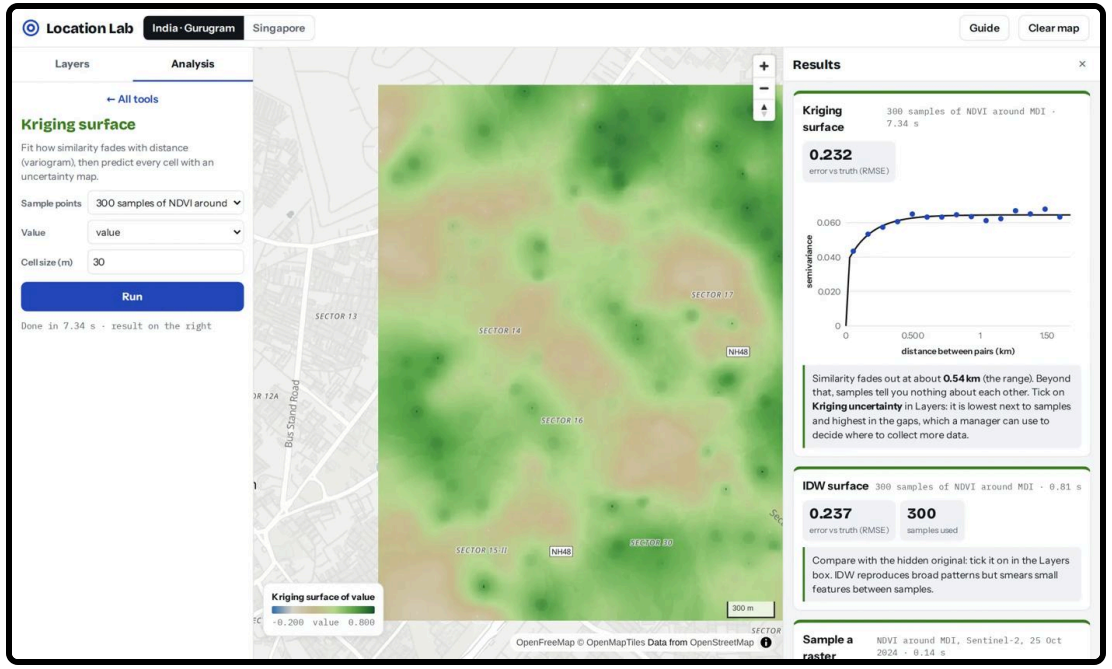


Kriging: the best guess, and how good it is

Kriging uses the variogram to choose the weights instead of a fixed power. Samples that are close and not redundant get more weight. It also returns a **standard error** for every cell: low near samples, high in the gaps.

That second map is a business tool: it tells you where to collect the next sample.

We hide a real NDVI image, keep 300 pixels and rebuild it. Hold-out error: IDW RMSE 0.237, kriging 0.232.



Where business already uses this

IN THE NEWS · VASSAR LABS · 13 AUG 2026

YES-TECH: satellites now carry 30% weight in India’s crop-insurance yield estimates

Canopy greenness from satellites plus ground samples, scaled up to whole districts under PMFBY: NDVI and interpolation deciding insurance pay-outs.

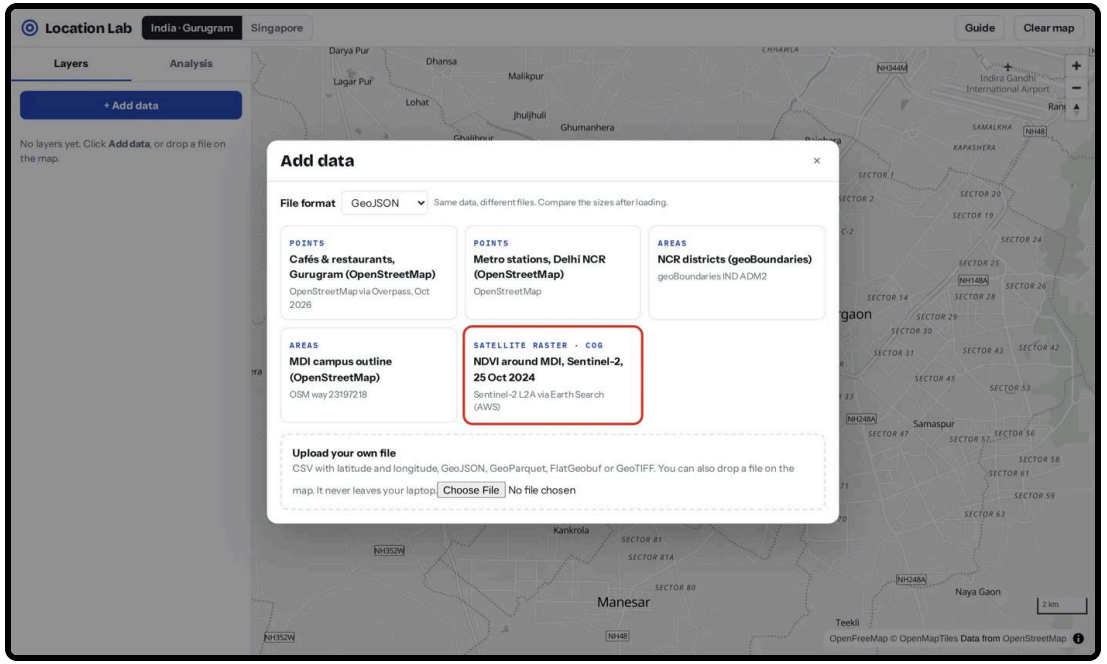
https://vassarlabs.com/yes-tech-explained-how-satellites-estimate-crop-yield-for-insurance/

6 · From samples to surfaces

Estimate a value everywhere from a few samples, then check against the truth.

Open **quantng.github.io/location-lab**, choose **India · Gurugram** in the top bar, then follow the steps.

1. Load the NDVI raster: this is the truth.
2. Keep only 300 random sample points.
3. IDW: rebuild the surface from the 300 samples.
4. Kriging: a surface plus its uncertainty.



Load the NDVI raster: this is the truth.

DO THIS

Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	NDVI around MDI, Sentinel-2, 25 Oct 2024
File format	Cloud-Optimised GeoTIFF

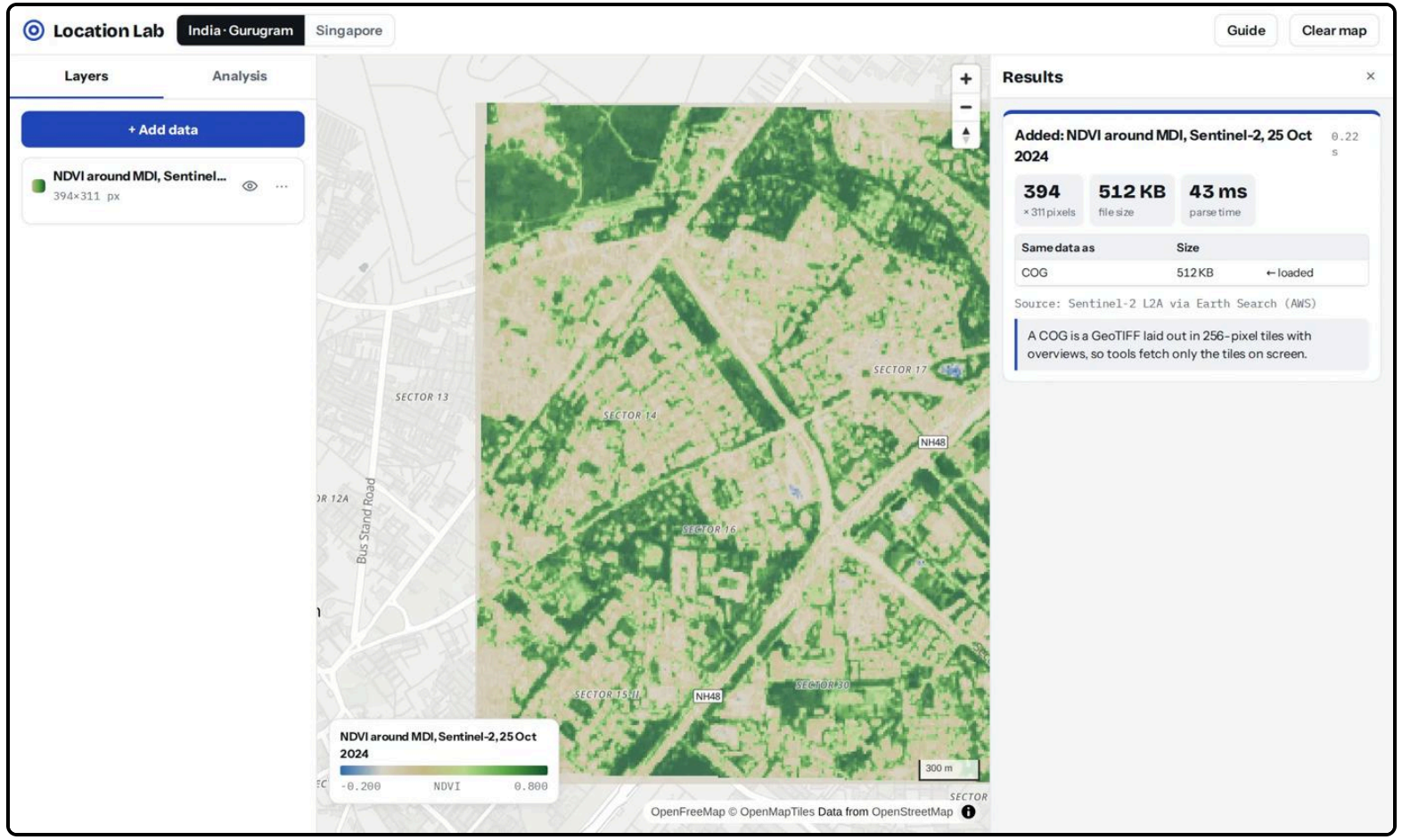
YOU SHOULD SEE

394 × 311 pixels

512 KB file size

43 ms parse time

> A COG is a GeoTIFF laid out in 256-pixel tiles with overviews, so tools fetch only the tiles on screen.



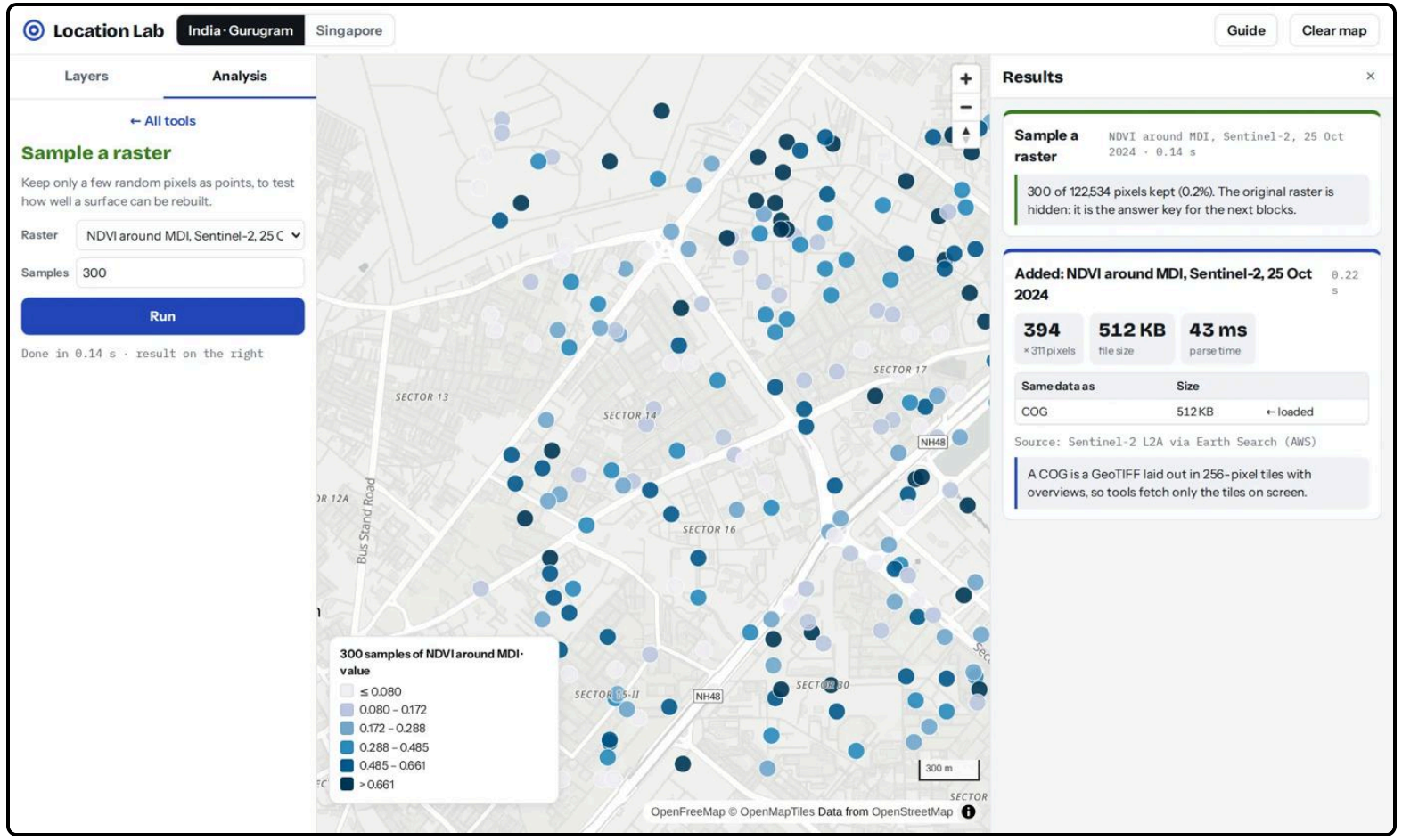
Keep only 300 random sample points.

DO THIS

Analysis tab → drag **Sample a raster** onto the map (or click it), check the settings, press **Run**.

Raster	NDVI around MDI, Sentinel-2, 25 Oct 2024
Samples	300

> 300 of 122,534 pixels kept (0.2%). The original raster is hidden: it is the answer key for the next blocks.



IDW: rebuild the surface from the 300 samples.

DO THIS

Analysis tab → drag **IDW surface** onto the map (or click it), check the settings, press **Run**.

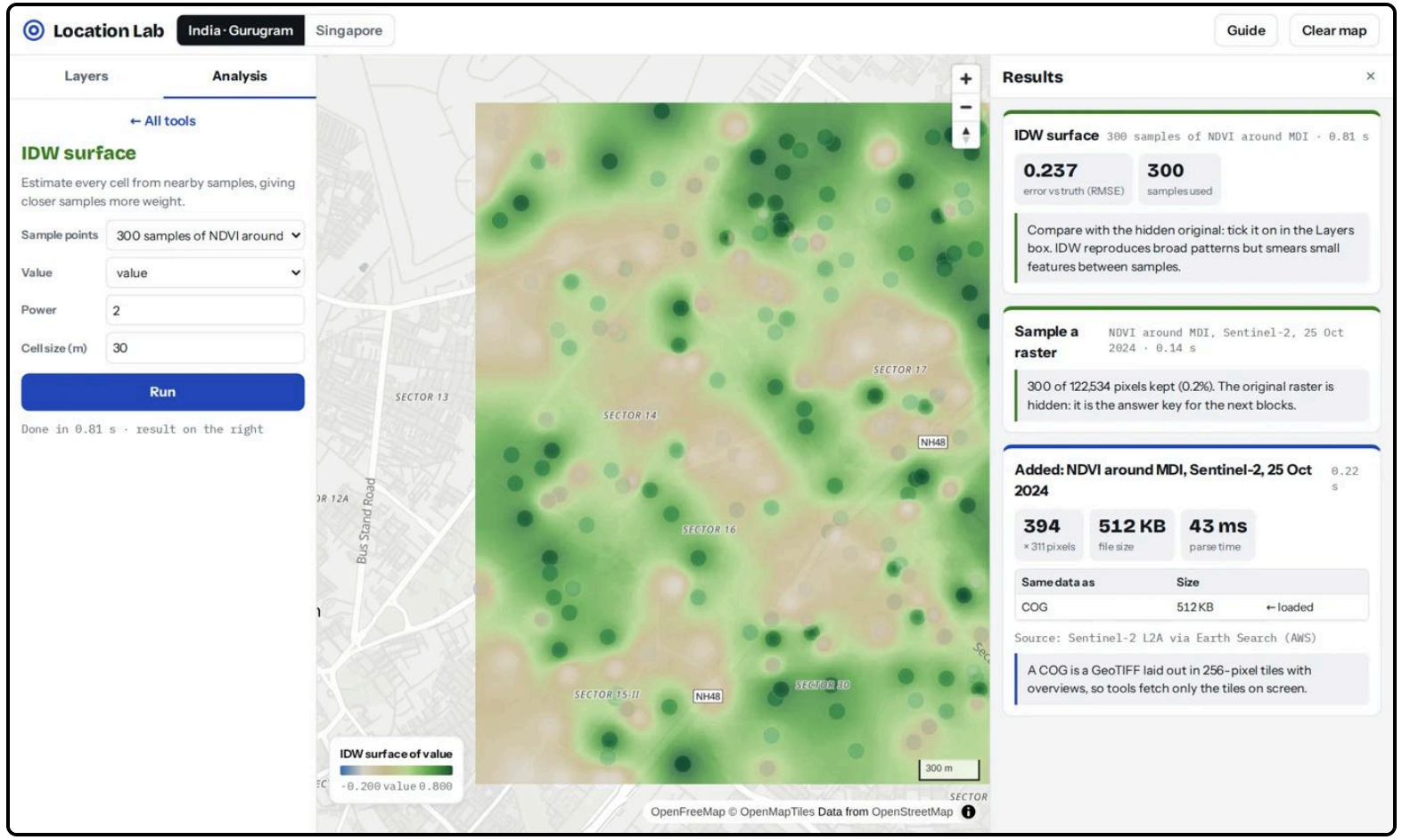
Sample points	300 samples of NDVI around MDI
Value	value
Power	2
Cell size (m)	30

YOU SHOULD SEE

0.237
error vs truth (RMSE)

300
samples used

> Compare with the hidden original: tick it on in the Layers box. IDW reproduces broad patterns but smears small features between samples.



Kriging: a surface plus its uncertainty.

DO THIS

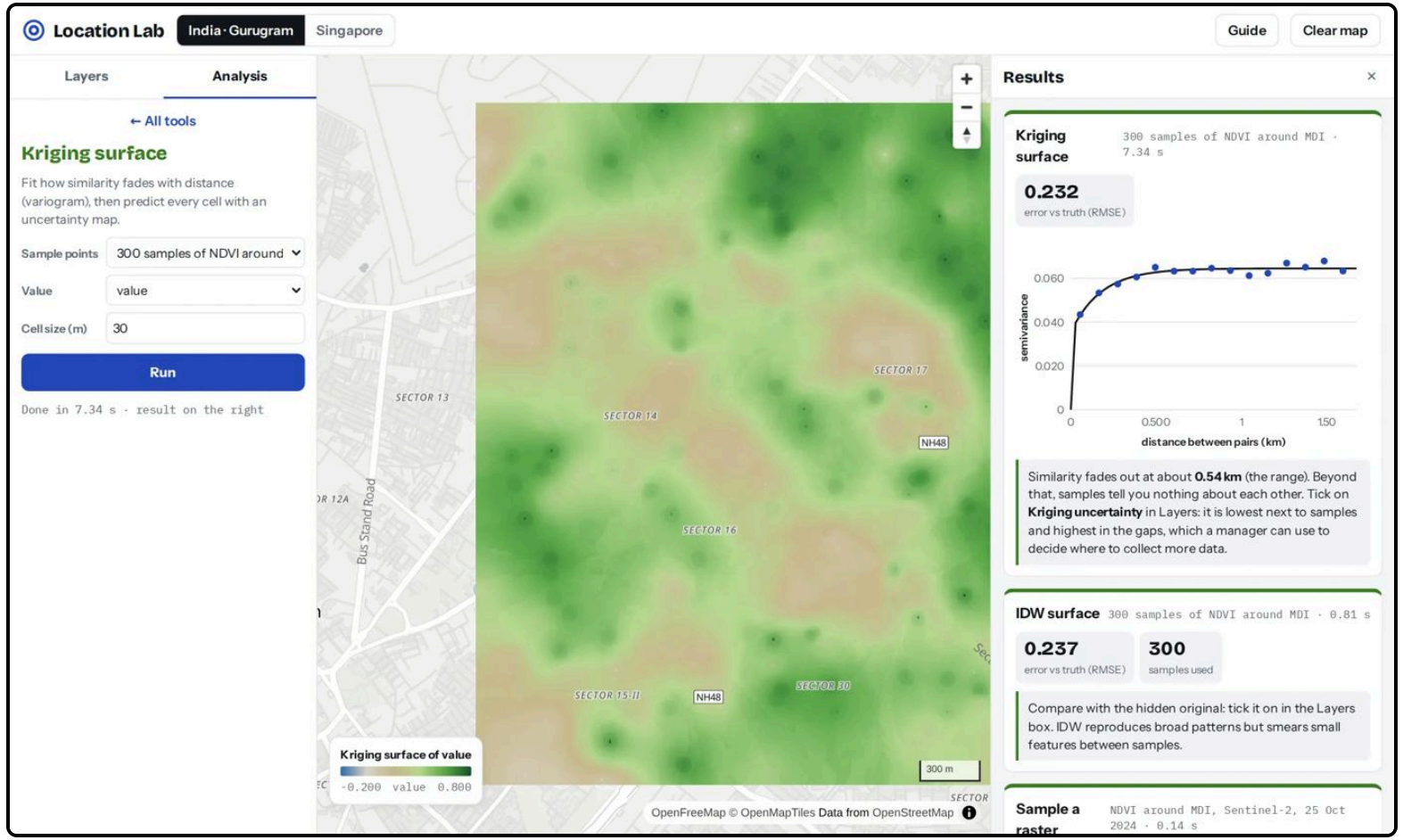
Analysis tab → drag **Kriging surface** onto the map (or click it), check the settings, press **Run**.

Sample points	300 samples of NDVI around MDI
Value	value
Cell size (m)	30

YOU SHOULD SEE

0.232
error vs truth (RMSE)

> Similarity fades out at about 0.54 km (the range). Beyond that, samples tell you nothing about each other. Tick on Kriging uncertainty in Layers: it is lowest next to samples and highest in...



Three questions

1. IDW with p = 1 gives 11.1 and with p = 2 gives 11.4. Why the rise?
2. The variogram range is 0.54 km. What does it tell a surveyor?
3. What does kriging give that IDW does not?

Answers

1. **IDW with p = 1 gives 11.1 and with p = 2 gives 11.4. Why the rise?**
   A higher power gives the nearest sample (12) even more weight.
2. **The variogram range is 0.54 km. What does it tell a surveyor?**
   Samples more than about half a kilometre apart tell little about each other; sample more densely than that.
3. **What does kriging give that IDW does not?**
   A standard-error map showing where the estimate is reliable.

PART 7

Explaining location

The path we take on the board:

OLS → **Independence** → **Residual Moran** → **Lag vs error** → **AIC** → **GWR**

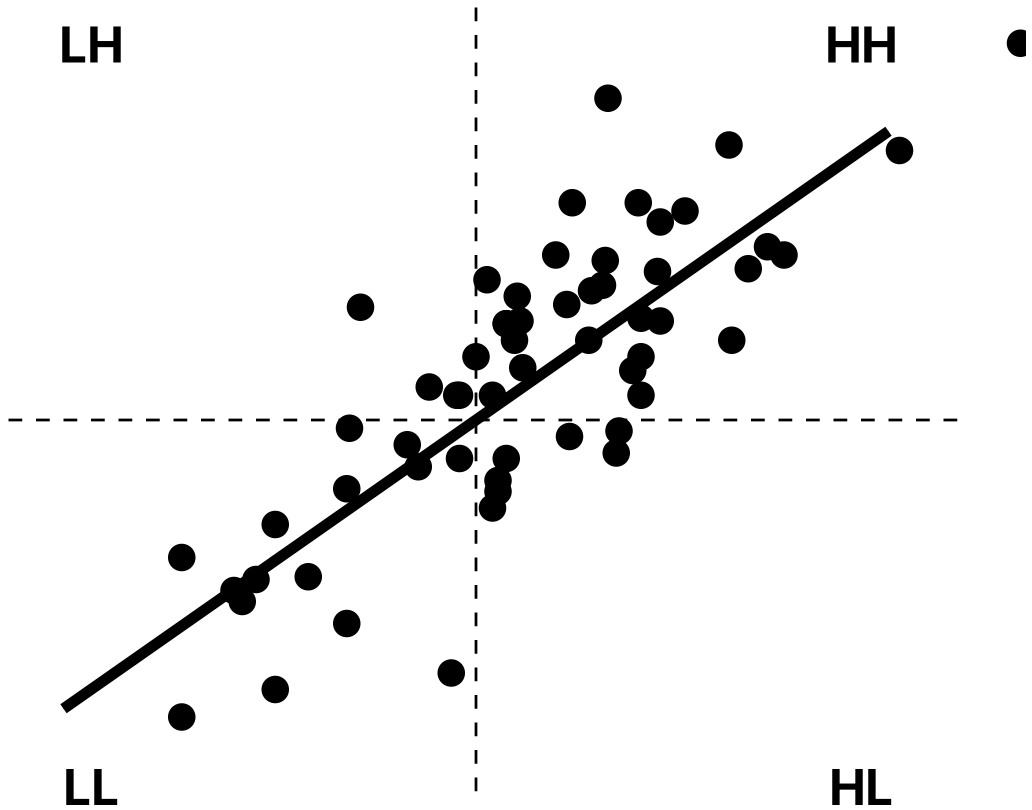
EXPLAIN · STEP 1

OLS: one line for the whole map

Ordinary least squares fits $y = \beta_0 + \beta_1 x_1 + \ldots + \varepsilon$ and reports one coefficient per variable for the whole map. Its inference assumes the errors ε are **independent**.

On a map that assumption is suspect: if an unmeasured factor (prestige, school zones, a new mall) is shared by neighbours, the errors of neighbours move together, standard errors are too small and coefficients may be biased.

Gurugram cafés per hexagon on distance to metro and to Cyber Hub: **$R^2 = 0.04$**.



one slope for all locations

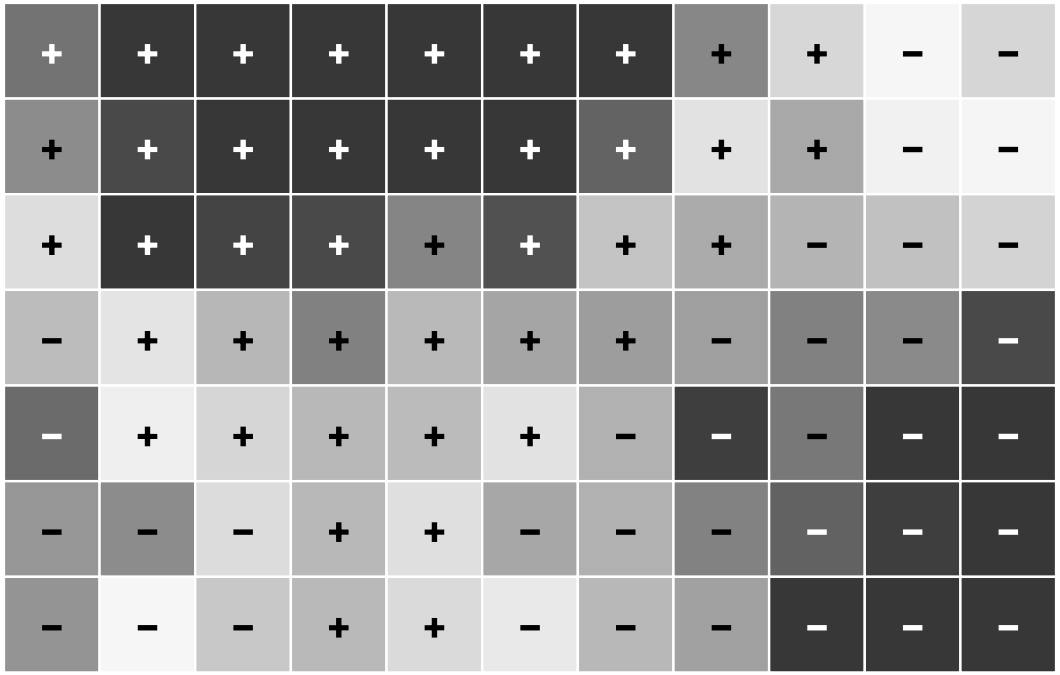
EXPLAIN · STEP 2

Test the errors: map the residuals

Map the residuals (actual – predicted). If OLS has captured everything spatial, the map looks like noise. If plus signs cluster with plus signs, something spatial is missing.

We already have the test: **Moran's I of the residuals**, using the same W.

Here the residuals still carry spatial structure, so a spatial model is needed.



Clustered residuals: the model misses a spatial factor

Two fixes: spillover or shared omissions

Spatial lag model: $y = \rho Wy + X\beta + \varepsilon$. My value depends on my neighbours' values: genuine spillover (a café draws footfall to the next one; a renovated block lifts its neighbours).

Spatial error model: $y = X\beta + u,\ u = \lambda Wu + \varepsilon$. Neighbours share unmeasured factors, but do not cause each other.

Fit both and compare **AIC** (fit penalised for complexity; lower is better). Gurugram: $\rho = 0.47$, $\lambda = 0.47$.

$$y = \rho Wy + X\beta + \varepsilon$$

$$y = X\beta + u,\ u = \lambda Wu + \varepsilon$$

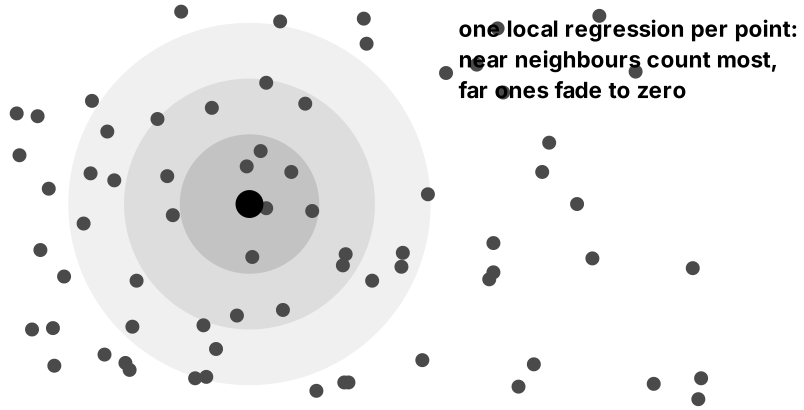
EXPLAIN · STEP 4

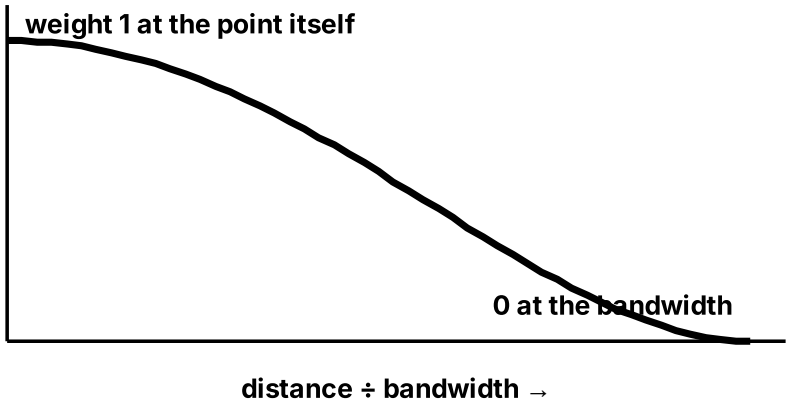
GWR: let the coefficients vary

OLS asks “what is the metro effect?” GWR asks “what is the metro effect **here**?” It fits a separate regression at every location, weighting nearby observations with a kernel that falls from 1 to 0 at the bandwidth.

Small bandwidth: very local, noisy. Large: smooth, close to OLS. The bandwidth is chosen automatically by minimising AICc.

Gurugram: R^2 rises from 0.04 to **0.45**: the factors matter, but differently by place.





Where business already uses this

IN THE NEWS · THE TRIBUNE · 16 FEB 2024

Old Gurugram metro: 28.5 km and 27 stations from Millennium City Centre

A new line changes every block's "distance to metro". Regression and GWR estimate how much that is worth, and where.

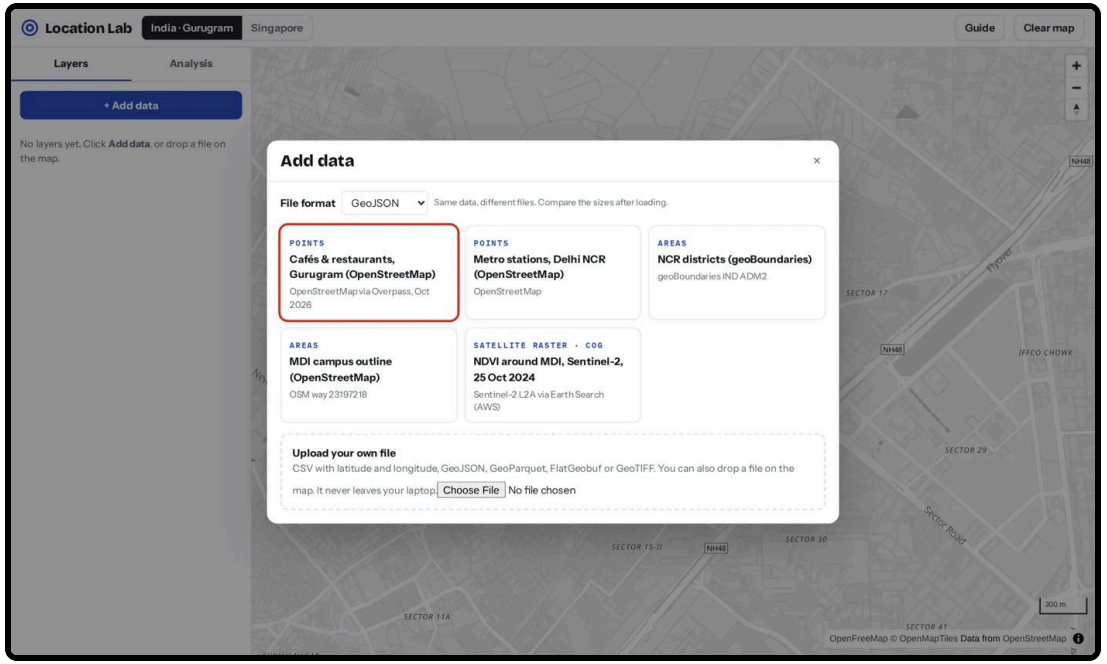
https://www.tribuneindia.com/news/haryana/old-gurugram-to-get-metro-service-pm-lays-foundation-591642

7 · What explains café density?

Regress café counts on access, then check whether space is still hiding in the errors.

Open **quantng.github.io/location-lab**, choose **India · Gurugram** in the top bar, then follow the steps.

1. Load the cafés.
2. Count per hexagon.
3. Load the metro stations.
4. Distance from each hexagon to its nearest metro station.
5. Distance to Cyber Hub, the business district.
6. Contiguity weights.
7. Ordinary regression, then Moran's I on its residuals.
8. Spatial lag model (spillover).
9. Spatial error model (clustered missing factors).
10. GWR: does the metro effect change across the city?



Load the cafés.

DO THIS

Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	Cafés & restaurants, Gurugram (OpenStreetMap)
File format	GeoJSON

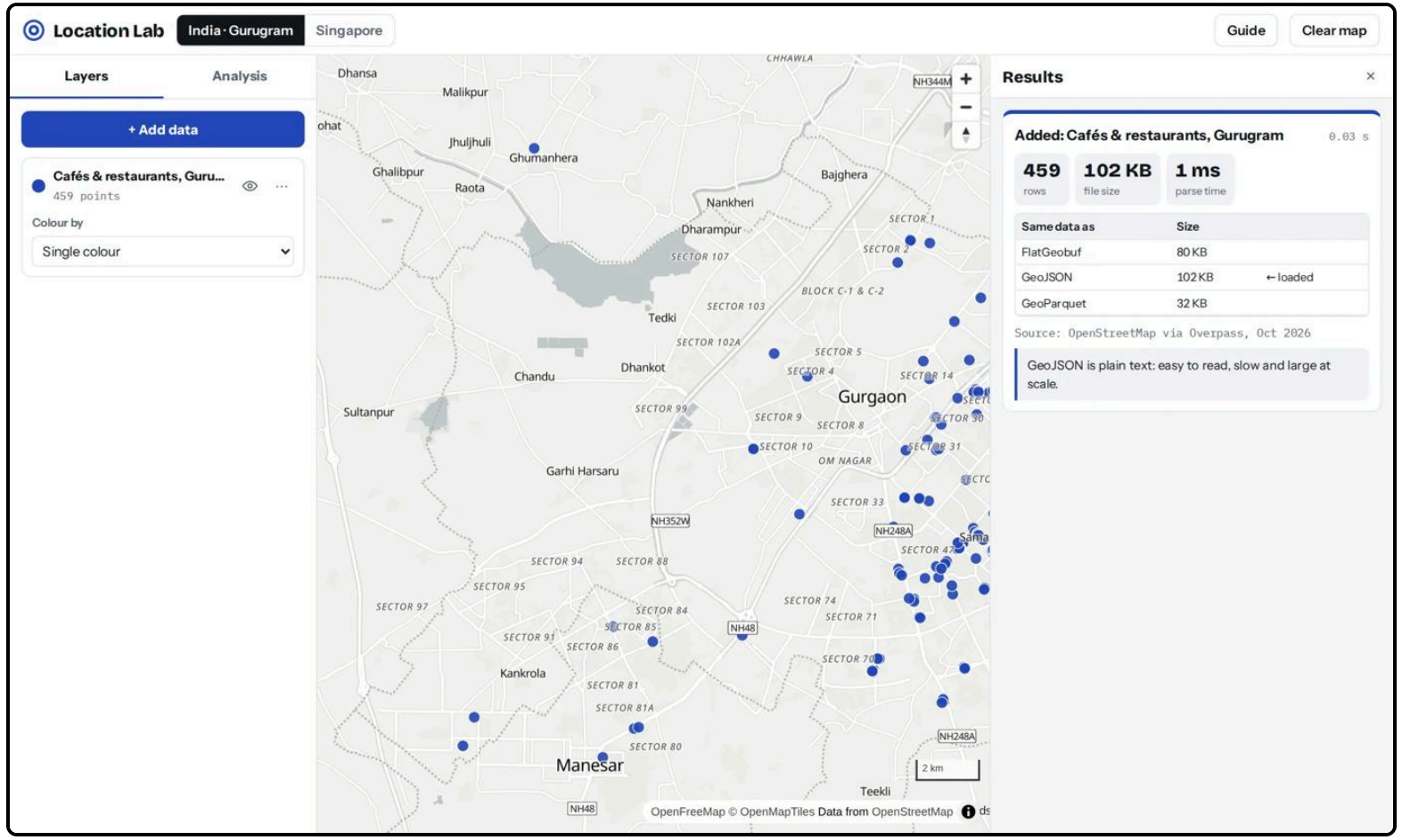
YOU SHOULD SEE

459 rows

102 KB file size

1 ms parse time

> GeoJSON is plain text: easy to read, slow and large at scale.



Count per hexagon.

DO THIS

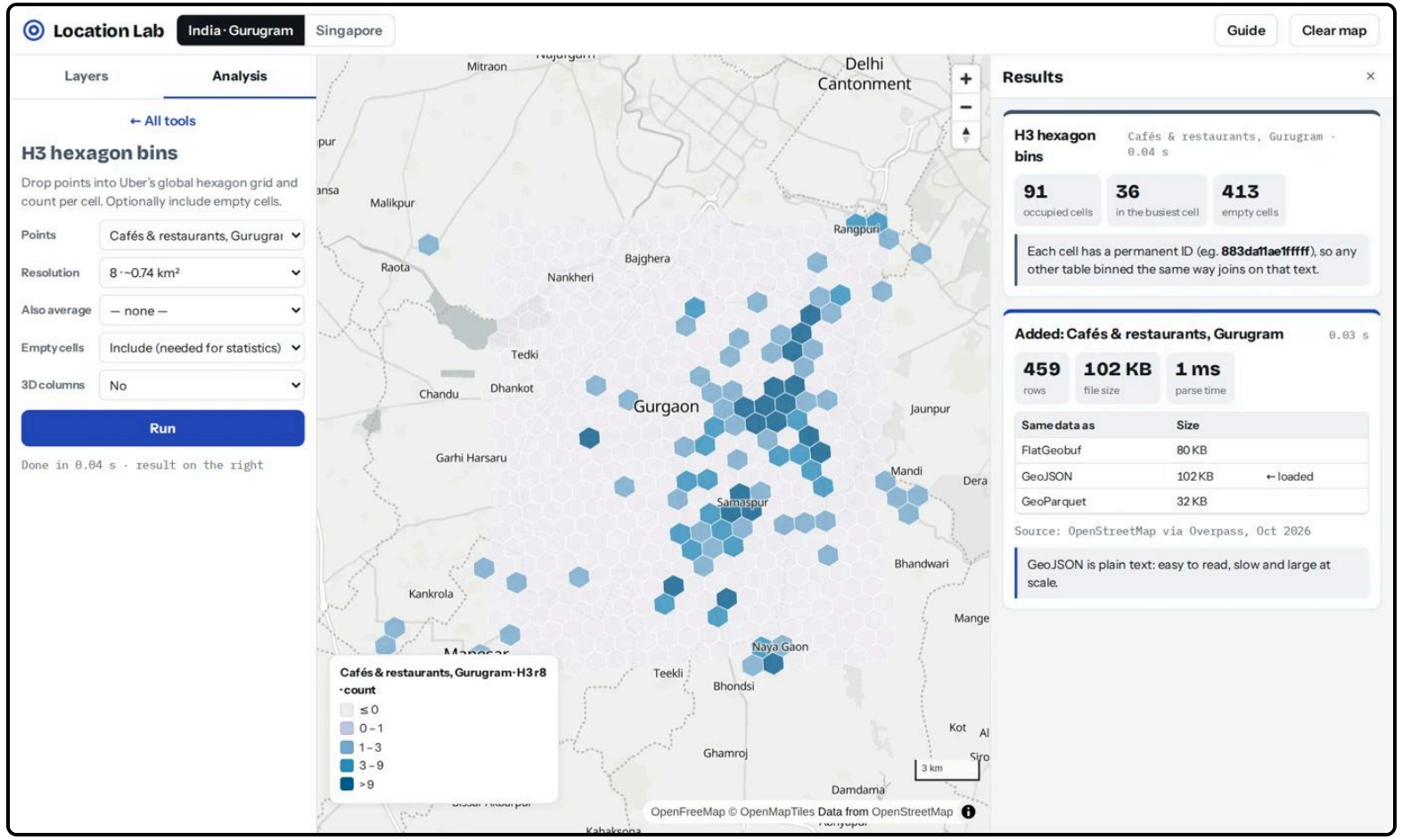
Analysis tab → drag **H3 hexagon bins** onto the map (or click it), check the settings, press **Run**.

Points	Cafés & restaurants, Gurugram (OpenStreetMap)
Resolution	8 · ~0.74 km²
Also average	— none —
Empty cells	Include (needed for statistics)
3D columns	No

YOU SHOULD SEE

91	36	413
occupied cells	in the busiest cell	empty cells

> Each cell has a permanent ID (e.g. 883da11ae1fffff), so any other table binned the same way joins on that text.



Load the metro stations.

DO THIS

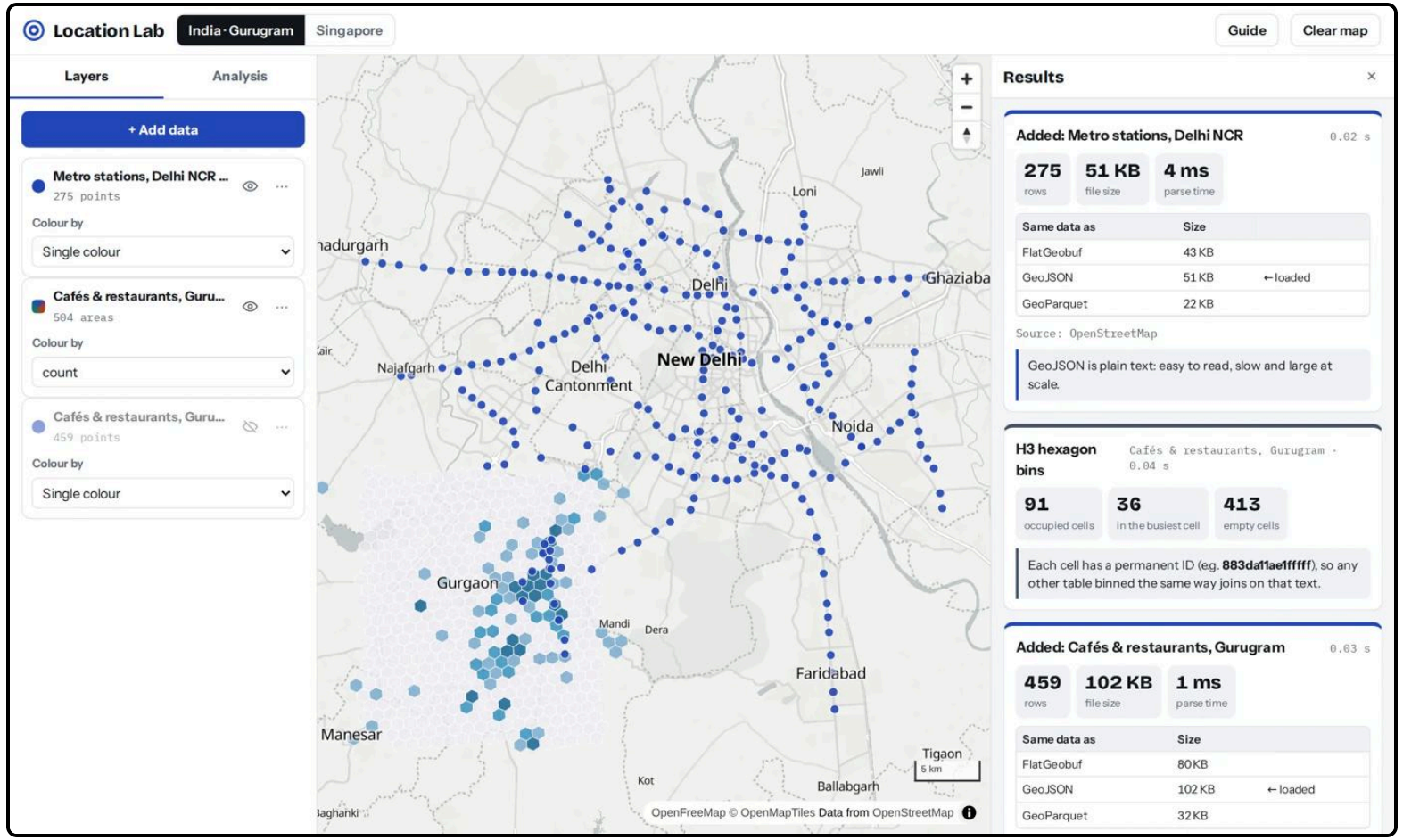
Layers tab → **+ Add data** → pick the format → click the dataset.

Dataset	Metro stations, Delhi NCR (OpenStreetMap)
File format	GeoJSON

YOU SHOULD SEE

275 rows | **51 KB** file size | **4 ms** parse time

> GeoJSON is plain text: easy to read, slow and large at scale.



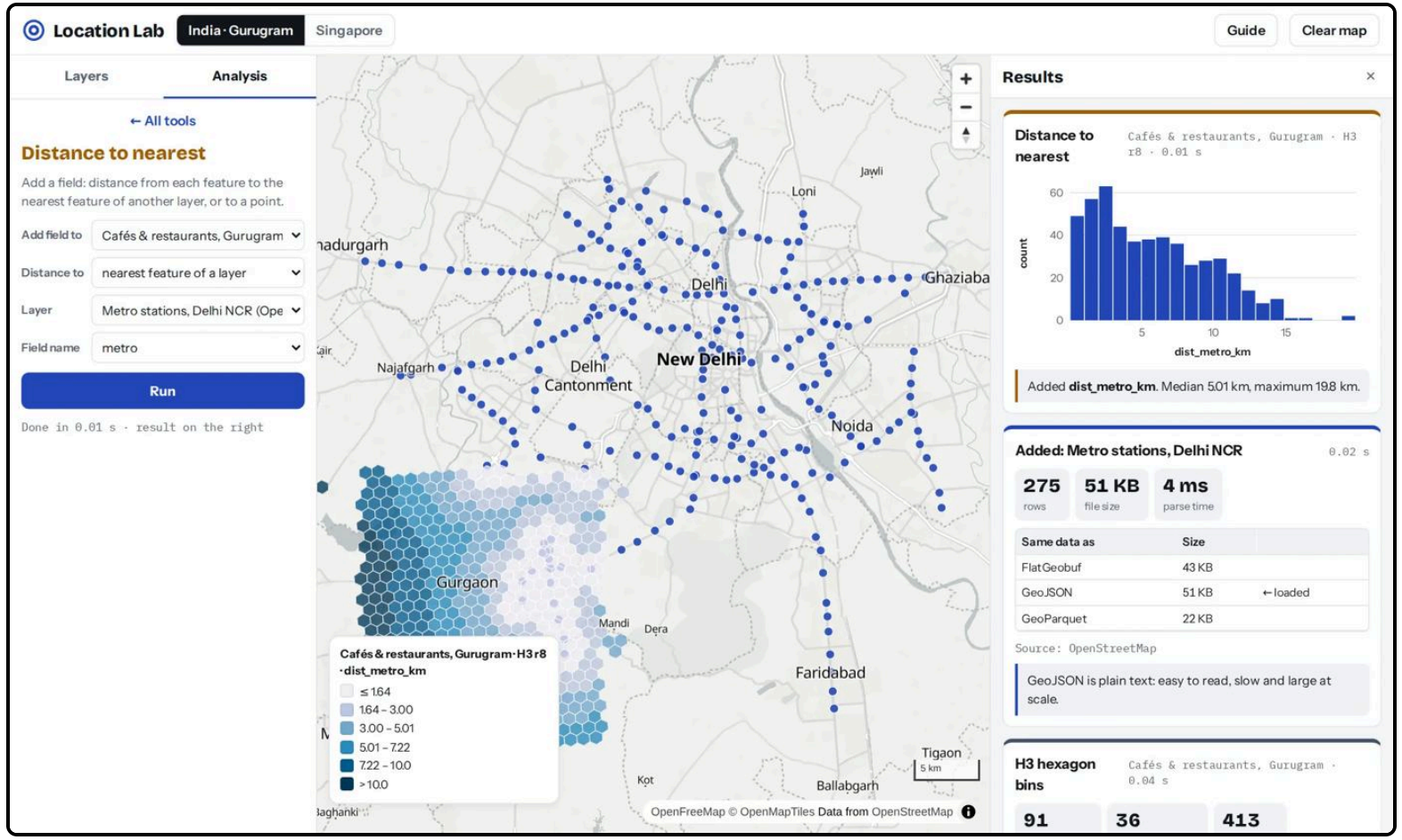
Distance from each hexagon to its nearest metro station.

DO THIS

Analysis tab → drag **Distance to nearest** onto the map (or click it), check the settings, press **Run**.

Add field to	Cafés & restaurants, Gurugram · H3 r8
Distance to	nearest feature of a layer
Layer	Metro stations, Delhi NCR (OpenStreetMap)
Field name	metro

> Added dist_metro_km. Median 5.01 km, maximum 19.8 km.



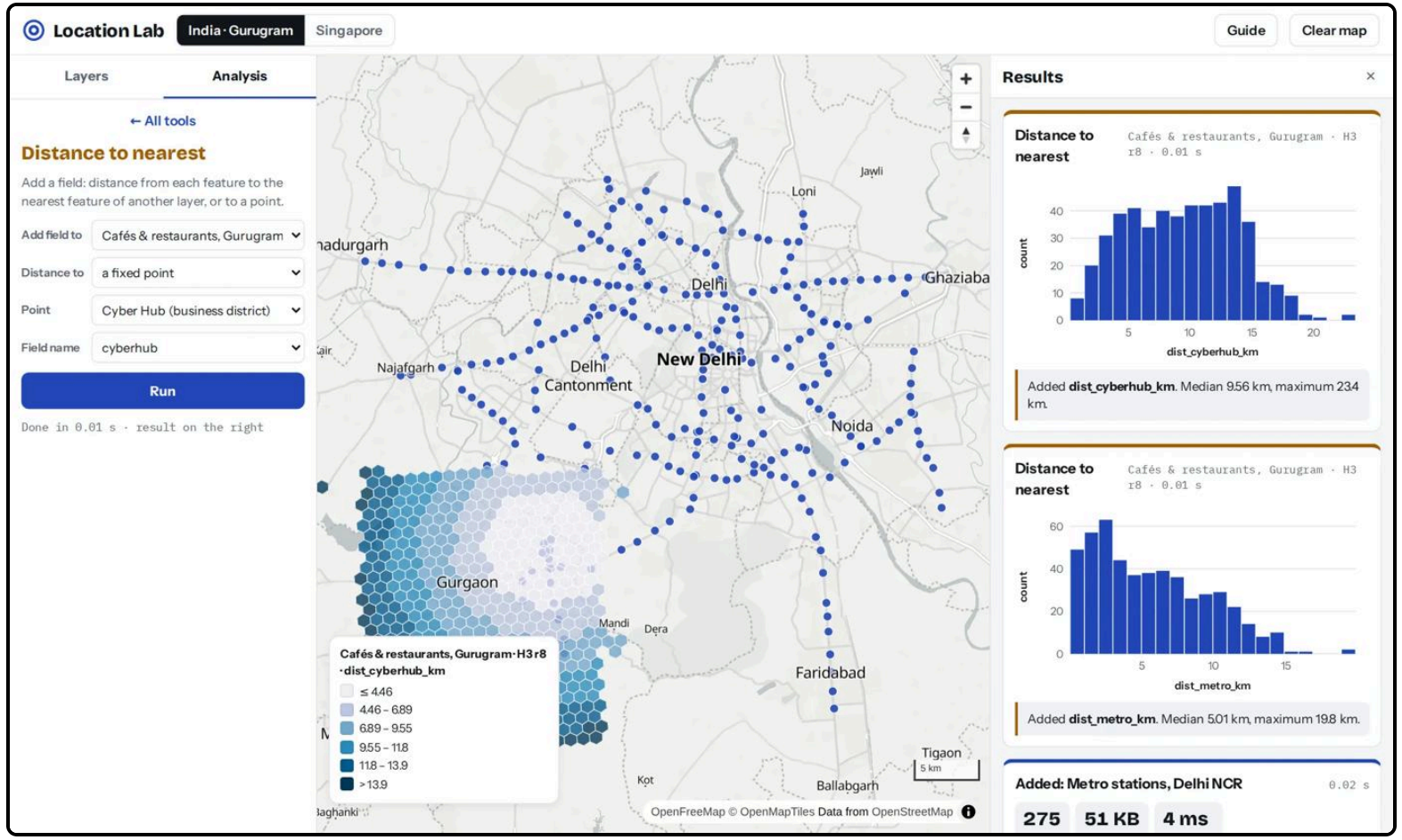
Distance to Cyber Hub, the business district.

DO THIS

Analysis tab → drag **Distance to nearest** onto the map (or click it), check the settings, press **Run**.

Add field to	Cafés & restaurants, Gurugram · H3 r8
Distance to	a fixed point
Point	Cyber Hub (business district)
Field name	cyberhub

> Added dist_cyberhub_km. Median 9.56 km, maximum 23.4 km.



Contiguity weights.

DO THIS

Analysis tab → drag **Spatial weights** onto the map (or click it), check the settings, press **Run**.

Layer	Cafés & restaurants, Gurugram · H3 r8
Neighbours are	touching H3 hexagons

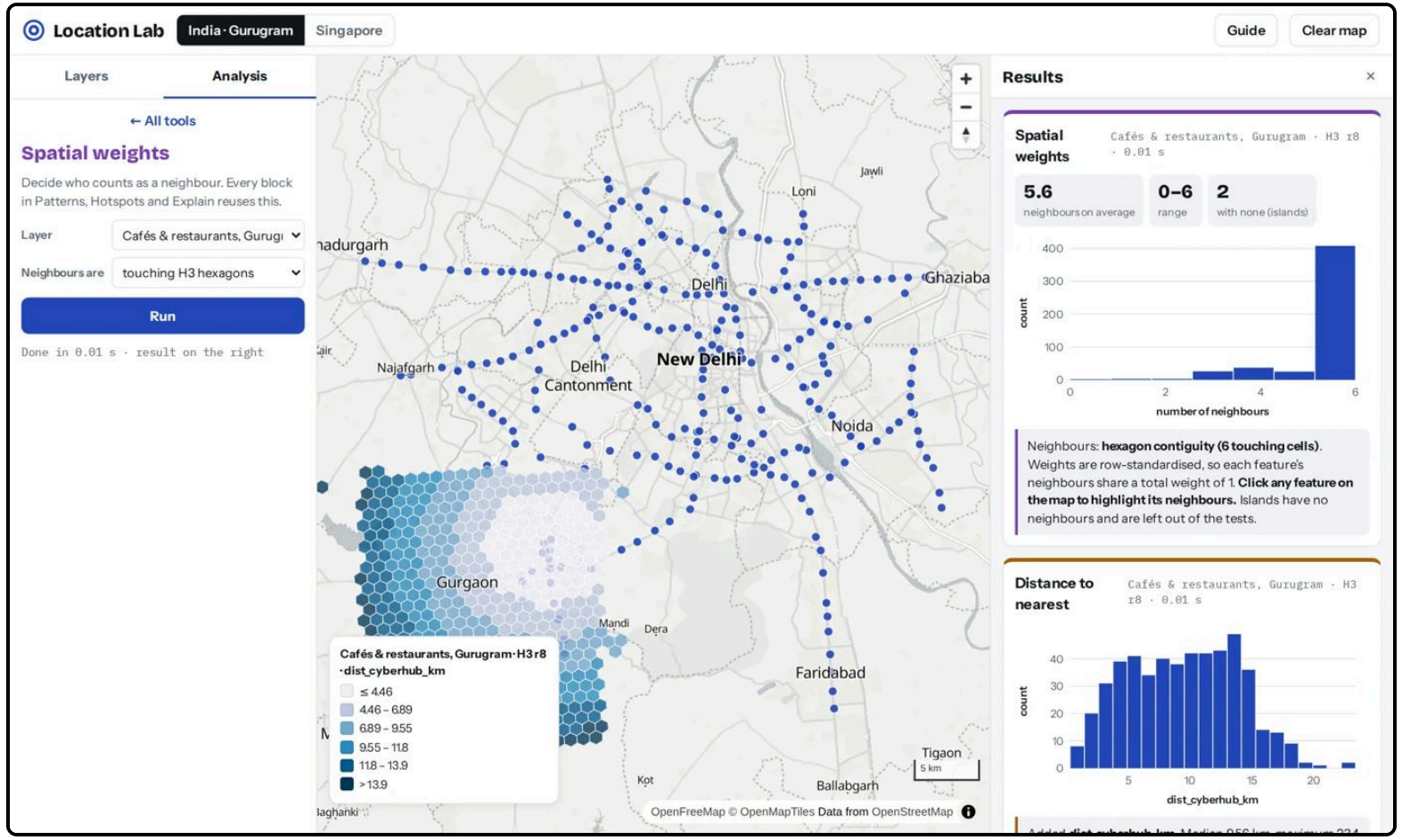
YOU SHOULD SEE

5.6
neighbours on average

0–6
range

2
with none (islands)

> Neighbours: hexagon contiguity (6 touching cells). Weights are row-standardised, so each feature's neighbours share a total weight of 1. Click any feature on the map to highlight its...



Ordinary regression, then Moran’s I on its residuals.

DO THIS

Analysis tab → drag **Ordinary regression (OLS)** onto the map (or click it), check the settings, press **Run**.

Layer	Cafés & restaurants, Gurugram · H3 r8
Explain (y)	count
Using (x)	dist_metro_km, dist_cyberhub_km

YOU SHOULD SEE

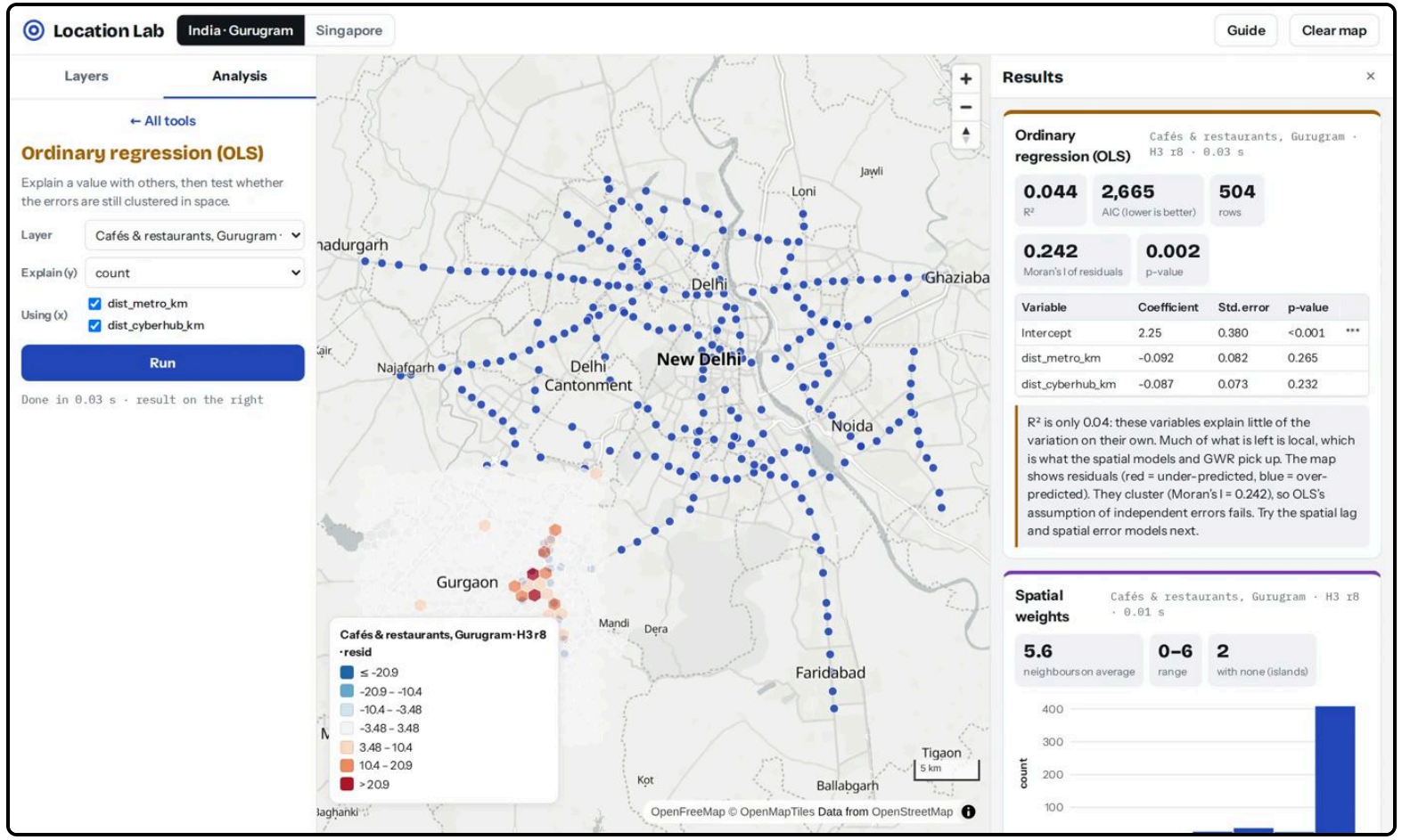
0.044
R^2

2,665
AIC (lower is better)

504
rows

0.242
Moran’s I of residuals

> R^2 is only 0.04: these variables explain little of the variation on their own. Much of what is left is local, which is what the spatial models and GWR pick up. The map shows residuals (red...



Spatial lag model (spillover).

DO THIS

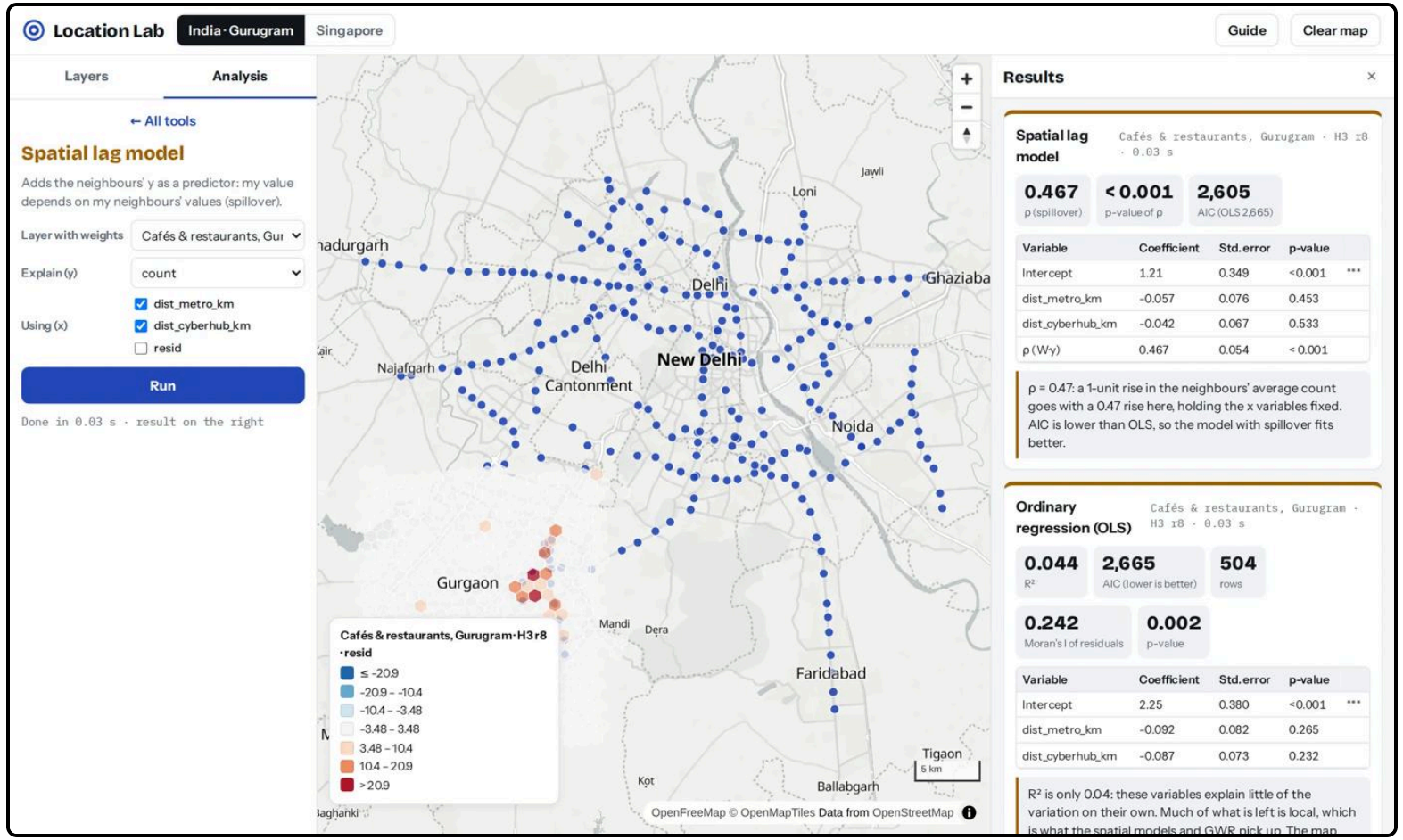
Analysis tab → drag **Spatial lag model** onto the map (or click it), check the settings, press **Run**.

Layer with weights	Cafés & restaurants, Gurugram · H3 r8
Explain (y)	count
Using (x)	dist_metro_km, dist_cyberhub_km

YOU SHOULD SEE

0.467	< 0.001	2,605
ρ (spillover)	p-value of ρ	AIC (OLS 2,665)

> ρ = 0.47: a 1-unit rise in the neighbours' average count goes with a 0.47 rise here, holding the x variables fixed. AIC is lower than OLS, so the model with spillover fits better.



Spatial error model (clustered missing factors).

DO THIS

Analysis tab → drag **Spatial error model** onto the map (or click it), check the settings, press **Run**.

Layer with weights	Cafés & restaurants, Gurugram · H3 r8
Explain (y)	count
Using (x)	dist_metro_km, dist_cyberhub_km

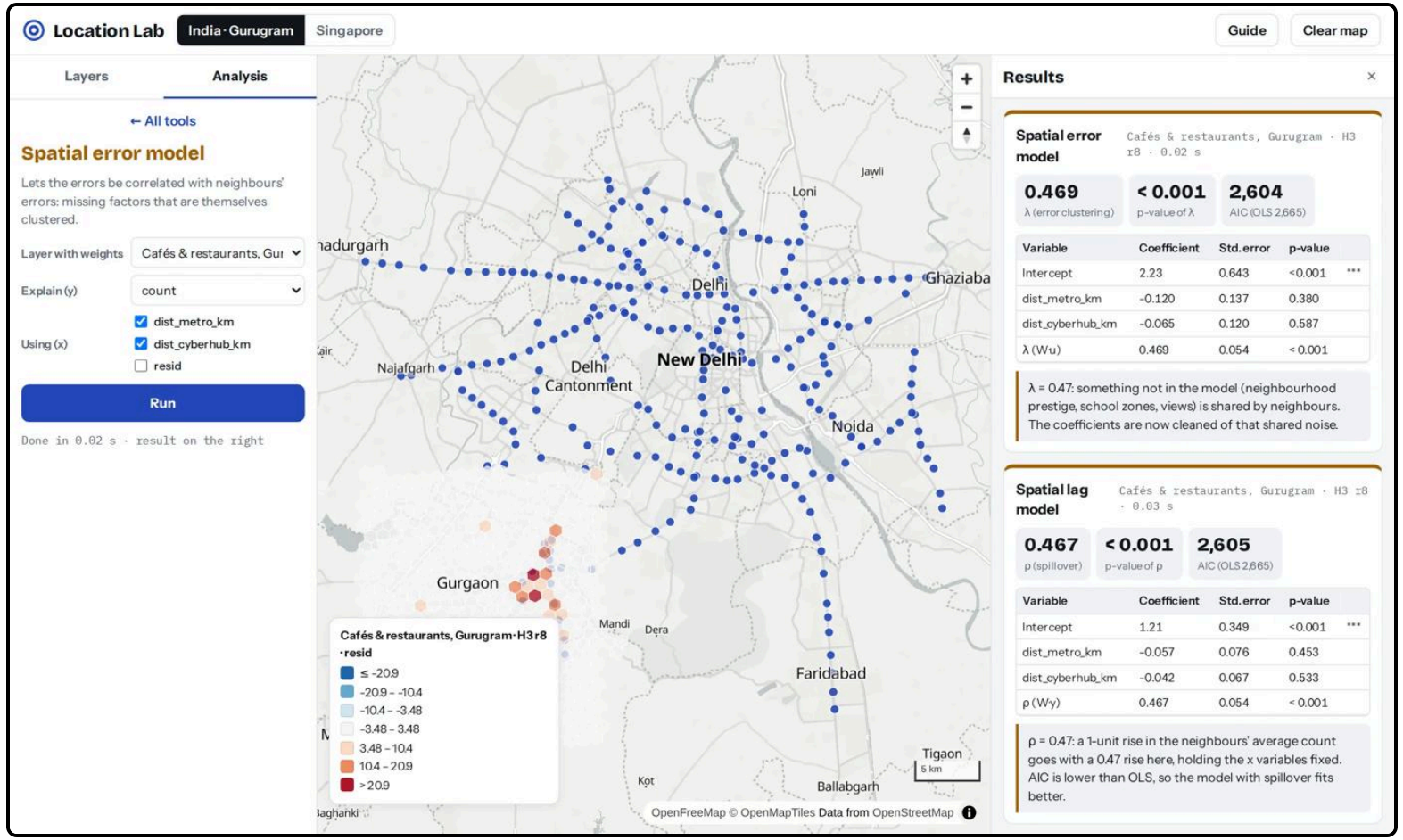
YOU SHOULD SEE

0.469
λ (error clustering)

< 0.001
p-value of λ

2,604
AIC (OLS 2,665)

> λ = 0.47: something not in the model (neighbourhood prestige, school zones, views) is shared by neighbours. The coefficients are now cleaned of that shared noise.



GWR: does the metro effect change across the city?

DO THIS

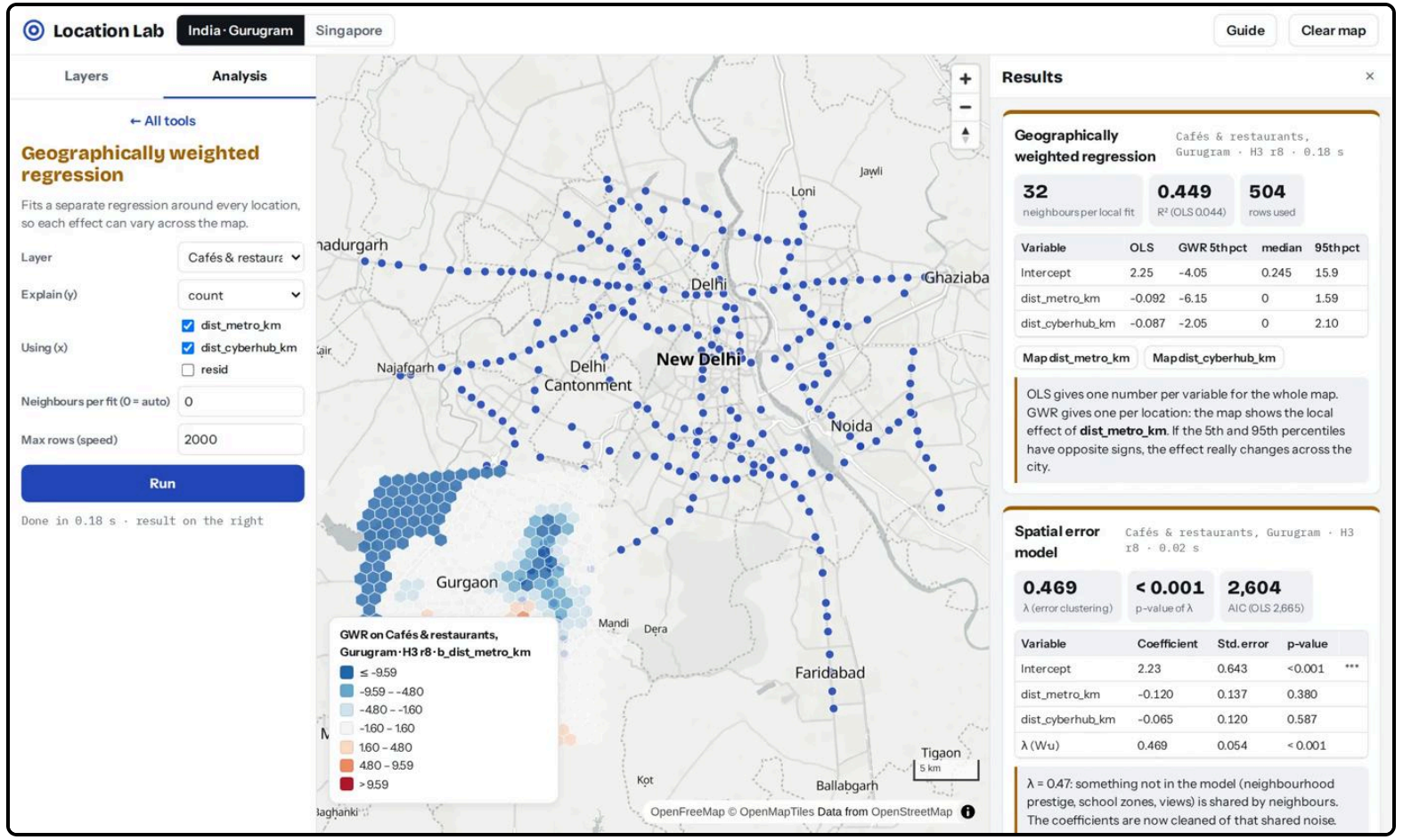
Analysis tab → drag **Geographically weighted regression** onto the map (or click it), check the settings, press **Run**.

Layer	Cafés & restaurants, Gurugram · H3 r8
Explain (y)	count
Using (x)	dist_metro_km, dist_cyberhub_km
Neighbours per fit (0 = auto)	0
Max rows (speed)	2000

YOU SHOULD SEE

32	0.449	504
neighbours per local fit	R^2 (OLS 0.044)	rows used

> OLS gives one number per variable for the whole map. GWR gives one per location: the map shows the local effect of dist_metro_km. If the 5th and 95th percentiles have opposite signs, the…



Three questions

1. OLS R^2 is 0.04 but GWR reaches 0.45. What does that suggest?
2. What is the difference between ρ and λ?
3. Which model do you prefer, and by what rule?

Answers

1. **OLS R^2 is 0.04 but GWR reaches 0.45. What does that suggest?**
   The factors matter, but their effects differ by area; one city-wide coefficient averages them away.
2. **What is the difference between ρ and λ?**
   ρ measures spillover from neighbours' outcomes; λ measures shared unmeasured factors in the errors.
3. **Which model do you prefer, and by what rule?**
   The one with the lower AIC, provided its residuals are no longer spatially clustered.

FINAL PRODUCT

A one-page decision brief

Question. A café chain wants its next three Gurugram outlets.

Evidence. Cafés are strongly clustered (R = 0.30) and café density is spatially autocorrelated (I = 0.273). LISA finds 24 High-High hexagons; DBSCAN finds 20 dining clusters. Metro and Cyber Hub distance alone explain little (R^2 0.04), but GWR lifts fit to 0.45: access matters differently by area.

Decision. Shortlist Low-High hexagons (a weak spot inside a strong area) next to existing clusters and within the local zones where GWR shows a strong metro effect. Validate footfall on site before signing.

Your deliverable

One page: the question, three maps or charts exported from the tool (layer card → download, or a screenshot), the numbers that matter, and a decision with its main risk. Use the same structure as the example on the left.

CLOSING

Glossary

Lag (time)	The series shifted by one period	**LISA**	Local Moran: hotspots, coldspots, outliers
Rook / queen	Neighbours share an edge / an edge or corner	**Gi***	z-score of the local total vs chance
Weights W	Who is a neighbour, row-standardised to sum to 1	**DBSCAN**	Density clusters from ε and minPts; noise flagged
Spatial lag Wy	Neighbours' weighted average	**IDW**	Weighted average, weight 1 ÷ distance^p
CSR · R	Randomness benchmark · observed ÷ expected NN distance	**Variogram**	Nugget, sill, range of fading similarity
Ripley's L	Clustering at many distances, with envelope	**Kriging**	Variogram-weighted prediction with standard error
Moran's I	Slope of z against its spatial lag	**SLM ρ · SEM λ**	Spillover · shared unmeasured factors
Permutation test	Compare with 999 shuffled maps	**GWR**	Local regressions; bandwidth by AICc